

1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

REV	ECN	DESCRIPTION OF REVISION	CK APPD DATE
C	0000734528	PRODUCTION RELEASED	2009-06-04

K24 MLB SCHEMATIC

6/12/2009

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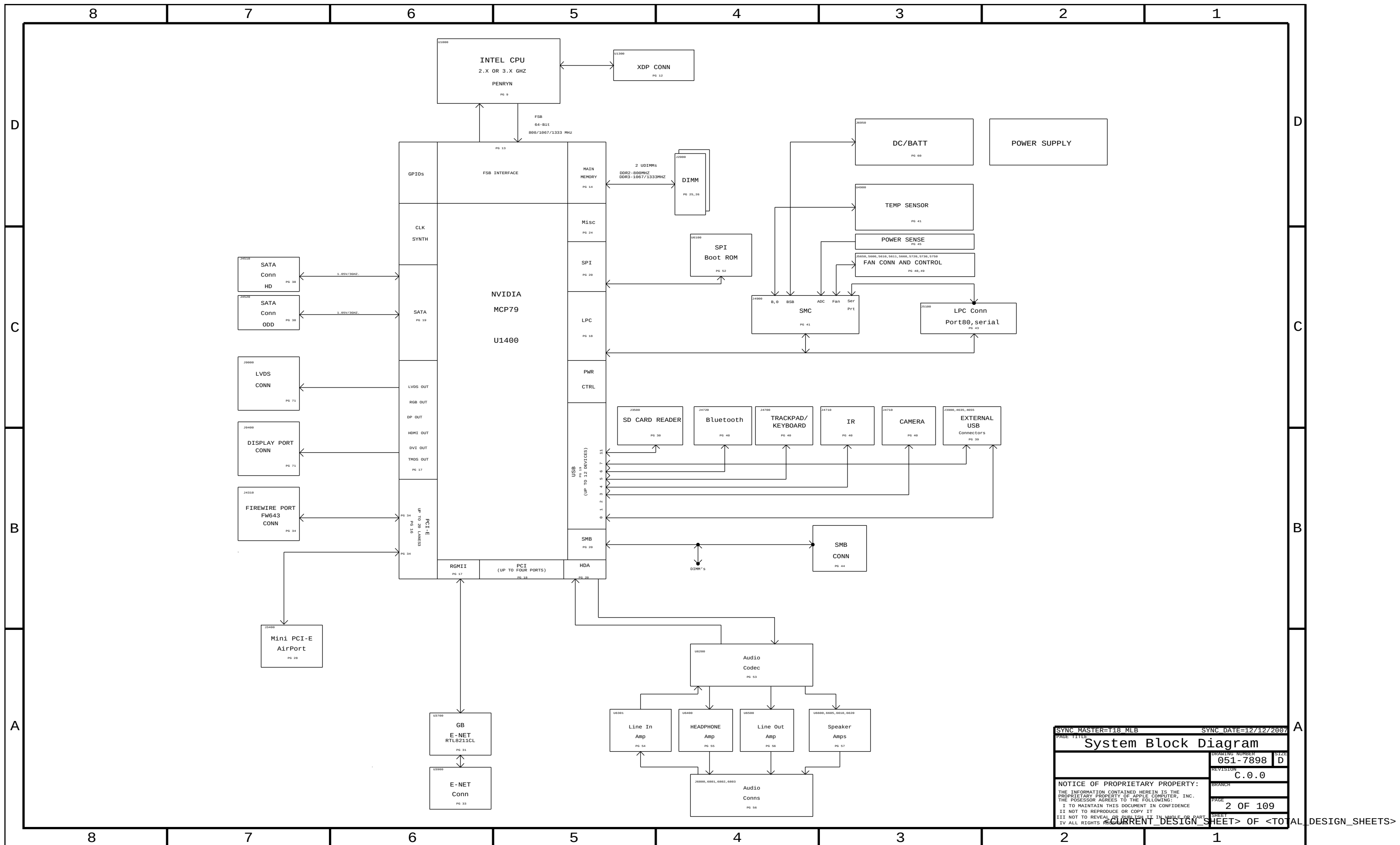
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Schematic / PCB #'s

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
051-7898	1	SCHEM, MLB, K24	SCH	CRITICAL	
820-2530	1	PCBF, MLB, K24	PCB	CRITICAL	

DRAWING TITLE				
SCHEM, MLB, K24				
	DRAWING NUMBER	051-7898	SIZE	D
	REVISION			
	C.0.0			
	BRANCH			
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K24 POWER SYSTEM ARCHITECTURE

This schematic diagram illustrates the power system architecture for the K24 platform. It details the power flow from the AC adapter and battery through various regulators and controllers to the CPU and other components.

Key Components and Connections:

- AC Adapter:** DCIN (16.5V) connected to the **PBUS SUPPLY/BATTERY CHARGER** (ISL6258A U7000).
- Battery:** BATT_POS_F (3S2P, 9 TO 12.6V) connected to the **PBUS SUPPLY/BATTERY CHARGER** and the **CHGR_BGATE** (Q7050).
- Power Management ICs:**
 - SMC (U4900):** Manages system power, including SLP_S3# and SLP_S3_L signals.
 - MCP79 (U1400):** Manages power for the CPU and other components.
 - TPS51125 (U7200):** 5V (RT) regulator for the CPU.
 - TPS51116 (U7300):** 1.5V regulator for the CPU.
 - TPS51117 (U7600):** 1.05V regulator for the CPU.
 - TPS51117 (U7600):** 1.05V regulator for the CPU.
 - TPS51117 (U7600):** 1.05V regulator for the CPU.
- Regulators and FETs:**
 - PP3V3_S5_REG (U7750):** 3.3V regulator for the CPU.
 - PP3V3_S3_REG (U7750):** 3.3V regulator for the CPU.
 - PP3V3_S0_FET (Q7930):** FET for the 3.3V S0 power rail.
 - PP3V3_S3_FET (Q7910):** FET for the 3.3V S3 power rail.
 - PP3V3_S5_FET (Q7910):** FET for the 3.3V S5 power rail.
 - PP3V3_S3_FET (Q7910):** FET for the 3.3V S3 power rail.
 - PP3V3_S5_FET (Q7910):** FET for the 3.3V S5 power rail.
- Control and Monitoring:**
 - Q5315:** Voltage sense for the CPU.
 - Q7050:** CHGR_BGATE FET.
 - Q7800:** SMC_PM_G2_EN FET.
 - Q7910:** P3V3S3_EN FET.
 - Q7930:** P3V3S0_EN FET.
 - Q7940:** P5V_S0_FET.
 - Q7910:** P3V3S3_EN FET.
 - Q7930:** P3V3S0_EN FET.
 - Q7940:** P5V_S0_FET.

Legend:

- 01:** AC Adapter
- 02:** Battery
- 03:** CPU
- 04:** SMC
- 05:** MCP79
- 06:** CPU
- 07:** CPU
- 08:** CPU
- 09:** CPU
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- 32:** CPU

Revision History:

REV	DESCRIPTION
001	Initial Release
002	Added SMC Adapter En
003	Added SMC Adapter En
004	Added SMC Adapter En
005	Added SMC Adapter En
006	Added SMC Adapter En
007	Added SMC Adapter En
008	Added SMC Adapter En
009	Added SMC Adapter En
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095	Added SMC Adapter En</

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BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
630 - 9923	PCBA, MLB, BETTER, K24	K24_COMMON, CPU_2_26GHZ, EEE_6GC, KB_BL
630 - 9924	PCBA, MLB, BEST, K24	K24_COMMON, CPU_2_53GHZ, EEE_6GD, KB_BL

Bar Code Labels / EEE #'s

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
826-4393	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEE:6G4]	CRITICAL	EEE_6G4
826-4393	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEE:6GC]	CRITICAL	EEE_6GC
826-4393	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEE:6GD]	CRITICAL	EEE_6GD

BOM Groups

BOM_GROUP	BOM_OPTIONS
K24_COMMON	COMMON, ALTERNATE, K24_MCP, K24_MISC, K24_DEBUG_PROD, K24_PROGPARTS
K24_MCP	MCP_B83, BOOT_MODE_USER, MCPSEQ_SMC
K24_MISC	ONEWIRE_PU, DP_ESD, MIKEY, BKLT_PROD, SUPERCAP_NO, LDO_NO
K24_PROGPARTS	BOOTROM_PROD, SMC_PROD, IR_PROD, WELLSPRING_PROD
K24_DEBUG_ENG	DEVEL_BOM, SMC_DEBUG_YES, XDP
K24_DEBUG_PVT	DEVEL_BOM, BMON_PROD, SMC_DEBUG_YES, XDP, NO_VREFMRGN
K24_DEBUG_PROD	BMON_PROD, SMC_DEBUG_YES, XDP, LPCPLUS_NOT, NO_VREFMRGN
K24_DEVEL_ENG	BMON_ENG, XDP_CONN, LPCPLUS, VREFMRGN, FWPHY_WAKE_YES
K24_DEVEL_PVT	LPCPLUS

Module Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
337S3646	1	PDC, SLGBE, PRQ, 2.8, 25W, 1066, M0, 3M, BGA	U1000	CRITICAL	CPU_2_6GHZ
337S3764	1	PDC, SLGE2, PRQ, 2.26, 25W, 1066, R0, 3M, BGA	U1000	CRITICAL	CPU_2_26GHZ
337S3639	1	PDC, SLB4N, PRQ, 2.4, 25W, 1066, M0, 3M, BGA	U1000	CRITICAL	CPU_2_4GHZ
337S3756	1	PDC, SLGF6, PRQ, 2.53, 25W, 1066, R0, 3M, BGA	U1000	CRITICAL	CPU_2_53GHZ
337S3761	1	PDC, SLGLA, PRQ, 2.66, 25W, 1066, E0, 3M, BGA	U1000	CRITICAL	CPU_2_66GHZ
338S0710	1	IC, GMPD, MCP79, 35X35MM, BGA1437, B03	U1400	CRITICAL	MCP_B03

Programmable Parts

338S0563	1	IC, SMC, H58/2117, 9X9MM, TLP, HF	U4900	CRITICAL	SMC_BLANK
341S2445		IC, SMC, K24	U4900	CRITICAL	SMC_PROG
335S0610	1	IC, FLASH, SPI, 32MBIT, 3-3V, 80MHZ, 8-SOP	U6100	CRITICAL	BOOTROM_BLANK
341S2441	1	IC, PRGRM, EFI BOOTROM, UNLOCK, K24	U6100	CRITICAL	BOOTROM_PROG
338S0375	1	IC, CY7C63833, ENCORE II, USB CONTROLLER	U4800	CRITICAL	IR_BLANK
341S2093	1	IC, IR CONTROLLER, M97	U4800	CRITICAL	IR_PROG
337S2983	1	IC, P50C+ / USB, 56 PIN, WLF, CY8C24794	U5701	CRITICAL	WELLSPRING_BLANK
341S2503	1	IC, PRGRM, WELLSPRING CONTROLLER	U5701	CRITICAL	WELLSPRING_PROG

LOCKED BOOTROM APN IS 341S2443

Alternate Parts

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
15250778	15250693		ALL	CYNTEC AS ALTERNATE
15250796	15250685		ALL	CYNTEC AS ALTERNATE
15750058	15750055		ALL	DELTA AS ALTERNATE
10450018	10450023		ALL	DALE/VISHAY AS ALTERNATE
128S0093	128S0218		ALL	KEMET AS ALTERNATE
15250874	15250516		ALL	MAGLAYERS AS ALTERNATE
15250847	15250586		ALL	MAGLAYERS AS ALTERNATE
15251025	15251024		ALL	TOKO AS ALTERNATE
337S3769	337S3704		ALL	INTEL P7550 CPU AS ALTERNATE
353S2718	353S2310		ALL	INTERSEL AS ALTERNATE

DEVELOPMENT BOM

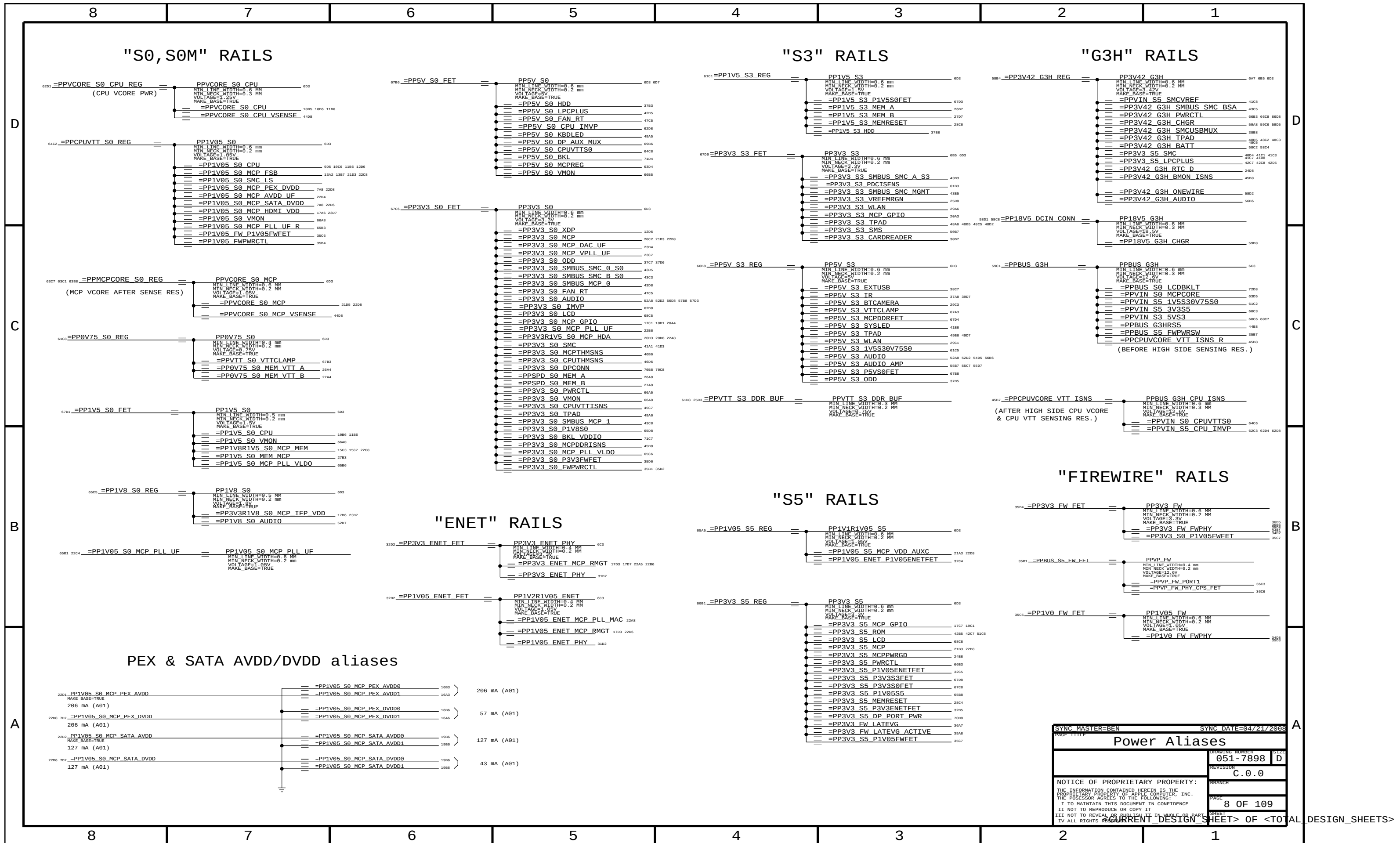
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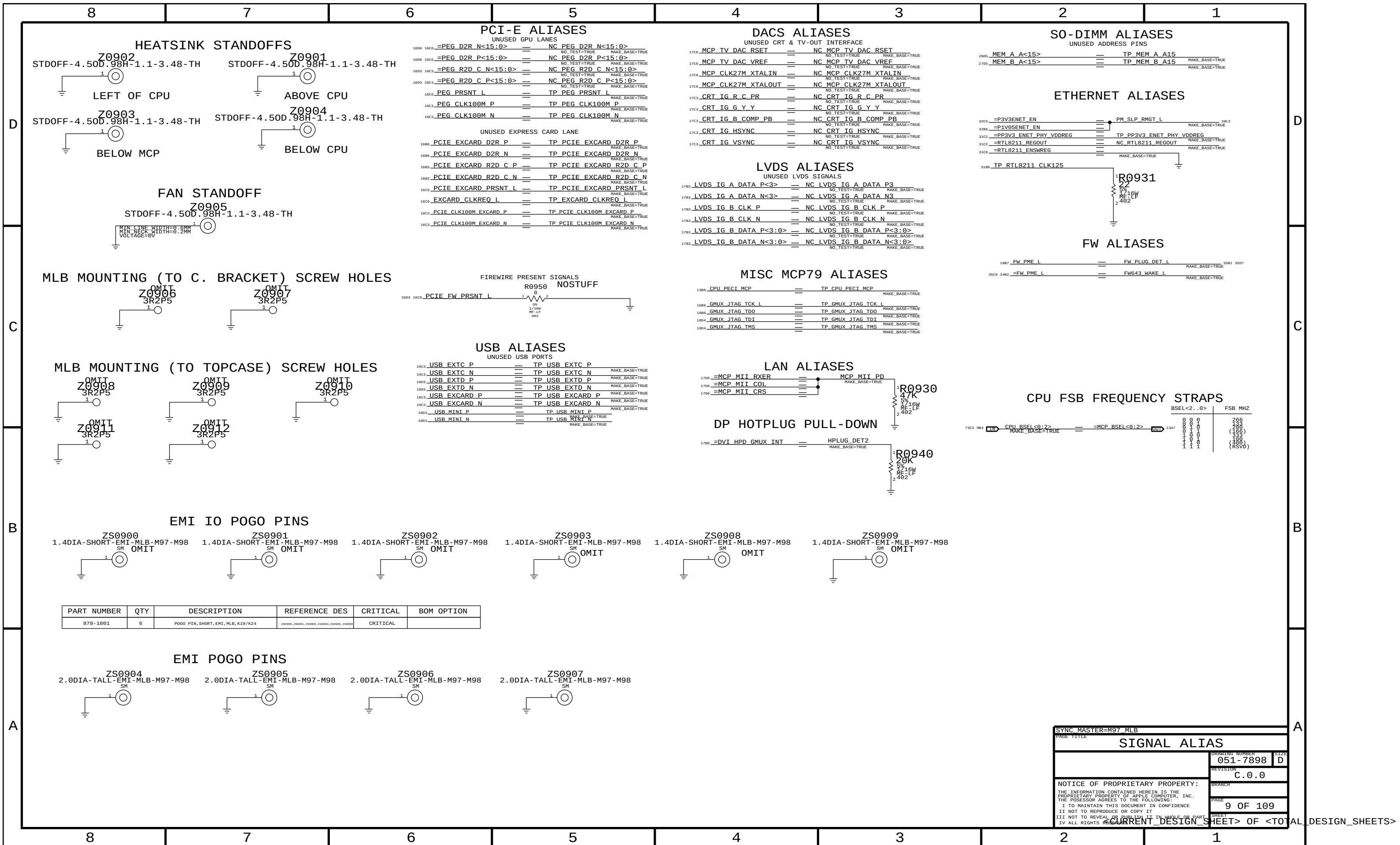
K24 BOARD STACK-UP

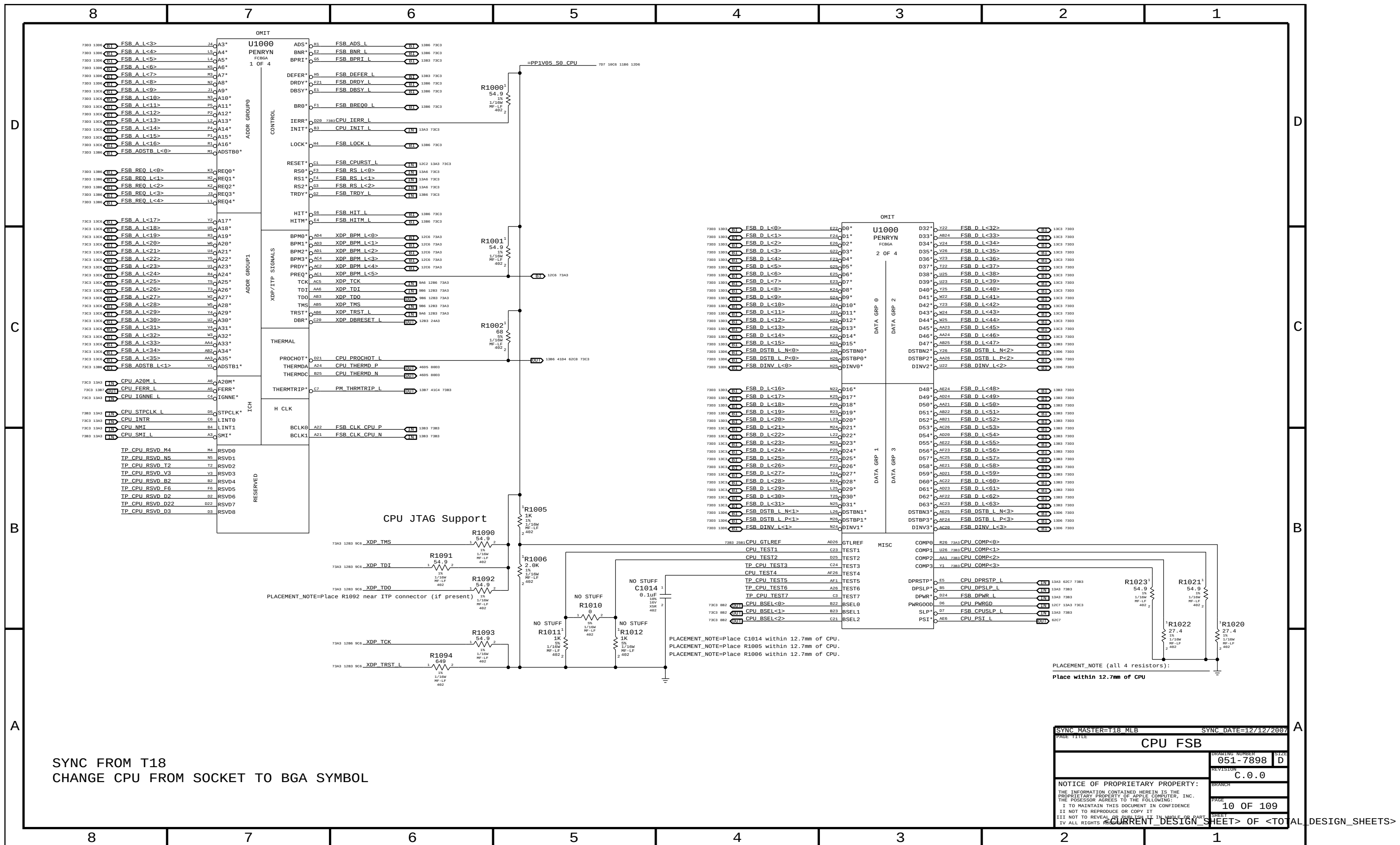
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2		GROUND
3		SIGNAL (High Speed)
4		SIGNAL (High Speed)
5		GROUND
6		POWER
7		POWER
8		GROUND
9		SIGNAL (High Speed)
10		SIGNAL (High Speed)
11		GROUND
BOTTOM		SIGNAL

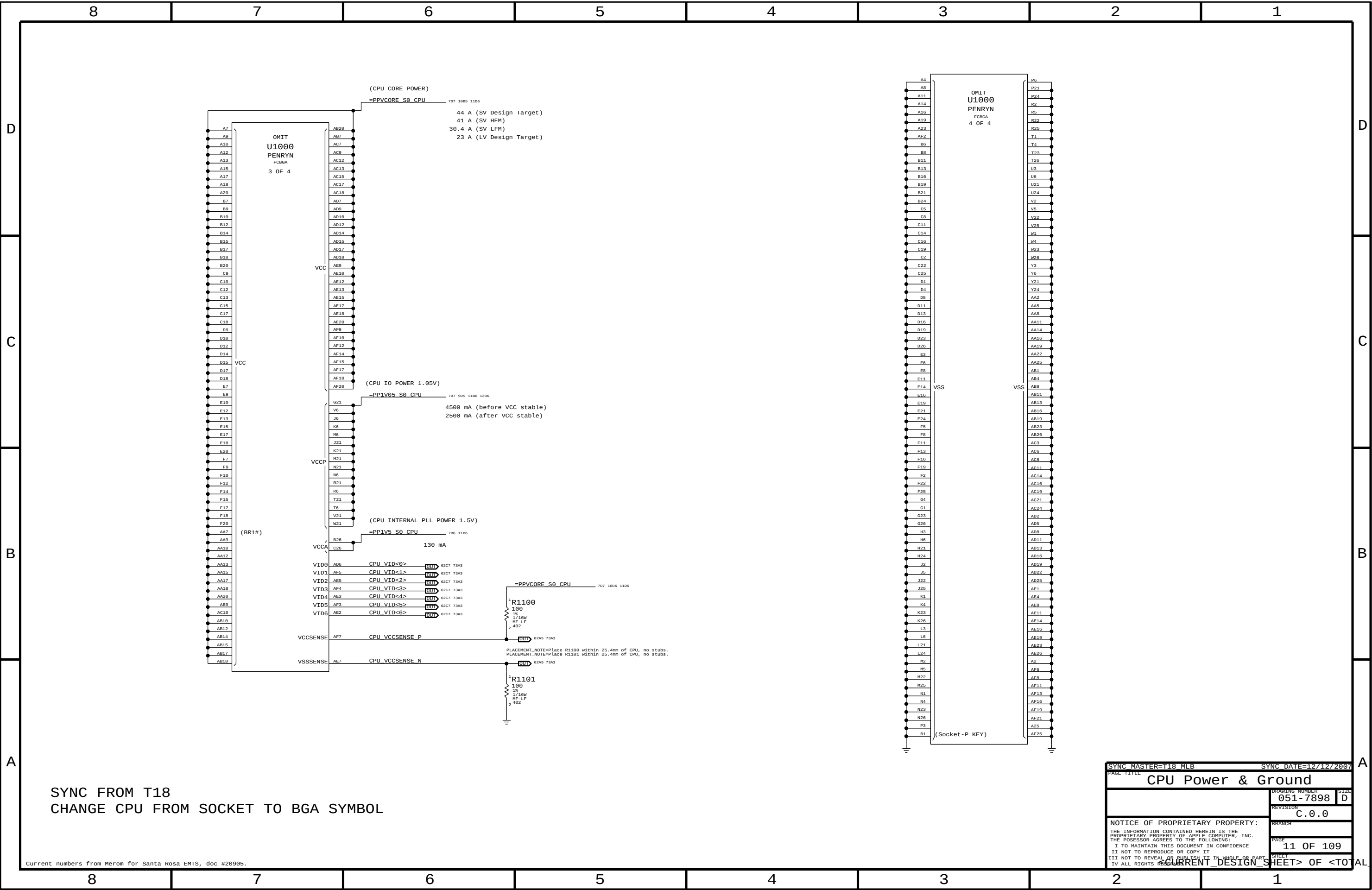
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	Functional Test Points								
D	Fan Connectors		RIGHT CLUTCH CONN		DEBUG VOLTAGE				D
	TRUE PP5V S0 (NEED 3 TP) TRUE FAN RT PWM TRUE FAN RT TACH (NEED TO ADD 3 GND TP)		TRUE PP5V S3 BITCAMERA F TRUE PCIE MINI D2R P TRUE PCIE MINI D2R N TRUE PCIE MINI R2D P TRUE PCIE MINI R2D N TRUE PCIE CLK100M MINI CONN P TRUE PCIE CLK100M MINI CONN N TRUE USB CAMERA CONN P TRUE USB CAMERA CONN N TRUE PP5V WLAN TRUE PCIE WAKE L TRUE SMBUS SMC A S3 SCL TRUE SMBUS SMC A S3 SDA TRUE CONN USB2 BT P TRUE CONN USB2 BT N TRUE MINI CLKREQ 0 L TRUE MINI RESET CONN L (NEED TO ADD 6 GND TP)		TRUE PPVCORE S0 CPU TRUE PPVCORE S0 MCP TRUE PP0V75 S0 TRUE PP1V05 S0 TRUE PP1V5 S0 TRUE PP1V8 S0 TRUE PP5V S0 TRUE PP3V3 S0 TRUE PP1V5 S3 TRUE PP3V3 S3 TRUE PP5V S3 TRUE PP1V1R1V05 S5 TRUE PP3V3 S5 TRUE PP3V42 G3H TRUE PPBUS G3H TRUE PP3V3 ENET PHY TRUE PP1V2R1V05 ENET TRUE PP3V3 G3 RTC TRUE PP5V WLAN TRUE PP5V SW ODD TRUE PP5V S0 HDD FLT TRUE PP3V3 S5 AVREF SMC TRUE PP18V5 S3 TRUE PP3V3 S3 LDO TRUE PP3V3 LCDVDD SW F TRUE PPVOUT S0 LCDBKLT TRUE PP4V5 AUDIO ANALOG TRUE SMC PM G2 EN TRUE PM SLP S4 L TRUE PM SLP S3 L (NEED TO ADD 4 GND TP)				C
	MIC FUNC_TEST TRUE BI MIC LO TRUE BI MIC HI TRUE BI MIC SHIELD		TRUE PP3V3 S3 LDO TRUE PP18V5 S3 TRUE Z2 CS L TRUE Z2 DEBUG3 TRUE Z2 MOST TRUE Z2 MISO TRUE Z2 SCLK TRUE Z2 BOOST EN TRUE Z2 HOST INTN TRUE Z2 CLKIN TRUE Z2 KEY ACT L TRUE Z2 RESET TRUE PSOC MISO TRUE PSOC MOSI TRUE PSOC SCLK TRUE SMBUS SMC A S3 SDA TRUE SMBUS SMC A S3 SCL TRUE PSOC F CS L TRUE PICKB L		TRUE PP3V3 S3 LDO TRUE PP18V5 S3 TRUE Z2 CS L TRUE Z2 DEBUG3 TRUE Z2 MOST TRUE Z2 MISO TRUE Z2 SCLK TRUE Z2 BOOST EN TRUE Z2 HOST INTN TRUE Z2 CLKIN TRUE Z2 KEY ACT L TRUE Z2 RESET TRUE PSOC MISO TRUE PSOC MOSI TRUE PSOC SCLK TRUE SMBUS SMC A S3 SDA TRUE SMBUS SMC A S3 SCL TRUE PSOC F CS L TRUE PICKB L				B
	SPEAKER FUNC_TEST TRUE SPKRAMP L N OUT TRUE SPKRAMP L P OUT TRUE SPKRAMP R N OUT TRUE SPKRAMP R P OUT TRUE SPKRAMP SUB N OUT TRUE SPKRAMP SUB P OUT		IPD_FLEX_CONN TRUE PP3V3 S3 LDO TRUE PP18V5 S3 TRUE Z2 CS L TRUE Z2 DEBUG3 TRUE Z2 MOST TRUE Z2 MISO TRUE Z2 SCLK TRUE Z2 BOOST EN TRUE Z2 HOST INTN TRUE Z2 CLKIN TRUE Z2 KEY ACT L TRUE Z2 RESET TRUE PSOC MISO TRUE PSOC MOSI TRUE PSOC SCLK TRUE SMBUS SMC A S3 SDA TRUE SMBUS SMC A S3 SCL TRUE PSOC F CS L TRUE PICKB L		TRUE PP3V3 S3 LDO TRUE PP18V5 S3 TRUE Z2 CS L TRUE Z2 DEBUG3 TRUE Z2 MOST TRUE Z2 MISO TRUE Z2 SCLK TRUE Z2 BOOST EN TRUE Z2 HOST INTN TRUE Z2 CLKIN TRUE Z2 KEY ACT L TRUE Z2 RESET TRUE PSOC MISO TRUE PSOC MOSI TRUE PSOC SCLK TRUE SMBUS SMC A S3 SDA TRUE SMBUS SMC A S3 SCL TRUE PSOC F CS L TRUE PICKB L				A
C	THERMAL FUNC_TEST TRUE MCPTHMSNS D2 P TRUE MCPTHMSNS D2 N		KEYBOARD CONN TRUE PP3V3 S3 TRUE PP3V42 G3H TRUE WS KBD1 TRUE WS KBD2 TRUE WS KBD3 TRUE WS KBD4 TRUE WS KBD5 TRUE WS KBD6 TRUE WS KBD7 TRUE WS KBD8 TRUE WS KBD9 TRUE WS KBD10 TRUE WS KBD11 TRUE WS KBD12 TRUE WS KBD13 TRUE WS KBD14 TRUE WS KBD15 CAP TRUE WS KBD16 NUM TRUE WS KBD17 TRUE WS KBD18 TRUE WS KBD19 TRUE WS KBD20 TRUE WS KBD21 TRUE WS KBD22 TRUE WS KBD23 TRUE WS KBD ONOFF L TRUE WS LEFT SHIFT KBD TRUE WS LEFT OPTION KBD TRUE WS CONTROL KBD (NEED TO ADD 1 GND TP)		TRUE PP3V3 S3 LDO TRUE PP18V5 S3 TRUE Z2 CS L TRUE Z2 DEBUG3 TRUE Z2 MOST TRUE Z2 MISO TRUE Z2 SCLK TRUE Z2 BOOST EN TRUE Z2 HOST INTN TRUE Z2 CLKIN TRUE Z2 KEY ACT L TRUE Z2 RESET TRUE PSOC MISO TRUE PSOC MOSI TRUE PSOC SCLK TRUE SMBUS SMC A S3 SDA TRUE SMBUS SMC A S3 SCL TRUE PSOC F CS L TRUE PICKB L				A
	LVDS FUNC_TEST TRUE PP3V3 LCDVDD SW F TRUE PP3V3 S0 LCD F TRUE PPVOUT S0 LCDBKLT TRUE LVDS IG DDC CLK TRUE LVDS IG DDC DATA TRUE LVDS IG A DATA N<0> TRUE LVDS IG A DATA P<0> TRUE LVDS IG A DATA N<1> TRUE LVDS IG A DATA P<1> TRUE LVDS IG A DATA N<2> TRUE LVDS IG A DATA P<2> TRUE LVDS IG A CLK F N TRUE LVDS IG A CLK F P TRUE LED RETURN 1 TRUE LED RETURN 2 TRUE LED RETURN 3 TRUE LED RETURN 4 TRUE LED RETURN 5 TRUE LED RETURN 6 TRUE TP BKL SYNC (NEED TO ADD 5 GND TP)		DC POWER CONN TRUE PP18V5 DCIN FUSE (NEED 3 TP) TRUE ADAPTER SENSE (NEED TO ADD 4 GND TP)		TRUE PP3V3 S3 LDO TRUE PP18V5 S3 TRUE Z2 CS L TRUE Z2 DEBUG3 TRUE Z2 MOST TRUE Z2 MISO TRUE Z2 SCLK TRUE Z2 BOOST EN TRUE Z2 HOST INTN TRUE Z2 CLKIN TRUE Z2 KEY ACT L TRUE Z2 RESET TRUE PSOC MISO TRUE PSOC MOSI TRUE PSOC SCLK TRUE SMBUS SMC A S3 SDA TRUE SMBUS SMC A S3 SCL TRUE PSOC F CS L TRUE PICKB L				A
	SATA ODD CONN TRUE PP5V SW ODD (NEED 4 TP) TRUE SMC ODD DETECT TRUE SATA ODD D2R C P TRUE SATA ODD D2R C N TRUE SATA ODD R2D P TRUE SATA ODD R2D N (NEED TO ADD 4 GND TP)		KBD BACKLIGHT CONN TRUE KBDLED ANODE (NEED 2 TP) TRUE SMC_KDBLED_PRESENT_L (NEED TO ADD 2 GND TP)		TRUE PP3V3 S3 LDO TRUE PP18V5 S3 TRUE Z2 CS L TRUE Z2 DEBUG3 TRUE Z2 MOST TRUE Z2 MISO TRUE Z2 SCLK TRUE Z2 BOOST EN TRUE Z2 HOST INTN TRUE Z2 CLKIN TRUE Z2 KEY ACT L TRUE Z2 RESET TRUE PSOC MISO TRUE PSOC MOSI TRUE PSOC SCLK TRUE SMBUS SMC A S3 SDA TRUE SMBUS SMC A S3 SCL TRUE PSOC F CS L TRUE PICKB L				A
	SATA HDD/IR/SIL TRUE PP5V S0 HDD FLT (NEED 4 TP) TRUE SATA HDD R2D P TRUE SATA HDD R2D N TRUE SATA HDD D2R C P TRUE SATA HDD D2R C N TRUE SYS LED ANODE R TRUE IR RX OUT TRUE PP5V S3 IR R (NEED TO ADD 4 GND TP)		BATT POWER CONN TRUE SMBUS SMC BSA SCL 						



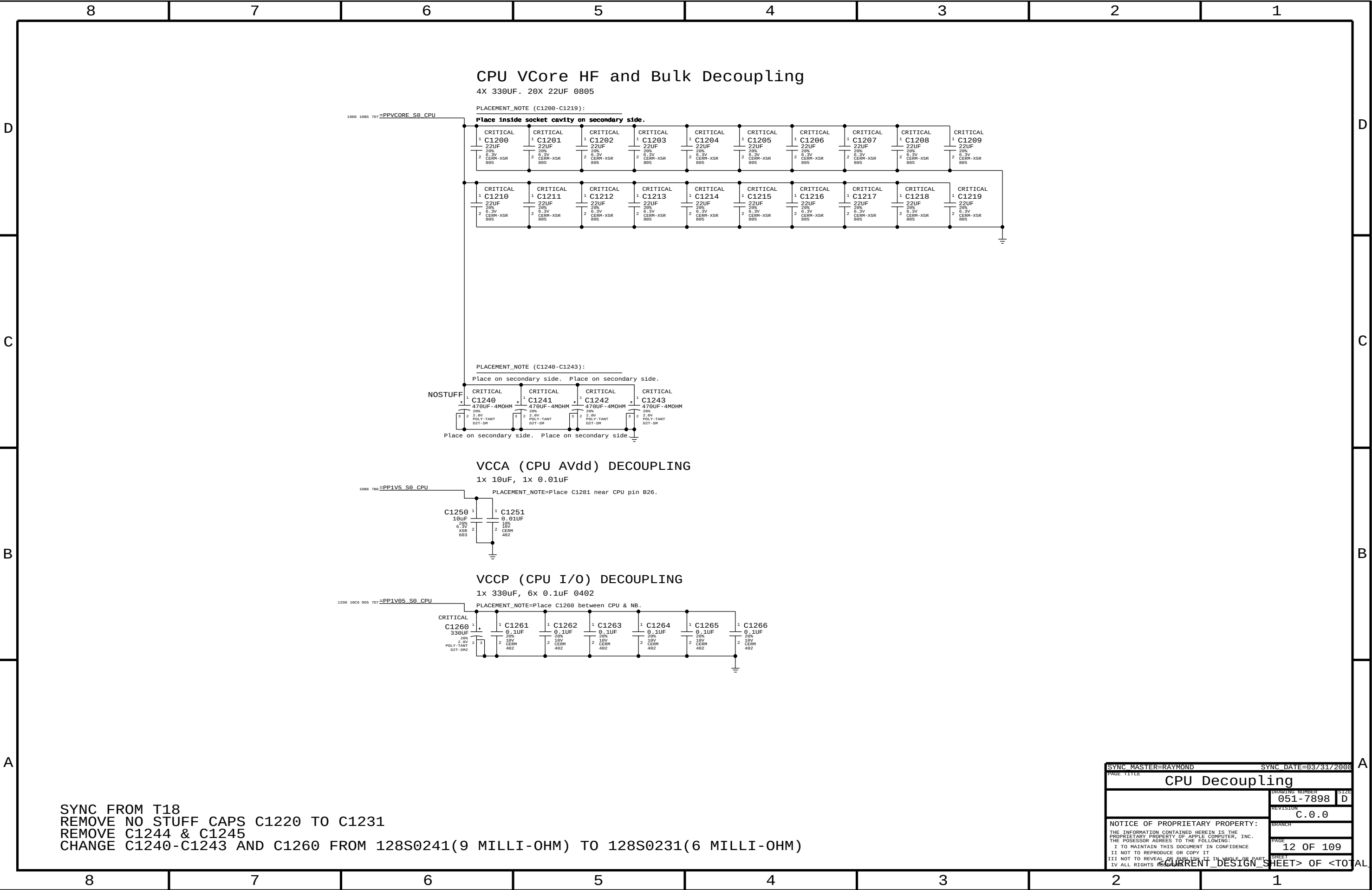






SYNC FROM T18
CHANGE CPU FROM SOCKET TO BGA SYMBOL

SYNC MASTER=T18 MLB		SYNC DATE=12/12/2007	
PAGE TITLE			
CPU Power & Ground			
		DRAWING NUMBER	SIZE
		051-7898	D
		REVISION	
		C.0.0	
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CURRENT DESIGN		SHEET> OF <TOTAL>	



1000 1885 707=PPVCORE_S0_CPU

1000 700=PP1V5_S0_CPU

1206 1806 905 707=PP1V05_S0_CPU

87654321

CPU Decoupling

051-7898 D

C.0.0

12 OF 109

CURRENT DESIGN SHEET> OF <TOTAL DESIGN SHEETS>

SYNC FROM T18

REMOVE NO STUFF CAPS C1220 TO C1231

REMOVE C1244 & C1245

CHANGE C1240-C1243 AND C1260 FROM 128S0241(9 MILLI-OHM) TO 128S0231(6 MILLI-OHM)

D

D

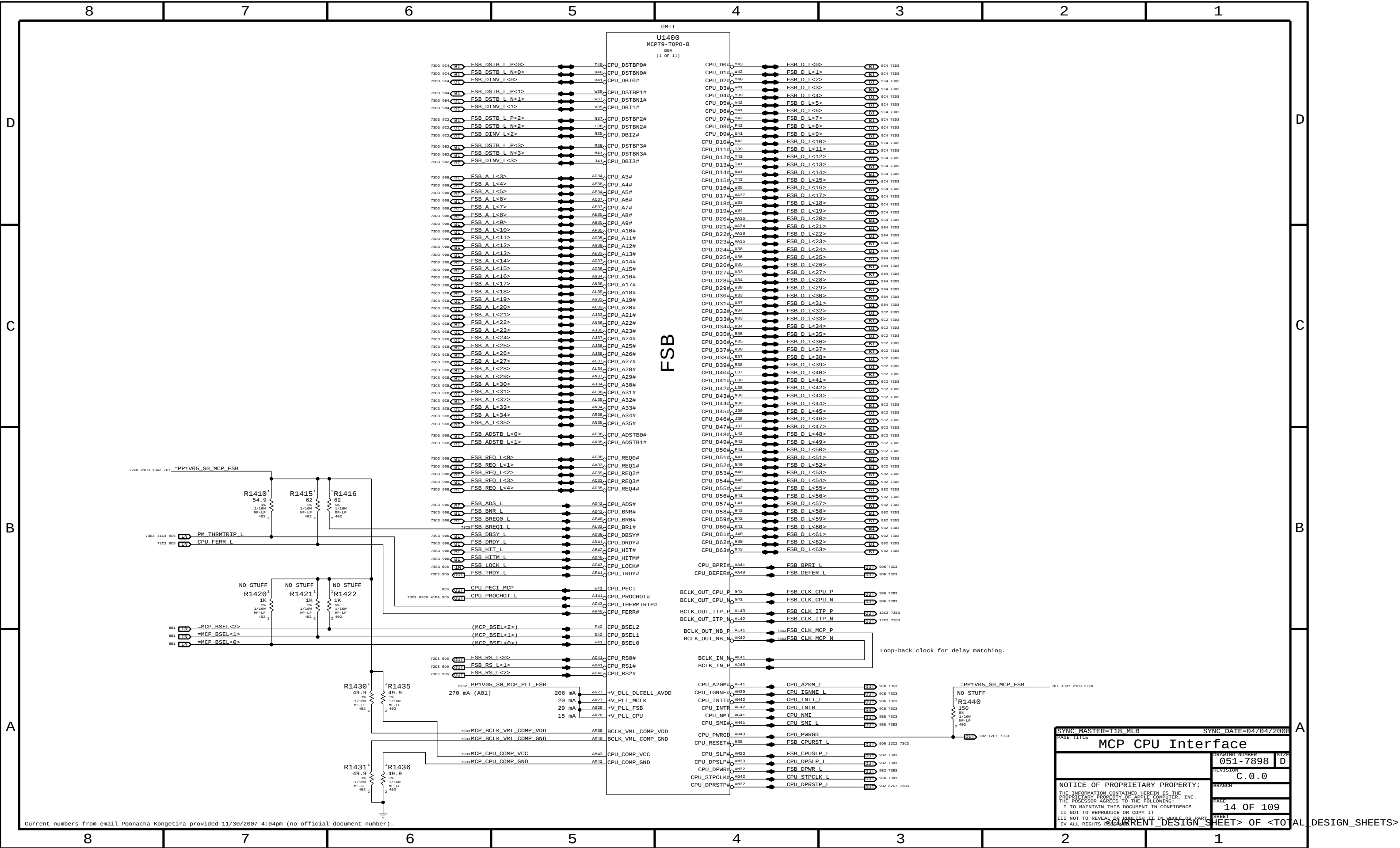
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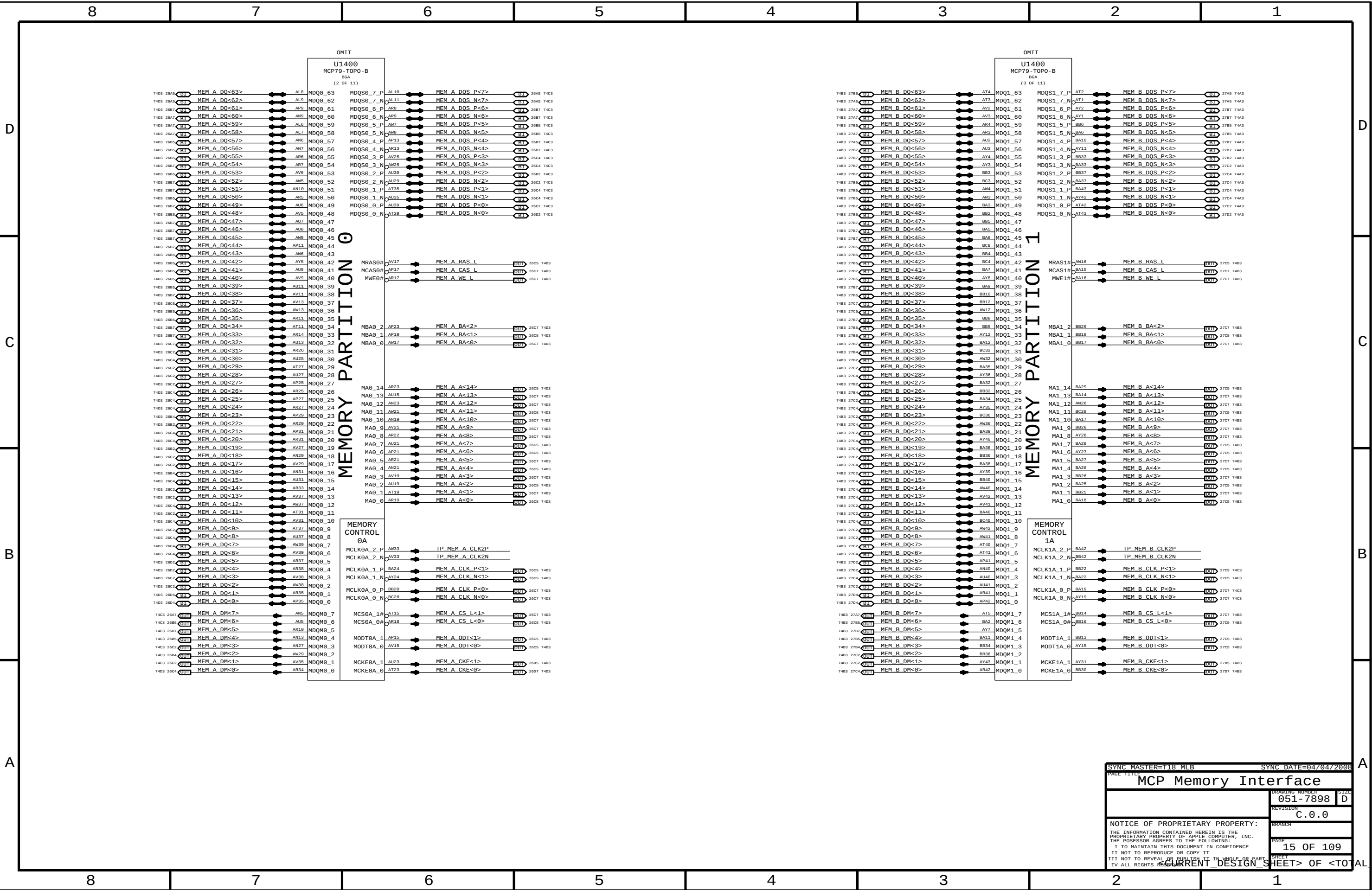


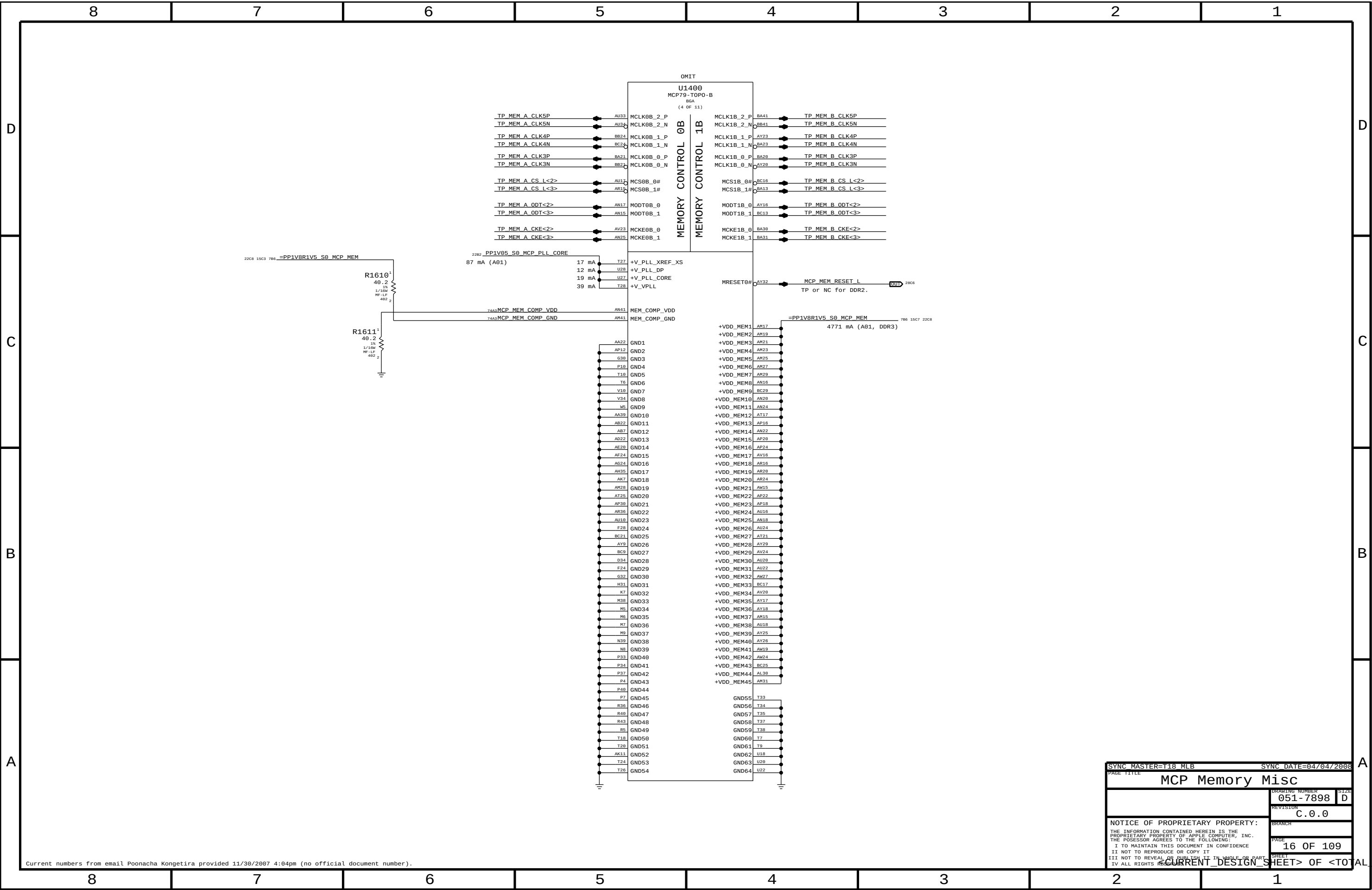
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B

A



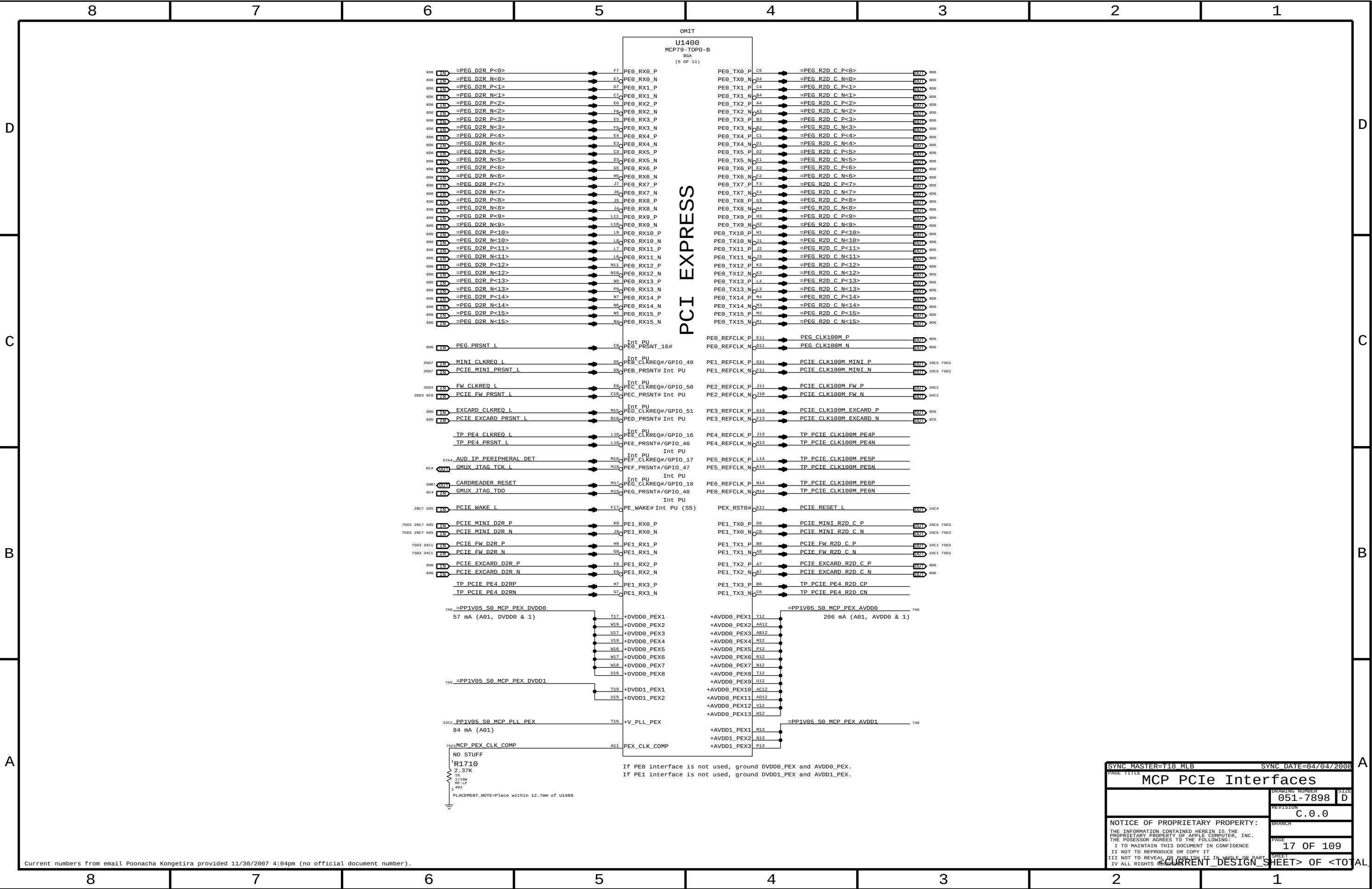




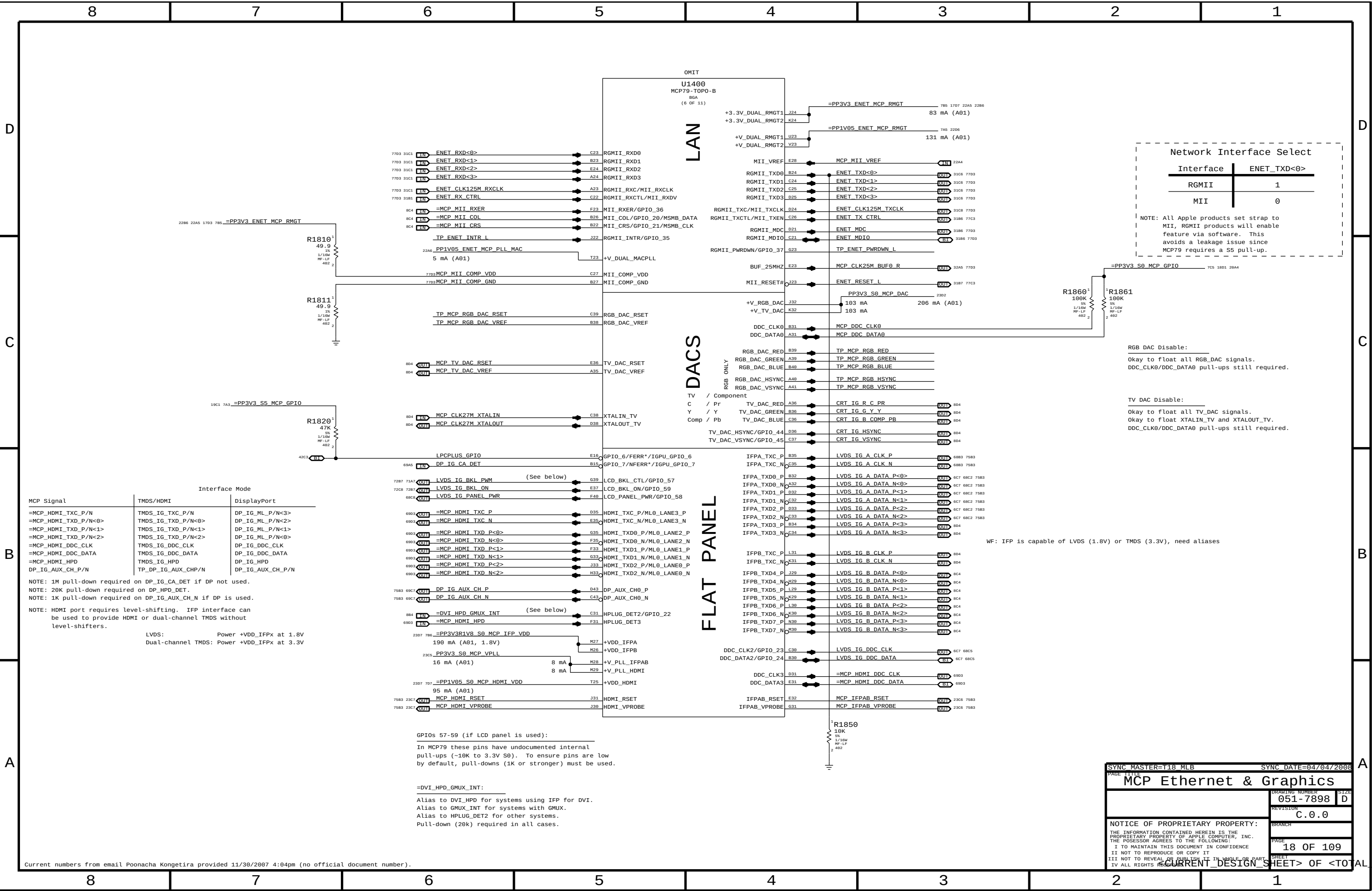
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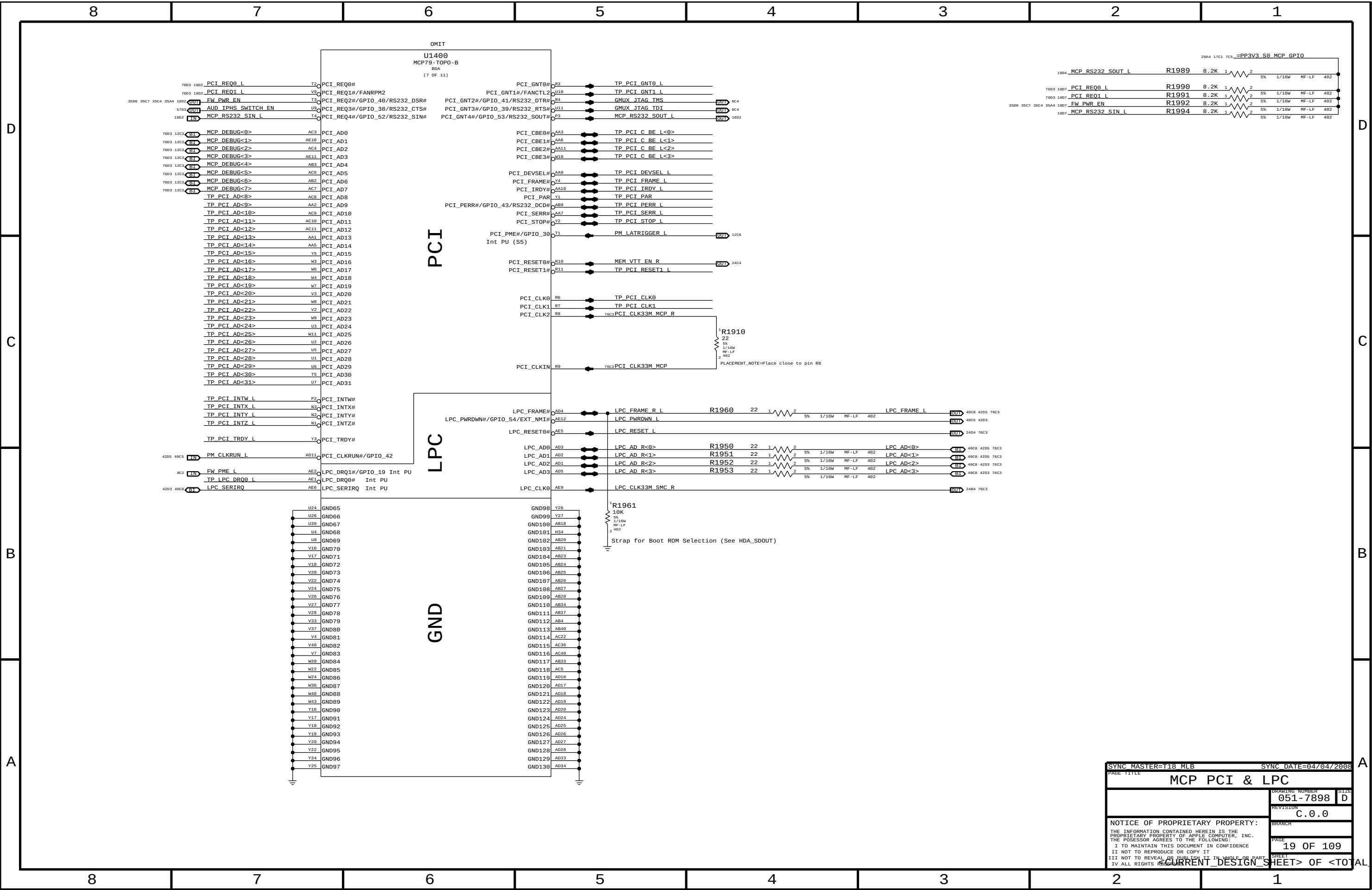
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PAGE TITLE			
MCP Memory Misc			
DRAWING NUMBER		SIZE	
051-7898		D	
REVISION		C.0.0	
BRANCH			
PAGE		16 OF 109	
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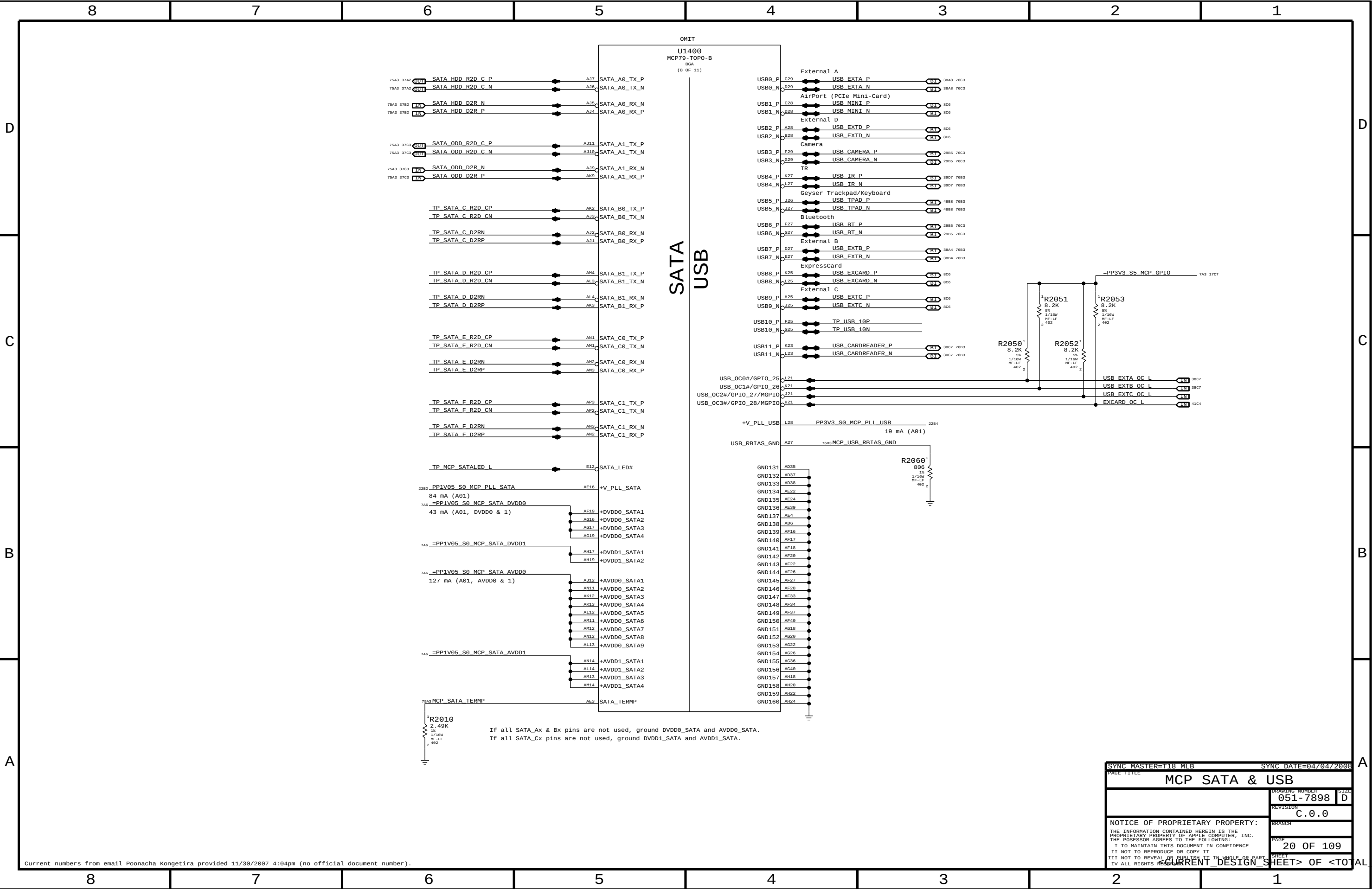
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		REVISION	C.0.0
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		PAGE	17 OF 109
		SHEET	
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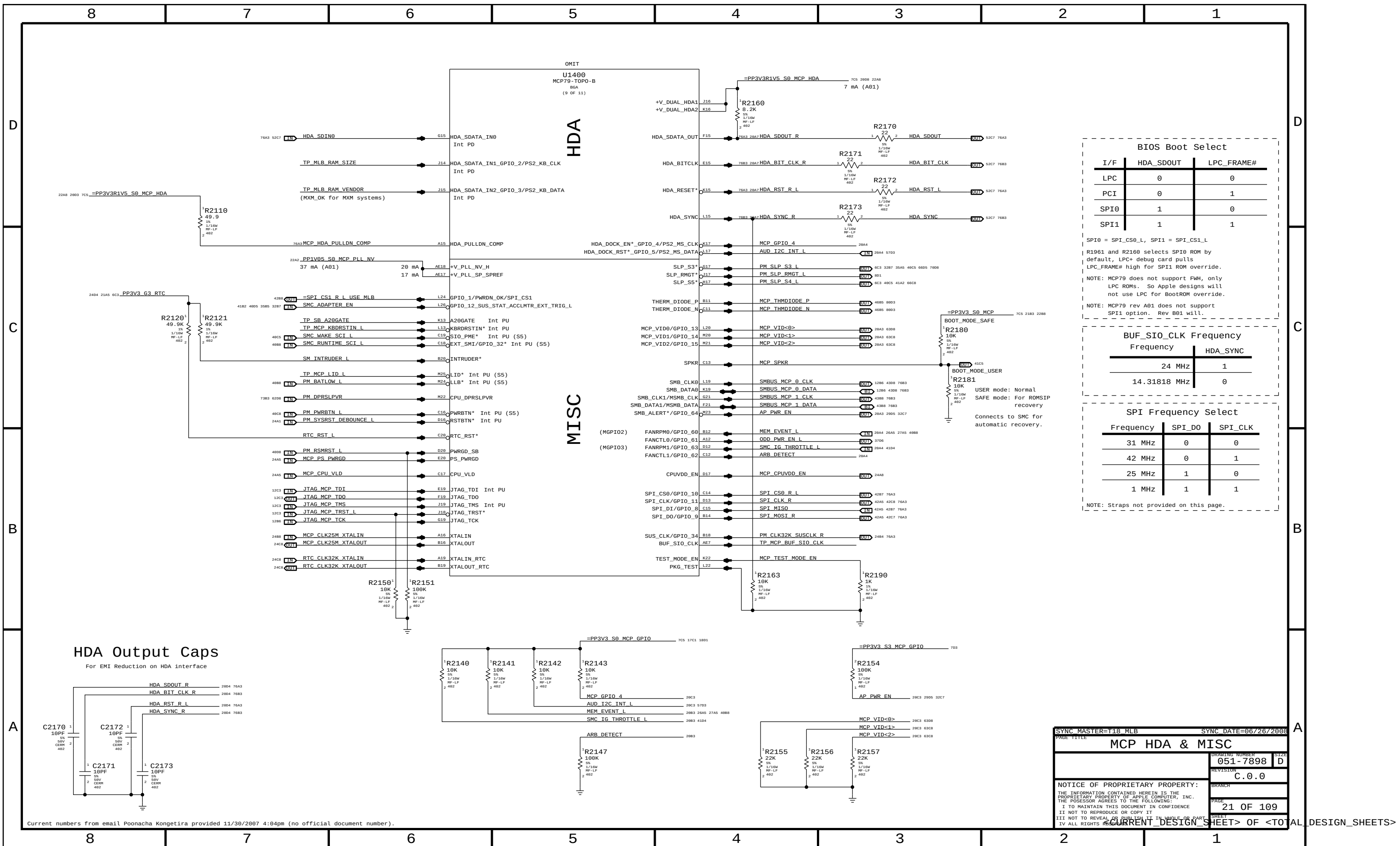


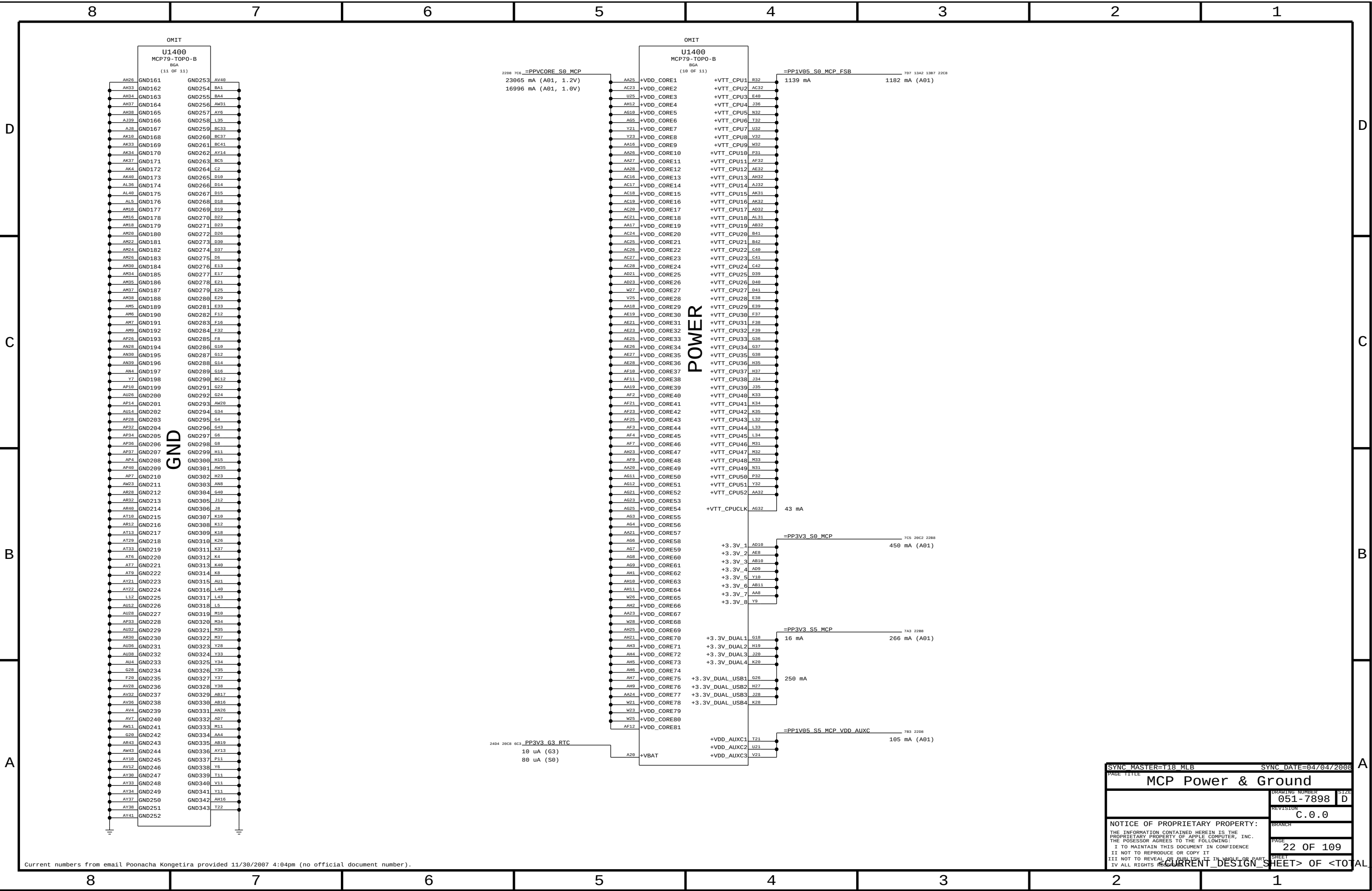




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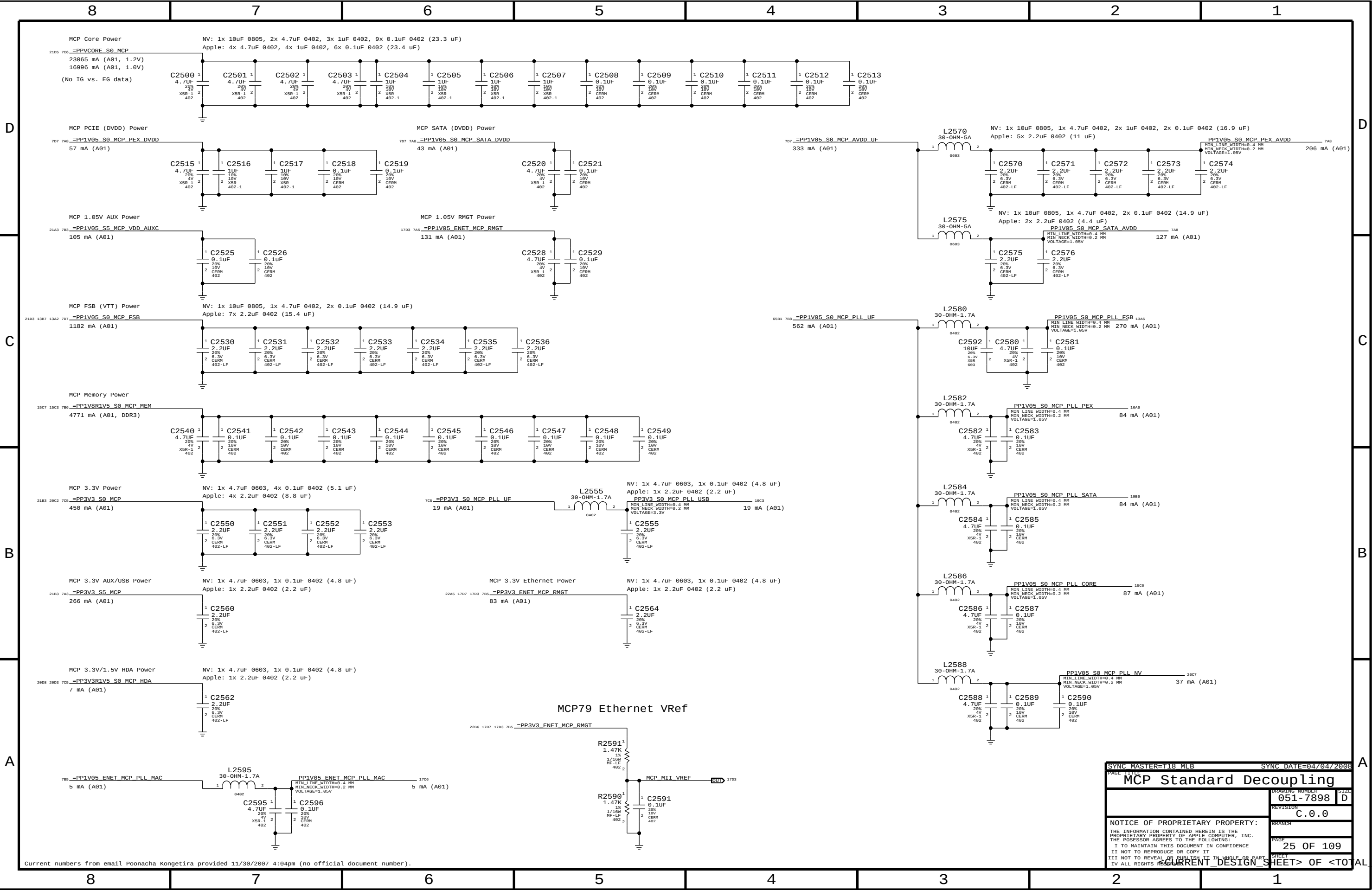
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REVISION		C.0.0	
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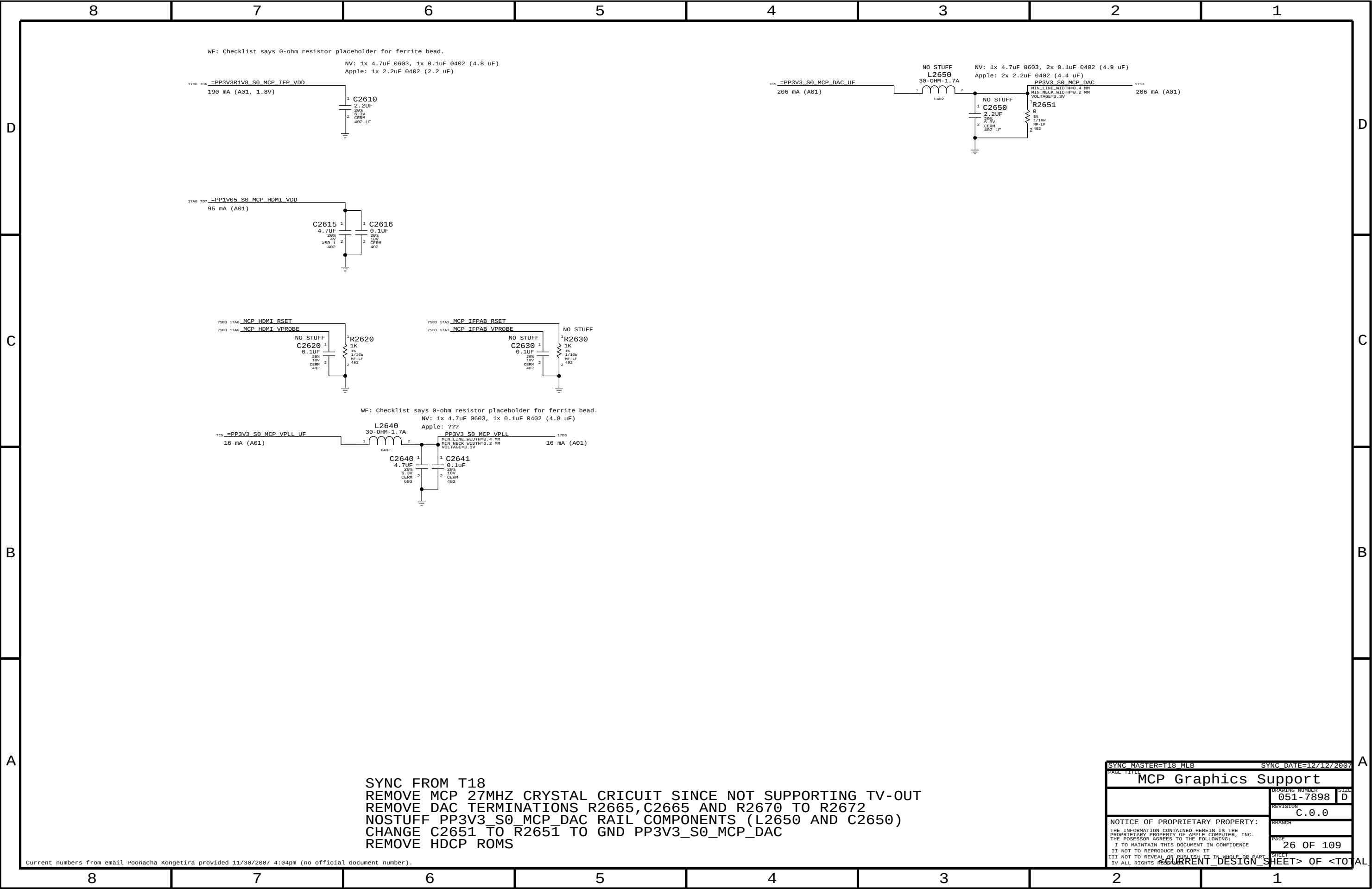


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SYNC MASTER=T18 MLB		SYNC DATE=04/04/2008	
PAGE TITLE			
MCP Power & Ground			
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		SIZE	D
		REVISION	C.0.0
		BRANCH	
		PAGE	22 OF 109
CURRENT DESIGN SHEET>		SHEET	OF <TOTAL>



PAGE TITLE		PAGE	
MCP Standard Decoupling		25 OF 109	
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REVISION		SHEET	
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PAGE TITLE			
MCP Graphics Support			
		DRAWING NUMBER	SIZE
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CURRENT DESIGN		SHEET> OF <TOTAL>	

Page Notes

Power aliases required by this page:

- =PP3V3_S3_VREFMRGN
- =PP3V3_S5_VREFMRGN
- =PPVTT_S3_DDR_BUF

Signal aliases required by this page:

- =I2C_VREFDACS_SCL
- =I2C_VREFDACS_SDA
- =I2C_PCA9557D_SCL
- =I2C_PCA9557D_SDA

BOM options provided by this page:

- VREFMRGN
- NO_VREFMRGN

DAC channel
Min DAC code
Max DAC code
Max sink I
Max source I
Nominal Vref
Min Vref
Max Vref
Vref Stepping
(per DAC LSB)

MEM A VREF DQ

MEM A VREF CA

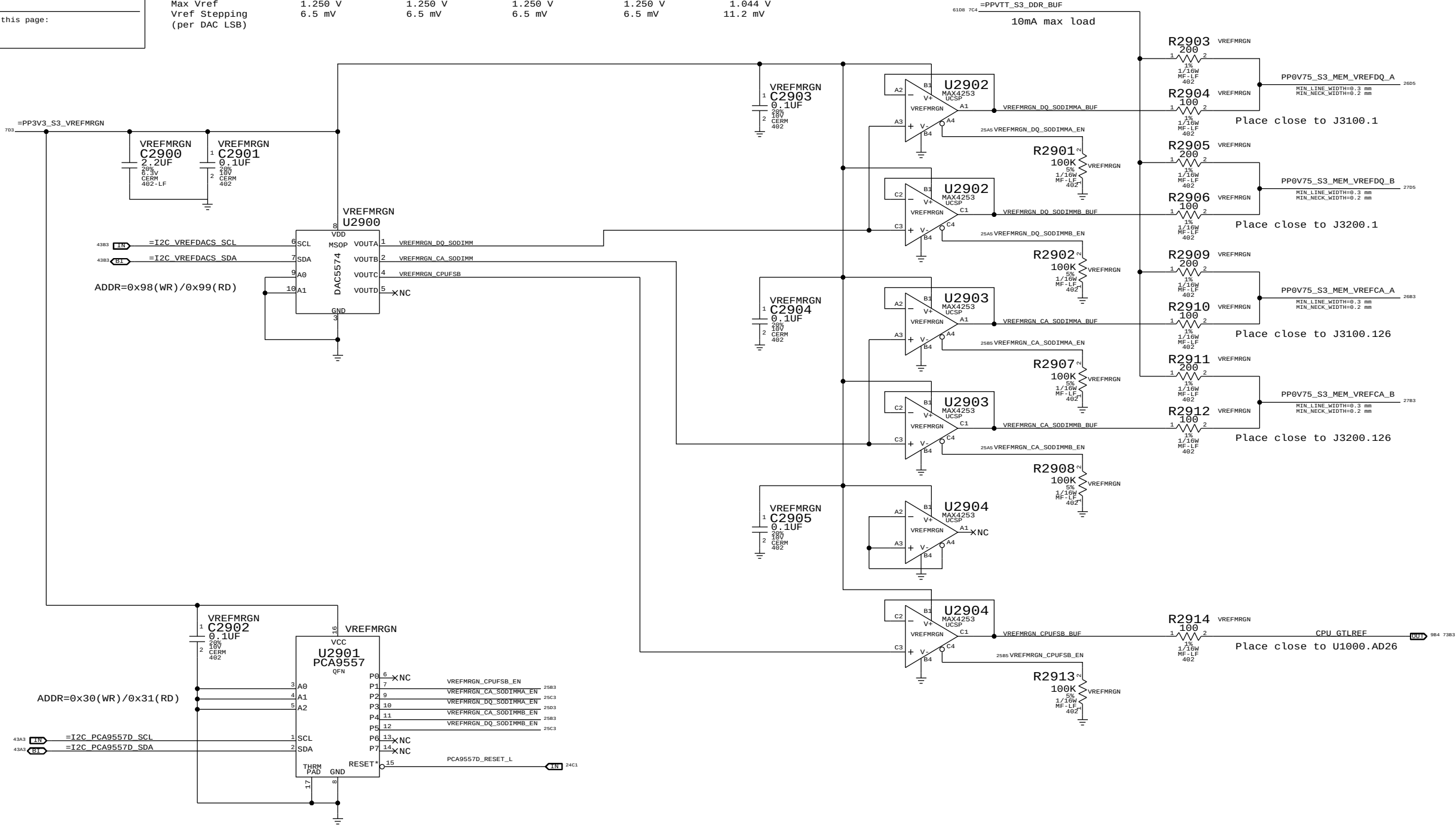
MEM B VREF DQ

MEM B VREF CA

CPU FSB VREF

	A	B	A	B	C
Min DAC code	0x00	0x00	0x00	0x00	0x00
Max DAC code	0x87	0x87	0x87	0x87	0x55
Max sink I	-3.75 mA	-3.75 mA	-3.75 mA	-3.75 mA	-0.91 mA
Max source I	5 mA	5 mA	5 mA	5 mA	0.52 mA
Nominal Vref	0.75 V	0.75 V	0.75 V	0.75 V	0.70 V
Min Vref	0.375 V	0.375 V	0.375 V	0.375 V	0.091 V
Max Vref	1.250 V	1.250 V	1.250 V	1.250 V	1.044 V
Vref Stepping (per DAC LSB)	6.5 mV	6.5 mV	6.5 mV	6.5 mV	11.2 mV

SO-DIMM A and SO-DIMM B Vref settings should be margined separately
(i.e. not simultaneously) due to current limitation of TPS51116 regulator.



Required zero ohm resistors when no VREF margining circuit stuffed

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
116S0004	1	RES,MTL FILM,0,5%,0402,SM,LF	R2903	CRITICAL	NO_VREFMRGN
116S0004	1	RES,MTL FILM,0,5%,0402,SM,LF	R2905	CRITICAL	NO_VREFMRGN
116S0004	1	RES,MTL FILM,0,5%,0402,SM,LF	R2909	CRITICAL	NO_VREFMRGN
116S0004	1	RES,MTL FILM,0,5%,0402,SM,LF	R2911	CRITICAL	NO_VREFMRGN

SYNC MASTER=BEN SYNC DATE=03/31/2008

FSB/DDR3 Vref Margining

DRAWING NUMBER 051-7898 SIZE D

REVISION C.0.0

BRANCH

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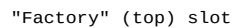
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DESIGN SHEET

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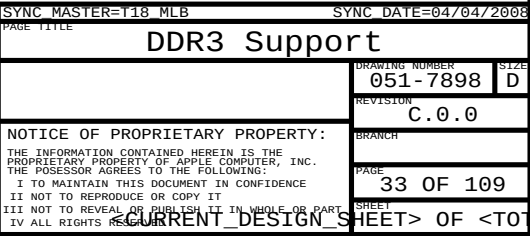


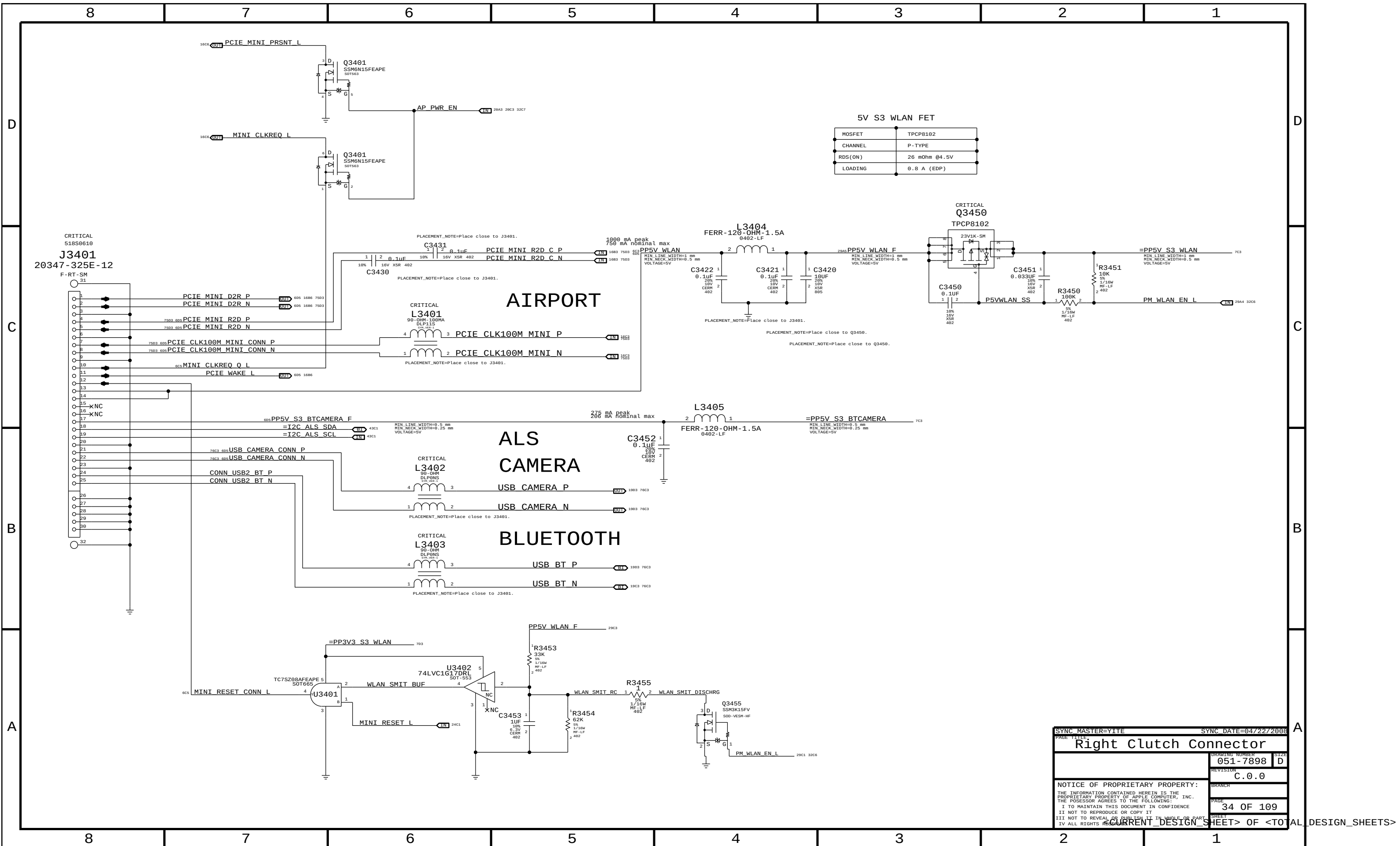
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DDR3 SO-DIMM Connector A			
		DRAWING NUMBER	SIZE
		051-7898	D
		REVISION	
		C.0.0	
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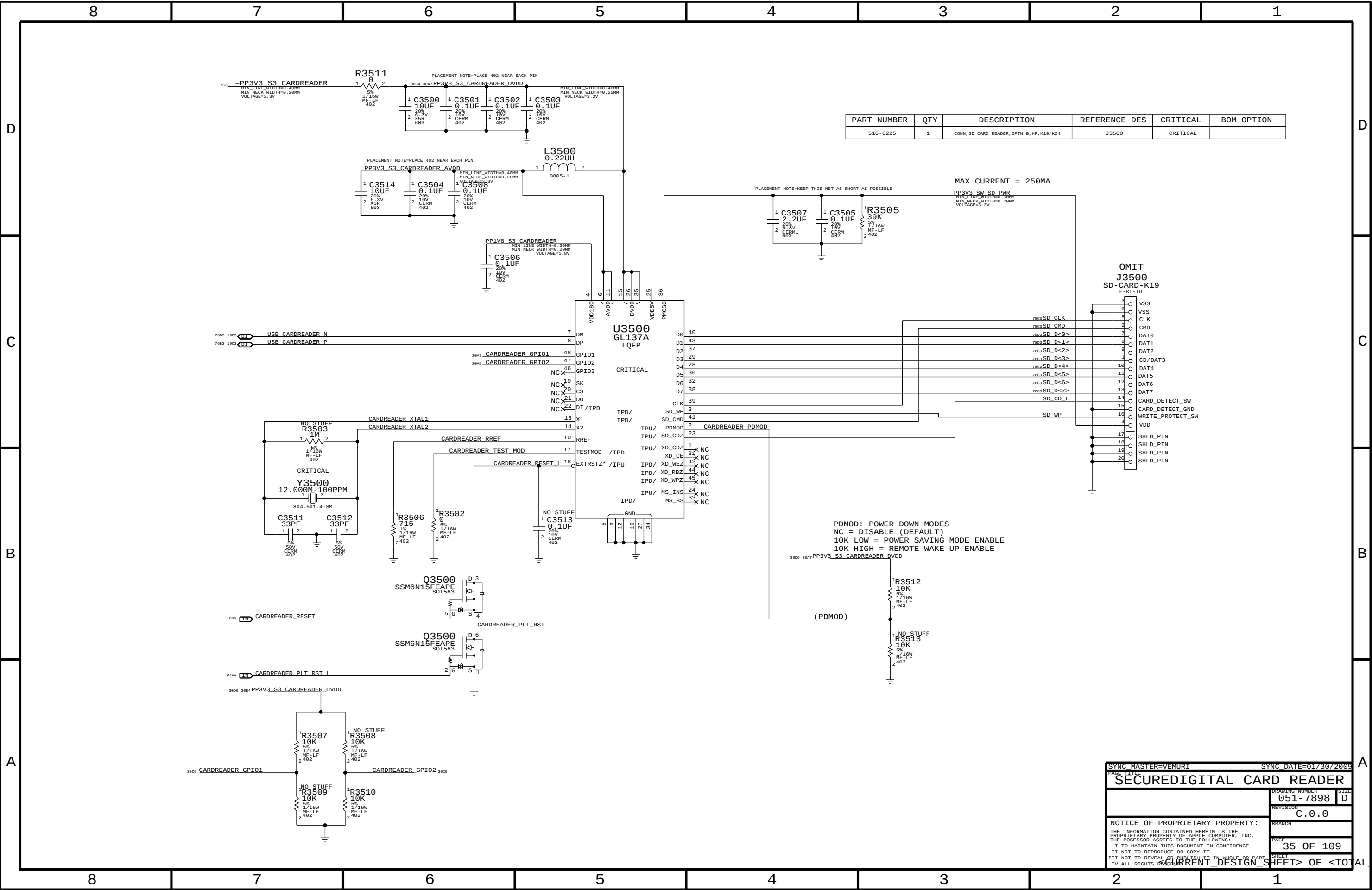
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Required because MCP79 does not meet DDR3 spec power-up reset timing requirement.

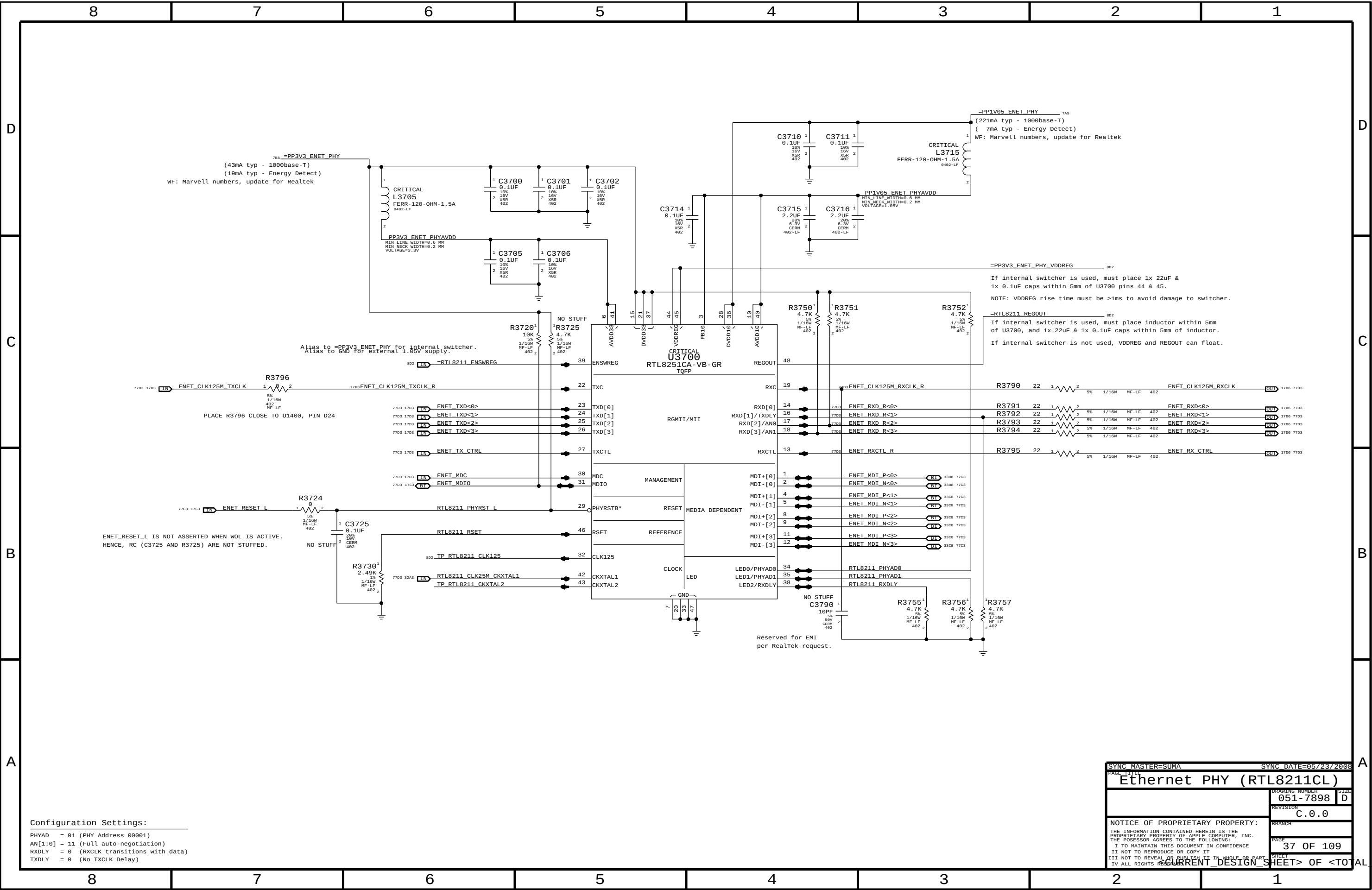


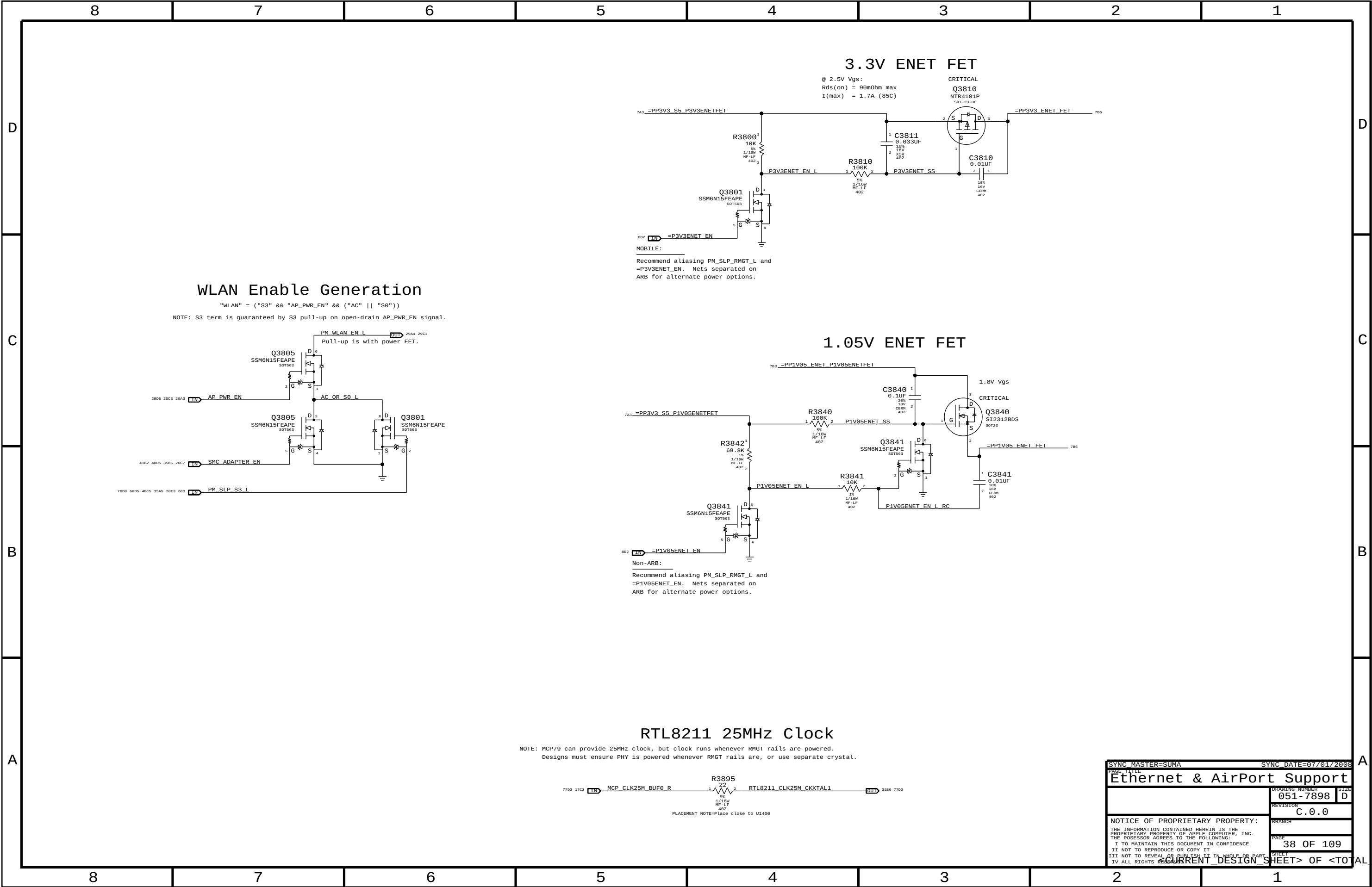




PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
516-0225	1	CONN, SD CARD READER, OPTN B, HF, K19/K24	J3500	CRITICAL	

PAGE TITLE		PAGE	
SECUREDIGITAL CARD READER		35 OF 109	
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051-7898		C.0.0	
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PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
114S0558	1	RES, 0.68 OHM, 1%, 0402, SMD	R4100	CRITICAL	

D

D

C

C

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A

A

SYNC MASTER=K19 MLB		SYNC DATE=11/02/2008	
PAGE 11111		FireWire LLC/PHY (FW643)	
DRAWING NUMBER		051-7898	D
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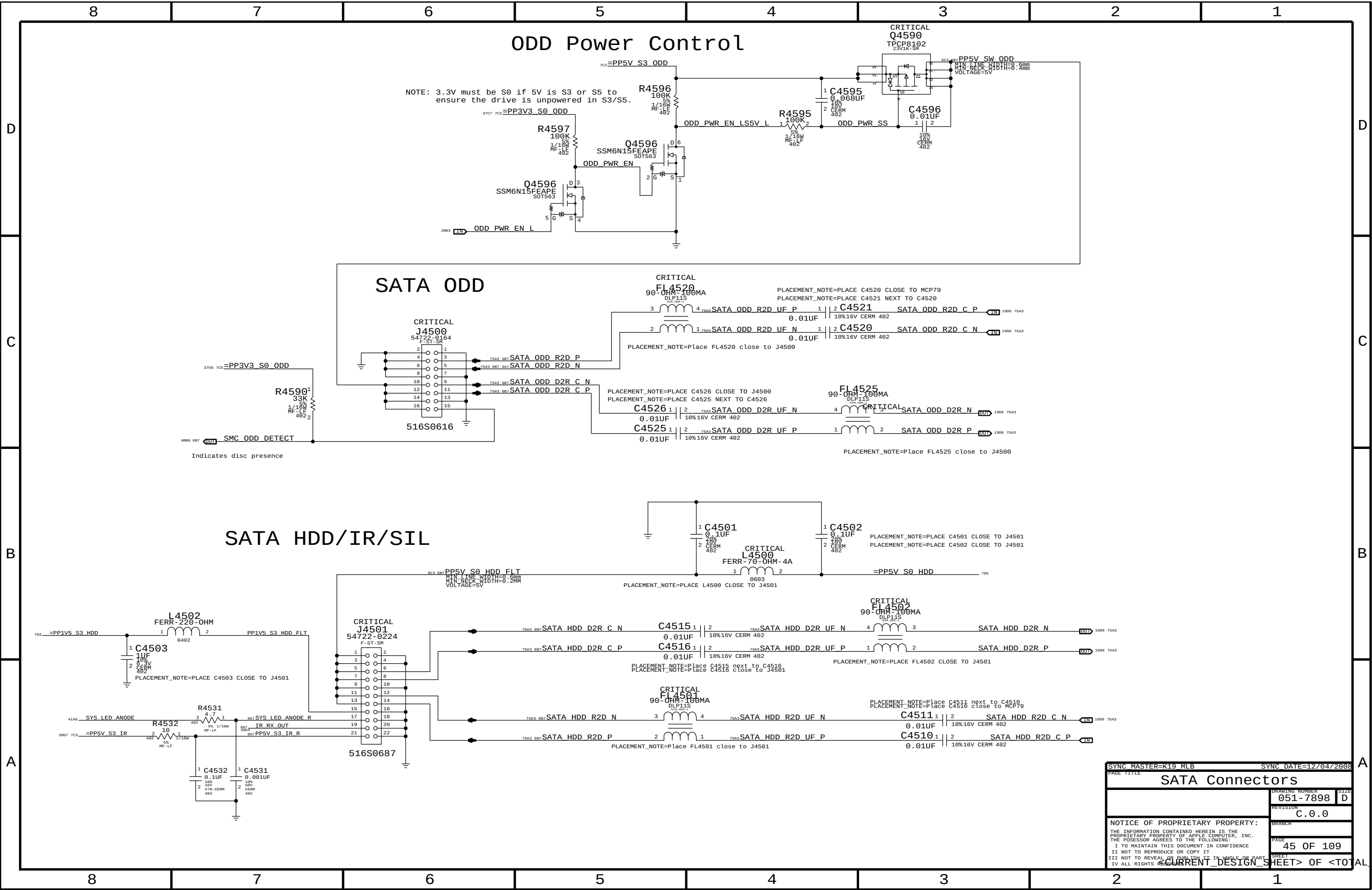
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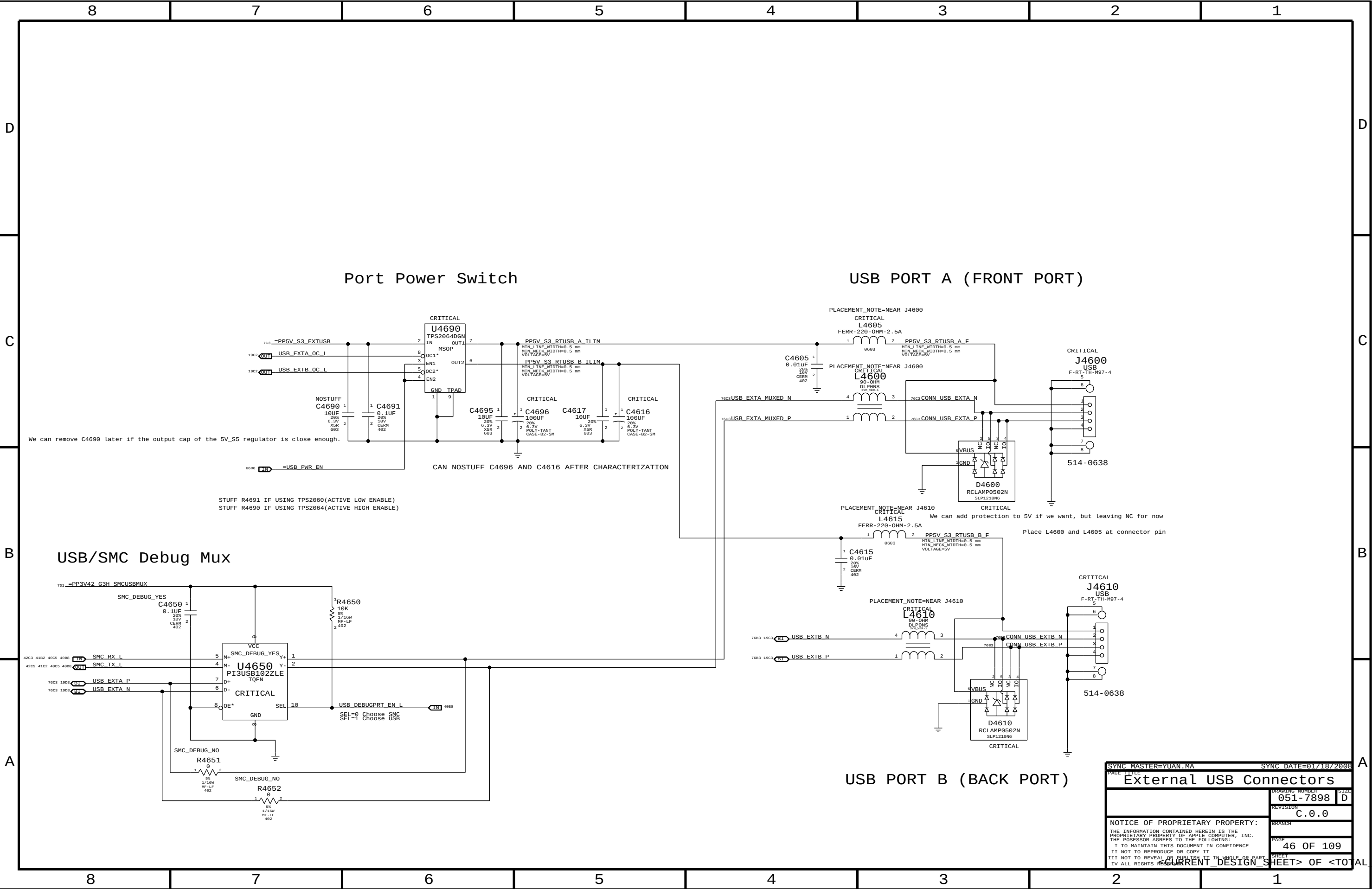
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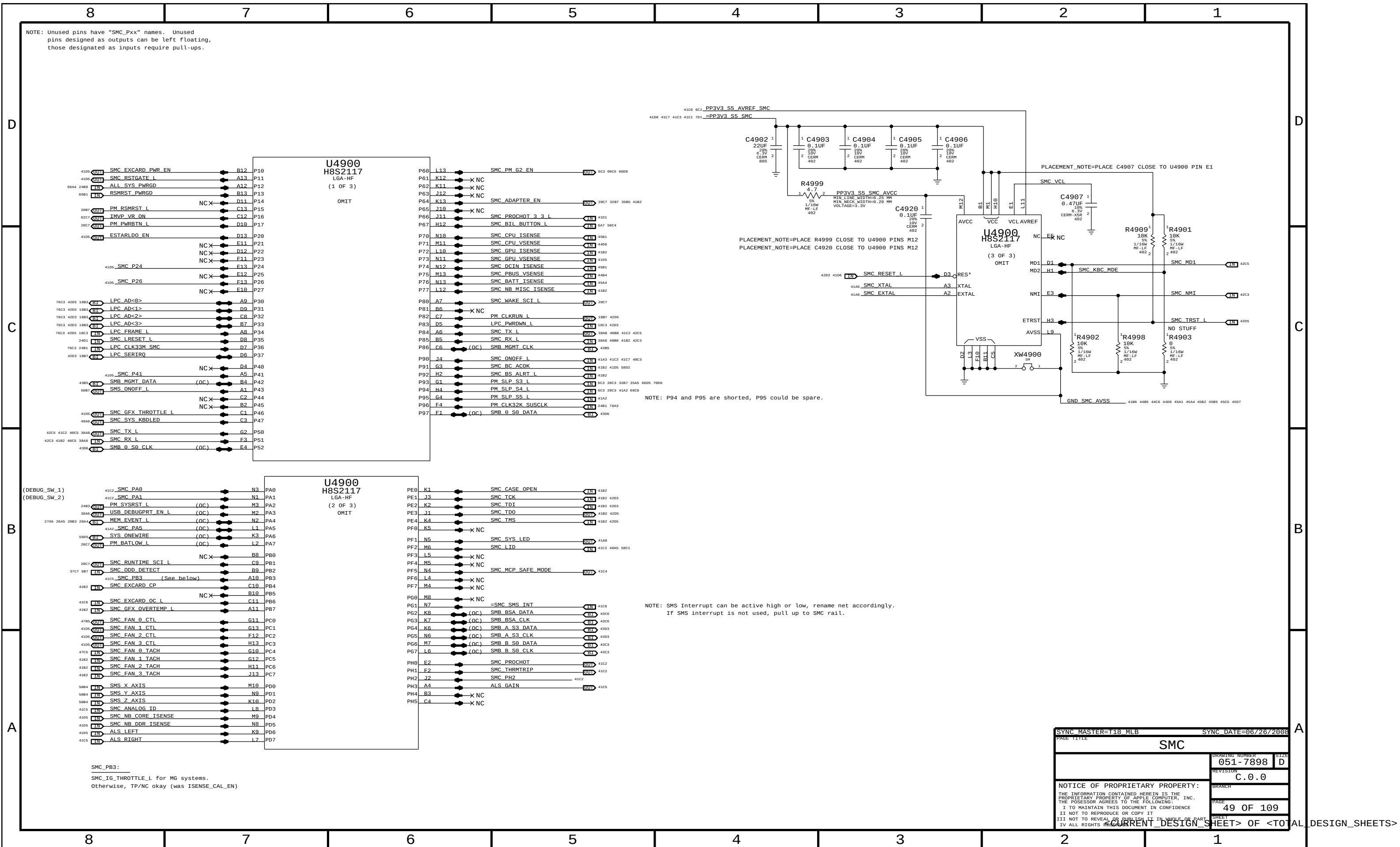
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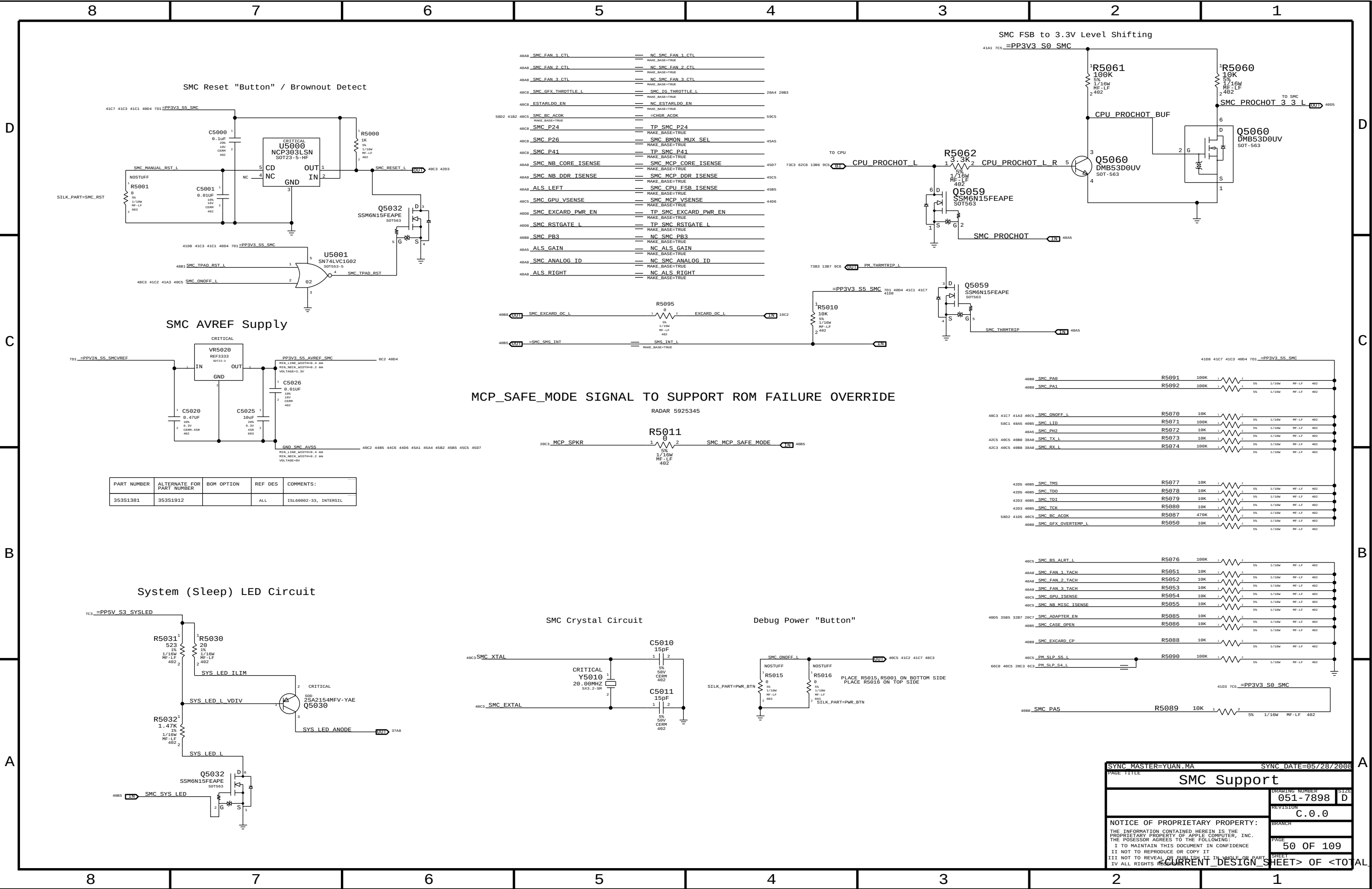


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SATA Connectors		DRAWING NUMBER	SIZE
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		REVISION	
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SYNC MASTER=YUAN.MA		SYNC DATE=01/18/2008			
PAGE TITLE		External USB Connectors			
DRAWING NUMBER		051-7898	D		
REVISION		C.0.0			
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Alternate SPI ROM Support

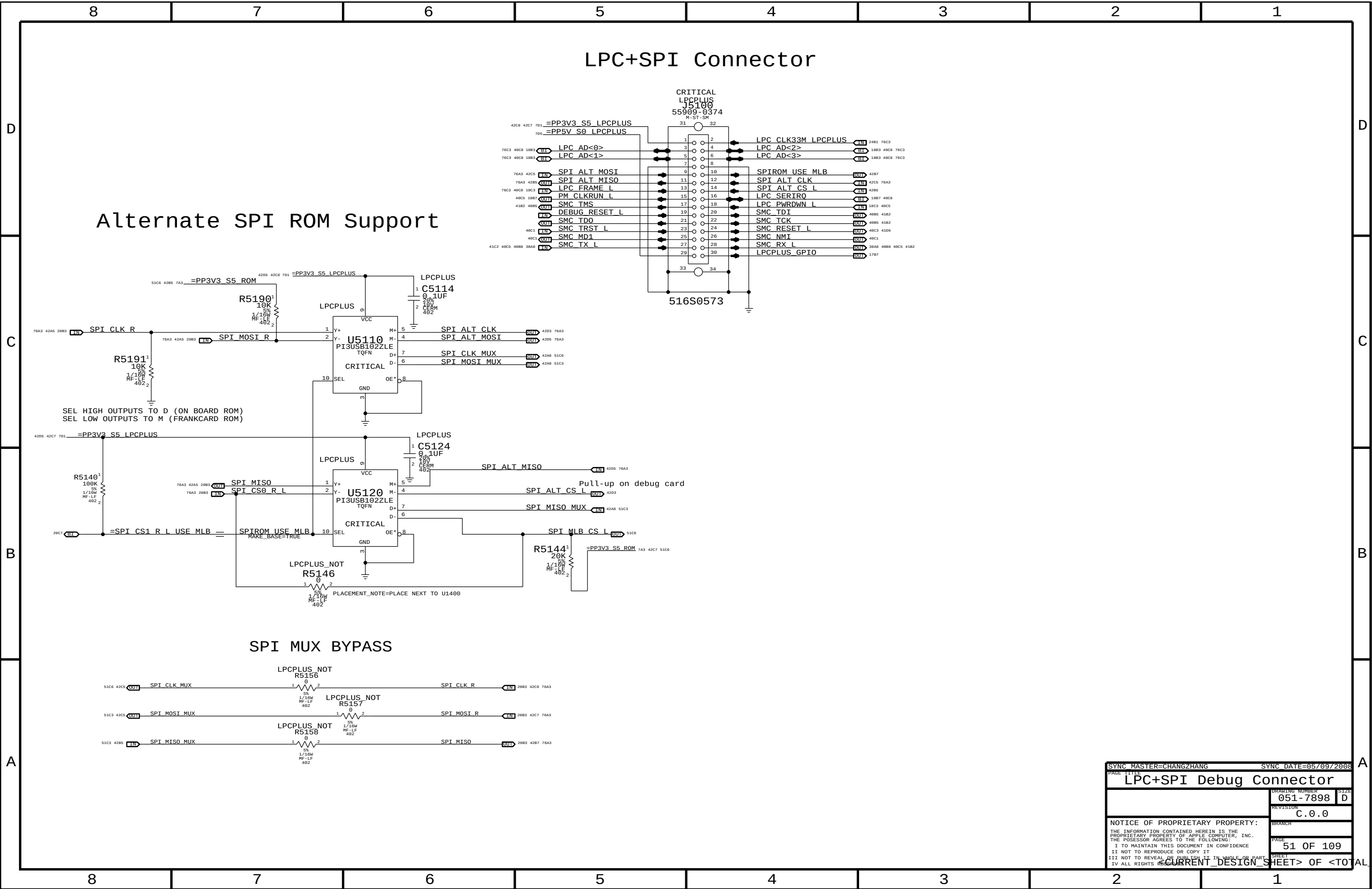
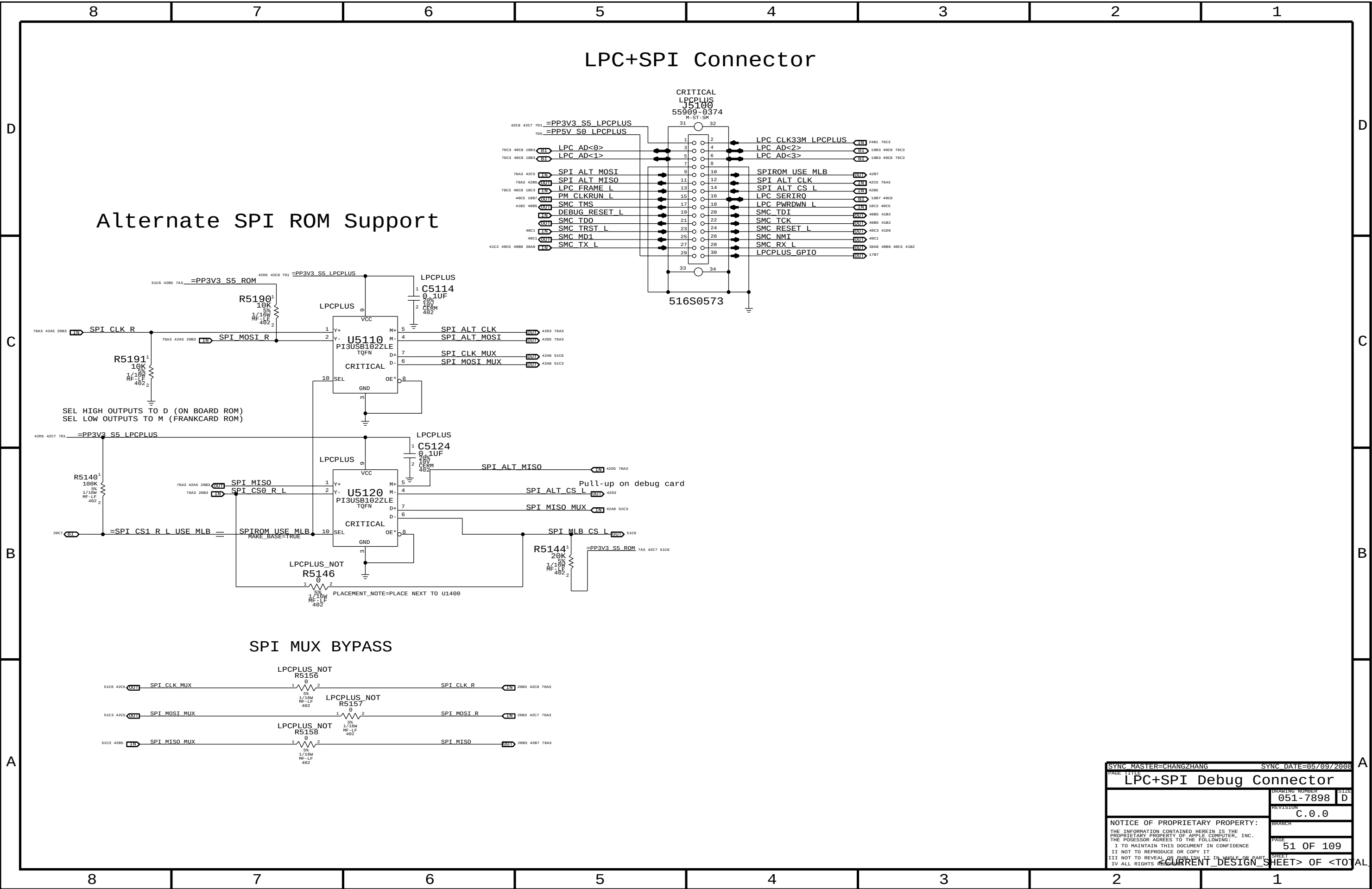
SEL HIGH OUTPUTS TO D (ON BOARD ROM)
SEL LOW OUTPUTS TO M (FRANKCARD ROM)

SPI MUX BYPASS

LPC+SPI Connector

Pin	Signal	Pin	Signal
1	LPC CLK33M LPCPLUS	24	SMC TCK
2	LPC AD<2>	25	SMC TRST L
3	LPC AD<1>	26	SMC MDI
4	LPC AD<0>	27	SMC TX L
5	SPI ALT MOSI	28	SMC RX L
6	SPI ALT MISO	29	LPCPLUS GPIO
7	LPC FRAME L	30	
8	PM CLKRUN L	31	
9	SMC TMS	32	
10	DEBUG RESET L	33	
11	SMC TD0	34	
12	SMC TRST L	35	
13	SMC MDI	36	
14	SMC TX L	37	
15	SPIROM USE MLB	38	
16	SPI ALT CLK	39	
17	SPI ALT CS L	40	
18	SPI CLK MUX	41	
19	SPI MOSI MUX	42	
20	SPI MISO MUX	43	
21	SPI MISO CS L	44	
22	SPI MISO CS L	45	
23	SPI MISO CS L	46	
24	SPI MISO CS L	47	
25	SPI MISO CS L	48	
26	SPI MISO CS L	49	
27	SPI MISO CS L	50	
28	SPI MISO CS L	51	
29	SPI MISO CS L	52	
30	SPI MISO CS L	53	
31	SPI MISO CS L	54	
32	SPI MISO CS L	55	
33	SPI MISO CS L	56	
34	SPI MISO CS L	57	
35	SPI MISO CS L	58	
36	SPI MISO CS L	59	
37	SPI MISO CS L	60	
38	SPI MISO CS L	61	
39	SPI MISO CS L	62	
40	SPI MISO CS L	63	
41	SPI MISO CS L	64	
42	SPI MISO CS L	65	
43	SPI MISO CS L	66	
44	SPI MISO CS L	67	
45	SPI MISO CS L	68	
46	SPI MISO CS L	69	
47	SPI MISO CS L	70	
48	SPI MISO CS L	71	
49	SPI MISO CS L	72	
50	SPI MISO CS L	73	
51	SPI MISO CS L	74	
52	SPI MISO CS L	75	
53	SPI MISO CS L	76	
54	SPI MISO CS L	77	
55	SPI MISO CS L	78	
56	SPI MISO CS L	79	
57	SPI MISO CS L	80	
58	SPI MISO CS L	81	
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61	SPI MISO CS L	84	
62	SPI MISO CS L	85	
63	SPI MISO CS L	86	
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65	SPI MISO CS L	88	
66	SPI MISO CS L	89	
67	SPI MISO CS L	90	
68	SPI MISO CS L	91	
69	SPI MISO CS L	92	
70	SPI MISO CS L	93	
71	SPI MISO CS L	94	
72	SPI MISO CS L	95	
73	SPI MISO CS L	96	
74	SPI MISO CS L	97	
75	SPI MISO CS L	98	
76	SPI MISO CS L	99	
77	SPI MISO CS L	100	

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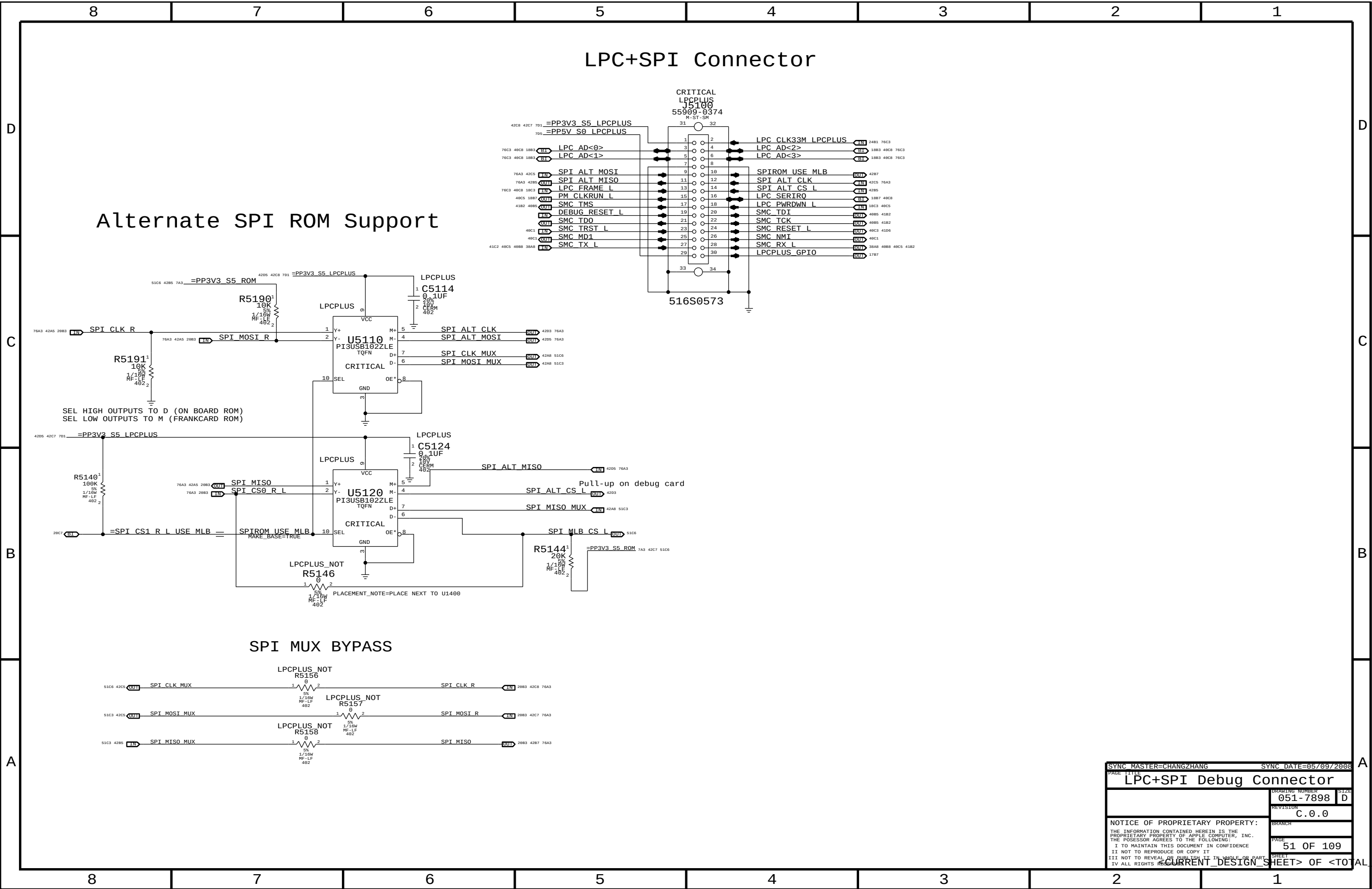
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1	LPC CLK33M LPCPLUS	24	SMC TCK
2	LPC AD<2>	25	SMC TRST L
3	LPC AD<1>	26	SMC MDI
4	LPC AD<0>	27	SMC TX L
5	SPI ALT MOSI	28	SMC RX L
6	SPI ALT MISO	29	LPCPLUS GPIO
7	LPC FRAME L	30	
8	PM CLKRUN L	31	
9	SMC TMS	32	
10	DEBUG RESET L	33	
11	SMC TD0	34	
12	SPI ALT CS L	35	
13	SPI ALT CS L	36	
14	SPI ALT CS L	37	
15	SPI ALT CS L	38	
16	SPI ALT CS L	39	
17	SPI ALT CS L	40	
18	SPI ALT CS L	41	
19	SPI ALT CS L	42	
20	SPI ALT CS L	43	
21	SPI ALT CS L	44	
22	SPI ALT CS L	45	
23	SPI ALT CS L	46	
24	SPI ALT CS L	47	
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26	SPI ALT CS L	49	
27	SPI ALT CS L	50	
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59	SPI ALT CS L	82	
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71	SPI ALT CS L	94	
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75	SPI ALT CS L	98	
76	SPI ALT CS L	99	
77	SPI ALT CS L	100	



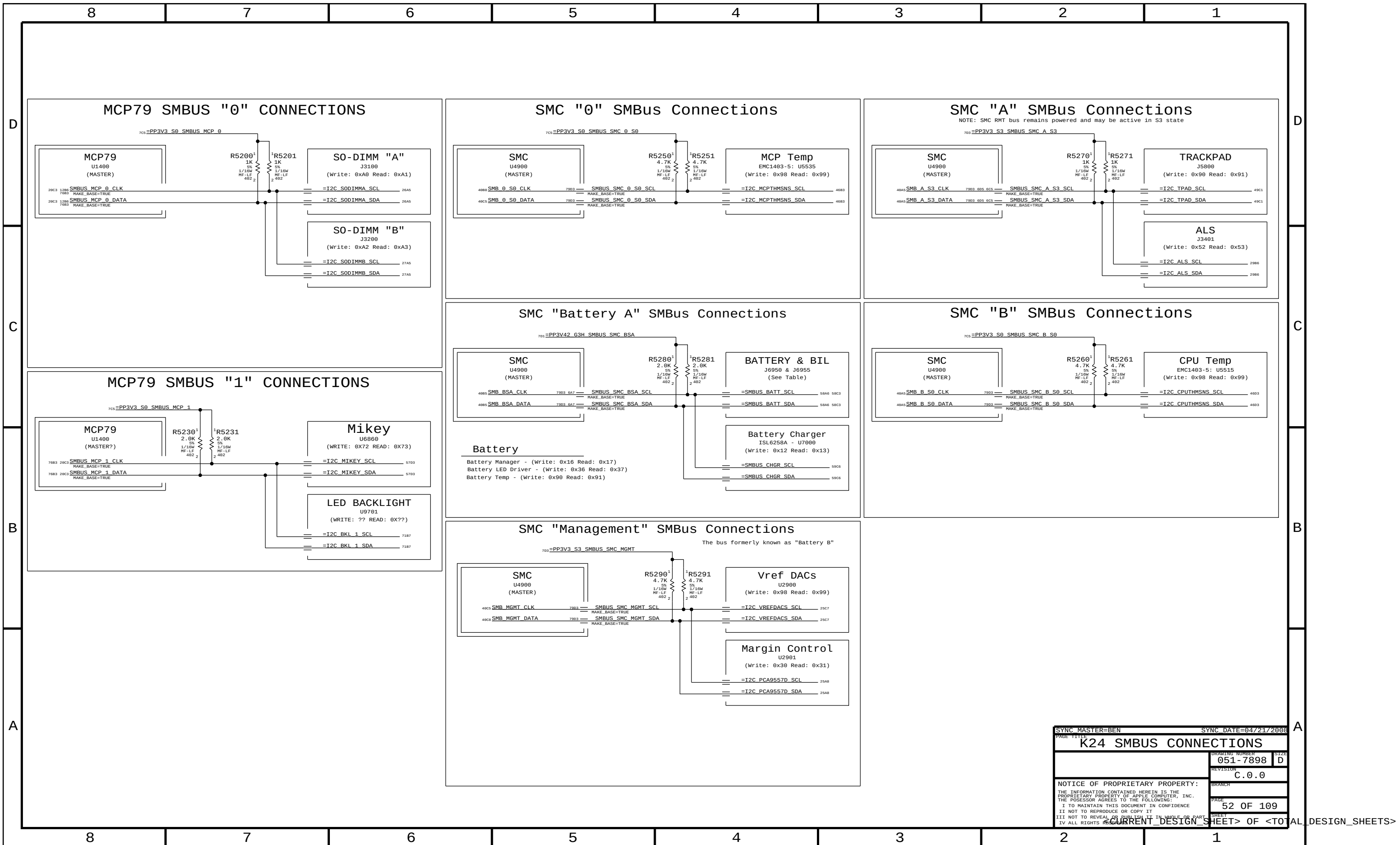
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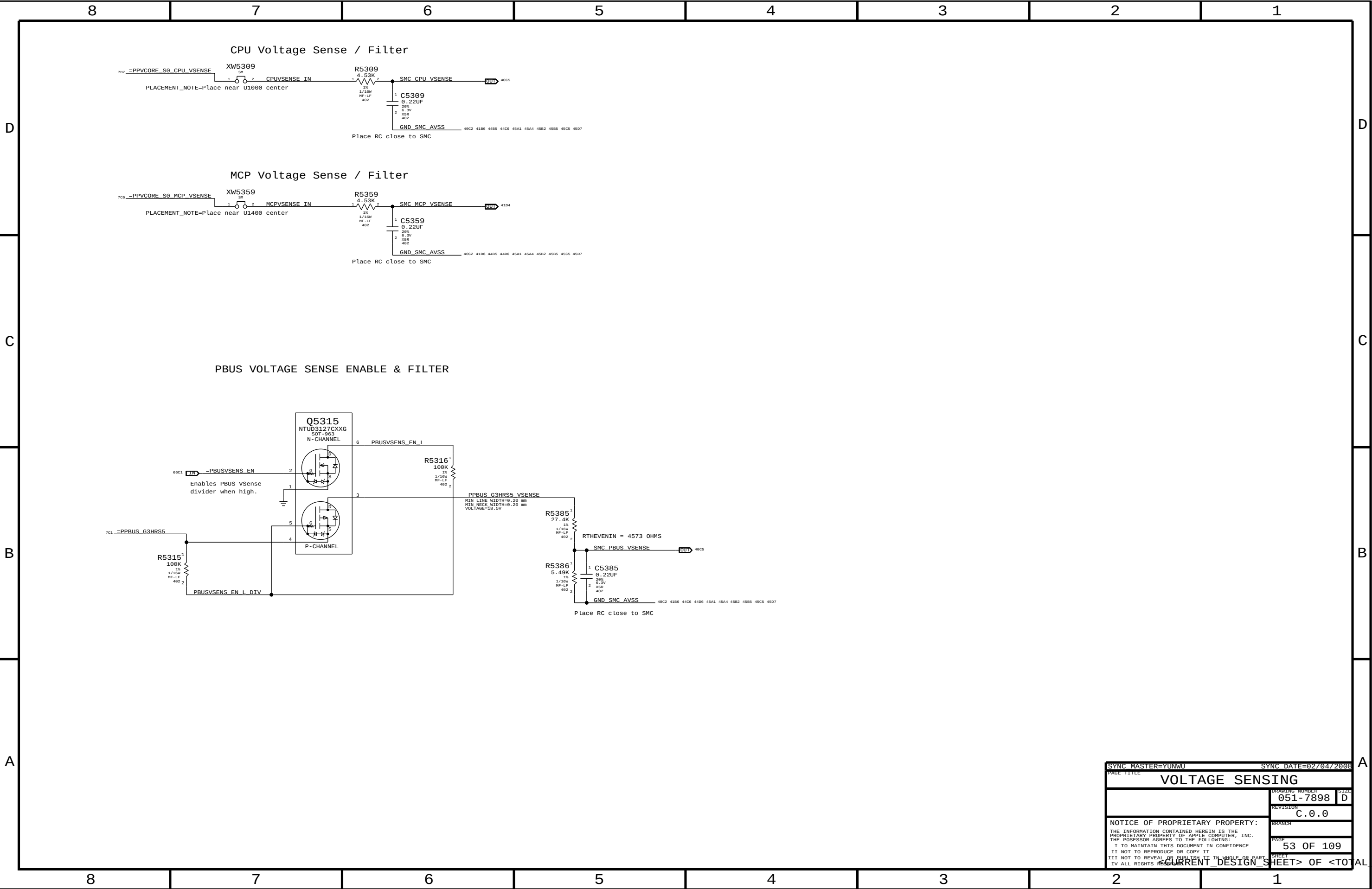
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SEL LOW OUTPUTS TO M (FRANKCARD ROM)

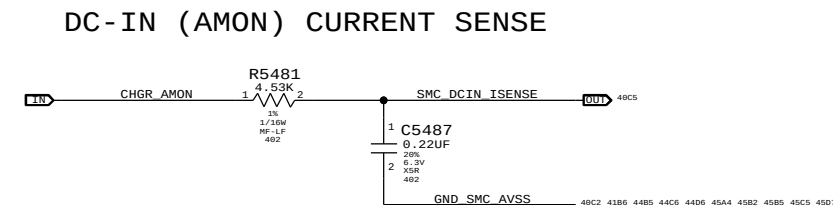
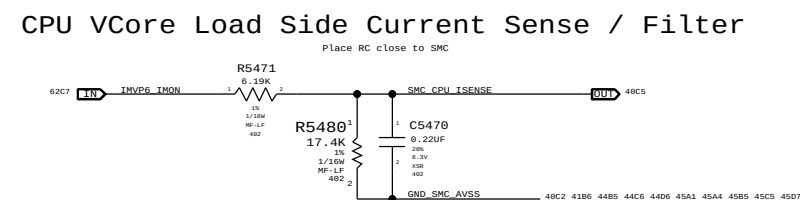
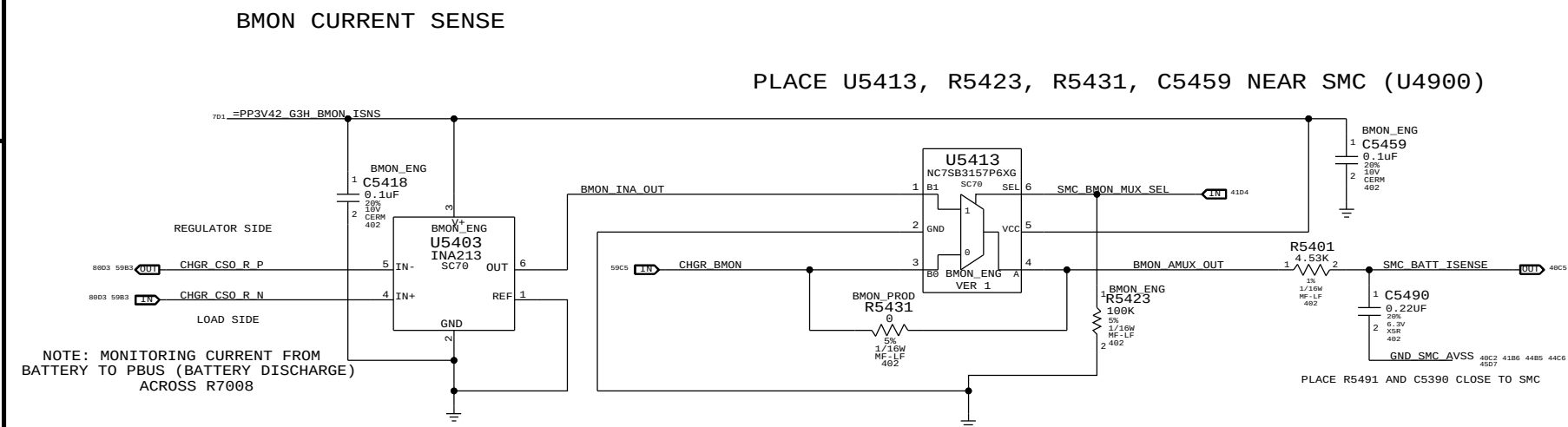
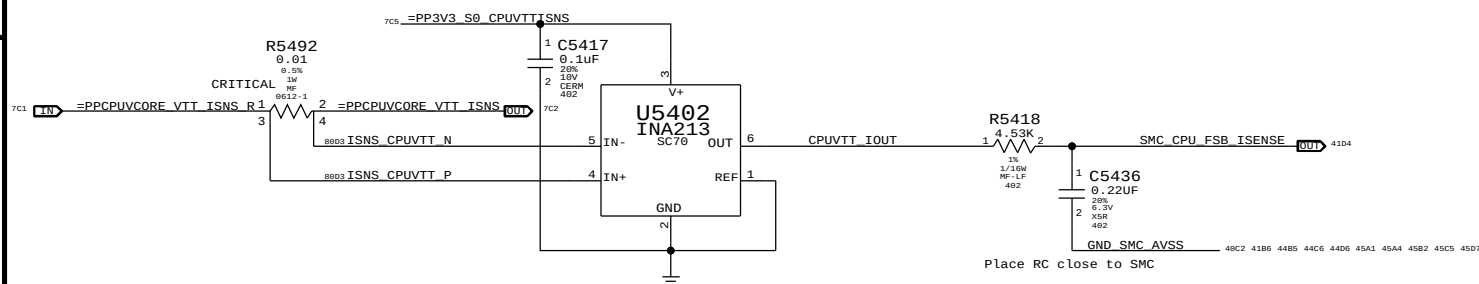
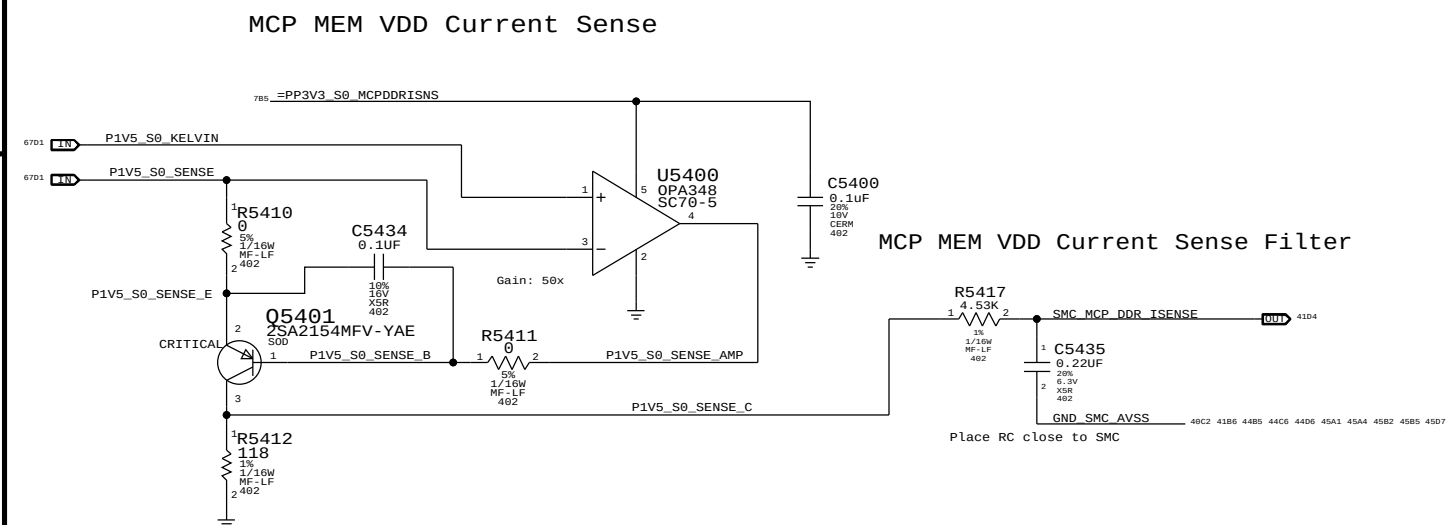
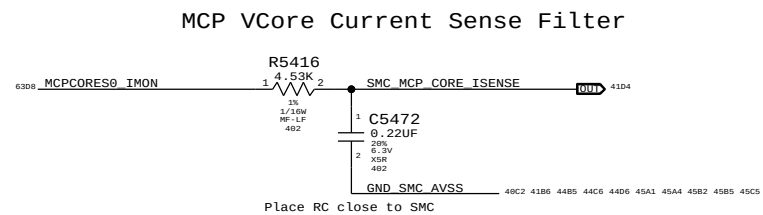
SPI MUX BYPASS

LPC+SPI Connector

Pin	Signal	Pin	Signal
1	LPC CLK33M LPCPLUS	24	SMC TCK
2	LPC AD<2>	25	SMC TRST L
3	LPC AD<1>	26	SMC MDI
4	LPC AD<0>	27	SMC TX L
5	SPI ALT MOSI	28	SMC RX L
6	SPI ALT MISO	29	LPCPLUS GPIO
7	LPC FRAME L	30	
8	PM CLKRUN L	31	
9	SMC TMS	32	
10	DEBUG RESET L	33	
11	SMC TD0	34	
12	SPI ALT CS L	35	
13	SPI ALT CS L	36	
14	SPI ALT CS L	37	
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77	SPI ALT CS L	100	







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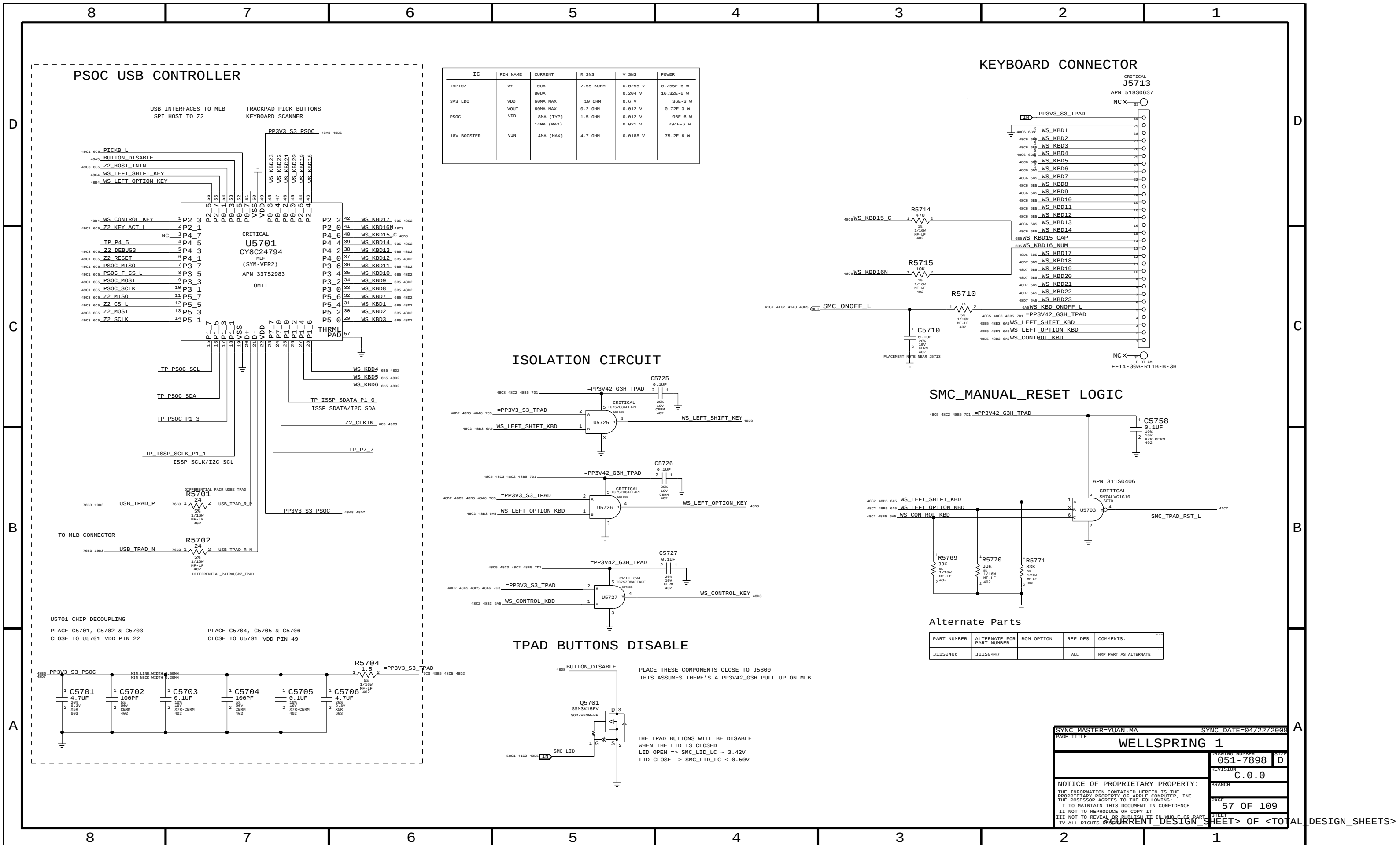
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This diagram illustrates the electrical design for the BOOSTER +18.5VDC FOR SENSORS, IPD FLEX CONNECTOR, 3V3 LDO FOR IPD, and KEYBOARD BACKLIGHT DRIVING AND DETECTION.

BOOSTER +18.5VDC FOR SENSORS

BOOSTER DESIGN CONSIDERATION:

- POWER CONSUMPTION
- DROOP LINE REGULATION
- RIPPLE TO MEET ERS
- 100-300 KHZ CLEAN SPECTRUM
- STARTUP TIME LESS THAN 2MS
- R5812, R5813, C5818 MODIFIED

The circuit includes components such as APN 152S0504 CRITICAL L5801, APN 353S1401 U5805, and various capacitors (C5800, C5816, C5817, C5818, C5819, C5812, C5813). It also features resistors R5805, R5806, R5811, R5812, and R5813.

IPD FLEX CONNECTOR

The connector interface includes signals like Z2_CS_L, Z2_DEBUG3, Z2_MOSI, Z2_SCLK, Z2_BOOST_EN, Z2_HOST_INTN, Z2_CLKIN, PP3V3_S3_LD0, Z2_KEY_ACT_L, Z2_RESET, PSOC_F_CS_L, PICKB_L, PSOC_MISO, PSOC_MOSI, PSOC_SCLK, I2C_TPAD_SDA, I2C_TPAD_SCL, and PP18V5_S3.

3V3 LDO FOR IPD

The LDO circuit uses APN 353S1364 VR5802 and includes capacitors C5853, C5838, and C5854, along with resistor R5836.

KEYBOARD BACKLIGHT DRIVING AND DETECTION

The backlight driving circuit includes APN 518S0691 CRITICAL J5815, APN 518S0691 CRITICAL L5850, and various capacitors (C5850, C5855) and resistors (R5853, R5854, R5855).

To detect Keyboard backlight, SMC will tristate SMC_SYS_KBDLED:

- LOW = keyboard backlight present
- HIGH= keyboard backlight not present

BOM OPTION: KBDLED_YES
TURNED ON FOR BEST MLB CONFIG
R5853 ALWAYS PRESENT

J5815 pin 1 is grounded on keyboard backlight flex

KBD BACKLIGHT CONNECTOR

The connector interface includes signals like SMC_KBDLED_PRESENT_L, KB_BL, and KBDLED_CAP.

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This diagram illustrates the electrical design for the BOOSTER +18.5VDC FOR SENSORS, IPD FLEX CONNECTOR, 3V3 LDO FOR IPD, and KEYBOARD BACKLIGHT DRIVING AND DETECTION.

BOOSTER +18.5VDC FOR SENSORS

BOOSTER DESIGN CONSIDERATION:

- POWER CONSUMPTION
- DROOP LINE REGULATION
- RIPPLE TO MEET ERS
- 100-300 KHZ CLEAN SPECTRUM
- STARTUP TIME LESS THAN 2MS
- R5812, R5813, C5818 MODIFIED

The circuit includes components such as APN 152S0504 CRITICAL L5801, APN 353S1401 U5805, and various capacitors (C5800, C5816, C5817, C5818, C5819, C5812, C5813) and resistors (R5805, R5811, R5812, R5813, R5806).

IPD FLEX CONNECTOR

The connector interface includes signals like Z2_CS_L, Z2_DEBUG3, Z2_MOSI, Z2_SCLK, Z2_BOOST_EN, Z2_HOST_INTN, Z2_CLKIN, PP3V3_S3_LD0, Z2_KEY_ACT_L, Z2_RESET, PSOC_F_CS_L, PICKB_L, PSOC_MISO, PSOC_MOSI, PSOC_SCLK, I2C_TPAD_SDA, I2C_TPAD_SCL, and PP18V5_S3.

3V3 LDO FOR IPD

The LDO circuit uses APN 353S1364 VR5802 and includes capacitors C5853, C5838, C5854 and resistor R5873.

KEYBOARD BACKLIGHT DRIVING AND DETECTION

The backlight driving circuit features APN 518S0691 CRITICAL J5815, APN 518S0691 CRITICAL L5850, and components like R5853, R5854, R5855, C5850, C5855, and LT3491.

To detect Keyboard backlight, SMC will tristate SMC_SYS_KBDLED:

- LOW = keyboard backlight present
- HIGH= keyboard backlight not present

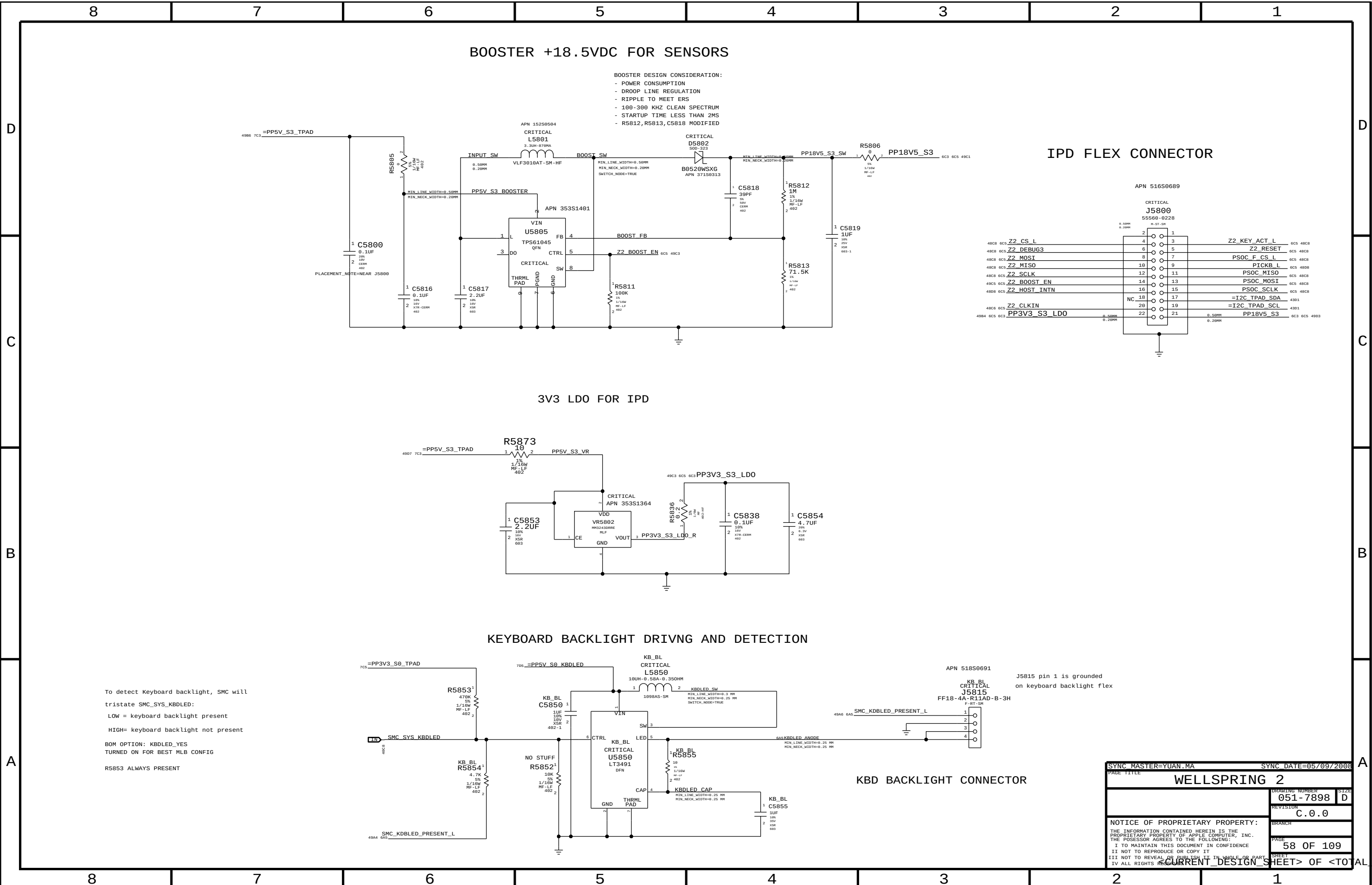
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TURNED ON FOR BEST MLB CONFIG
R5853 ALWAYS PRESENT

J5815 pin 1 is grounded on keyboard backlight flex

KBD BACKLIGHT CONNECTOR

The connector interface includes signals like SMC_KBDLED_PRESENT_L, KB_BL, and KBDLED_CAP.

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BOOSTER +18.5VDC FOR SENSORS

BOOSTER DESIGN CONSIDERATION:

- POWER CONSUMPTION
- DROOP LINE REGULATION
- RIPPLE TO MEET ERS
- 100-300 KHZ CLEAN SPECTRUM
- STARTUP TIME LESS THAN 2MS
- R5812, R5813, C5818 MODIFIED

IPD FLEX CONNECTOR

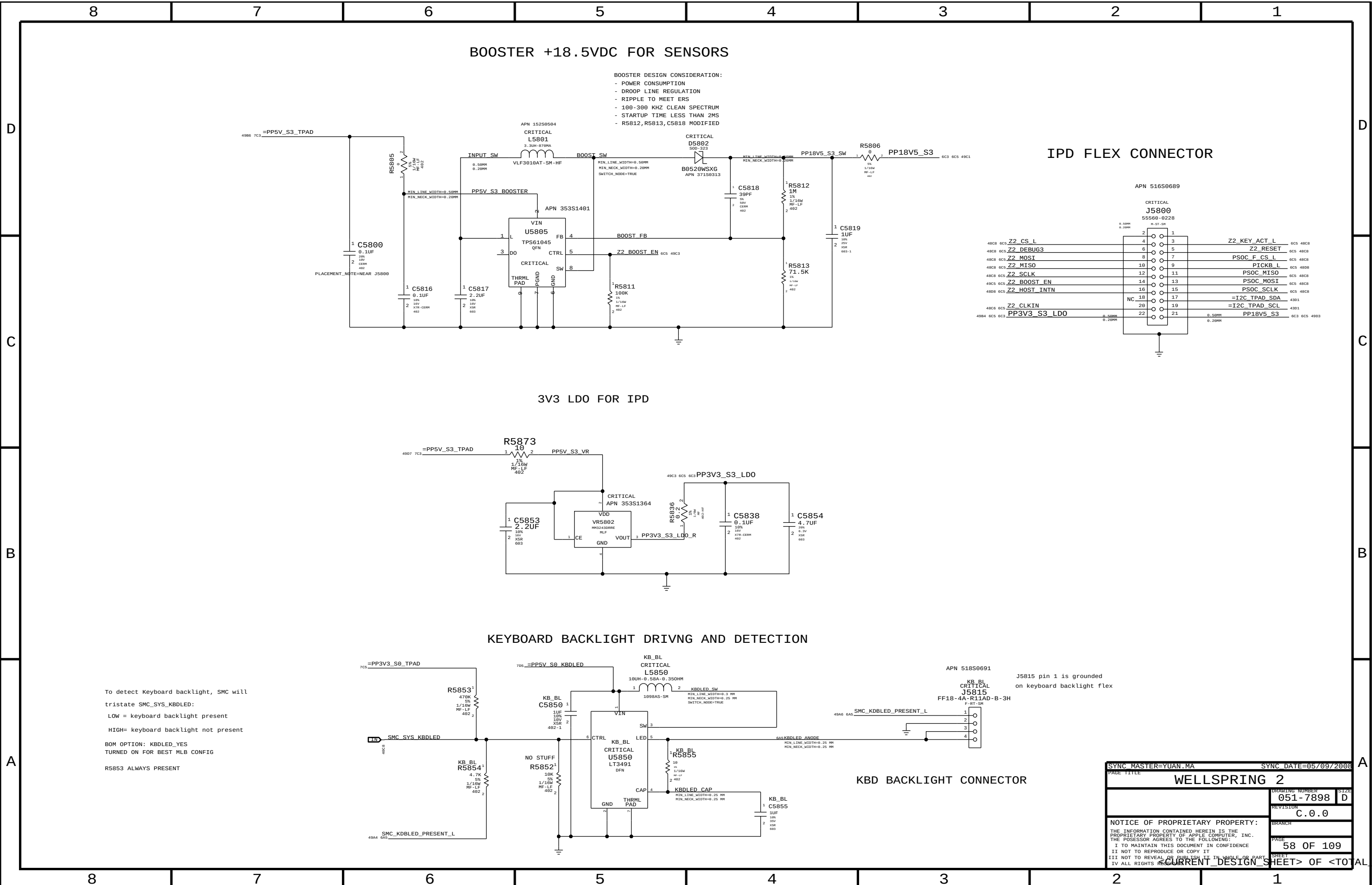
3V3 LDO FOR IPD

KEYBOARD BACKLIGHT DRIVING AND DETECTION

To detect Keyboard backlight, SMC will tristate SMC_SYS_KBDLED:
LOW = keyboard backlight present
HIGH= keyboard backlight not present
BOM OPTION: KBDLED_YES
TURNED ON FOR BEST MLB CONFIG
R5853 ALWAYS PRESENT

KBD BACKLIGHT CONNECTOR

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		TOTAL SHEETS		109



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BOOSTER +18.5VDC FOR SENSORS

BOOSTER DESIGN CONSIDERATION:

- POWER CONSUMPTION
- DROOP LINE REGULATION
- RIPPLE TO MEET ERS
- 100-300 KHZ CLEAN SPECTRUM
- STARTUP TIME LESS THAN 2MS
- R5812, R5813, C5818 MODIFIED

IPD FLEX CONNECTOR

3V3 LDO FOR IPD

KEYBOARD BACKLIGHT DRIVING AND DETECTION

KBD BACKLIGHT CONNECTOR

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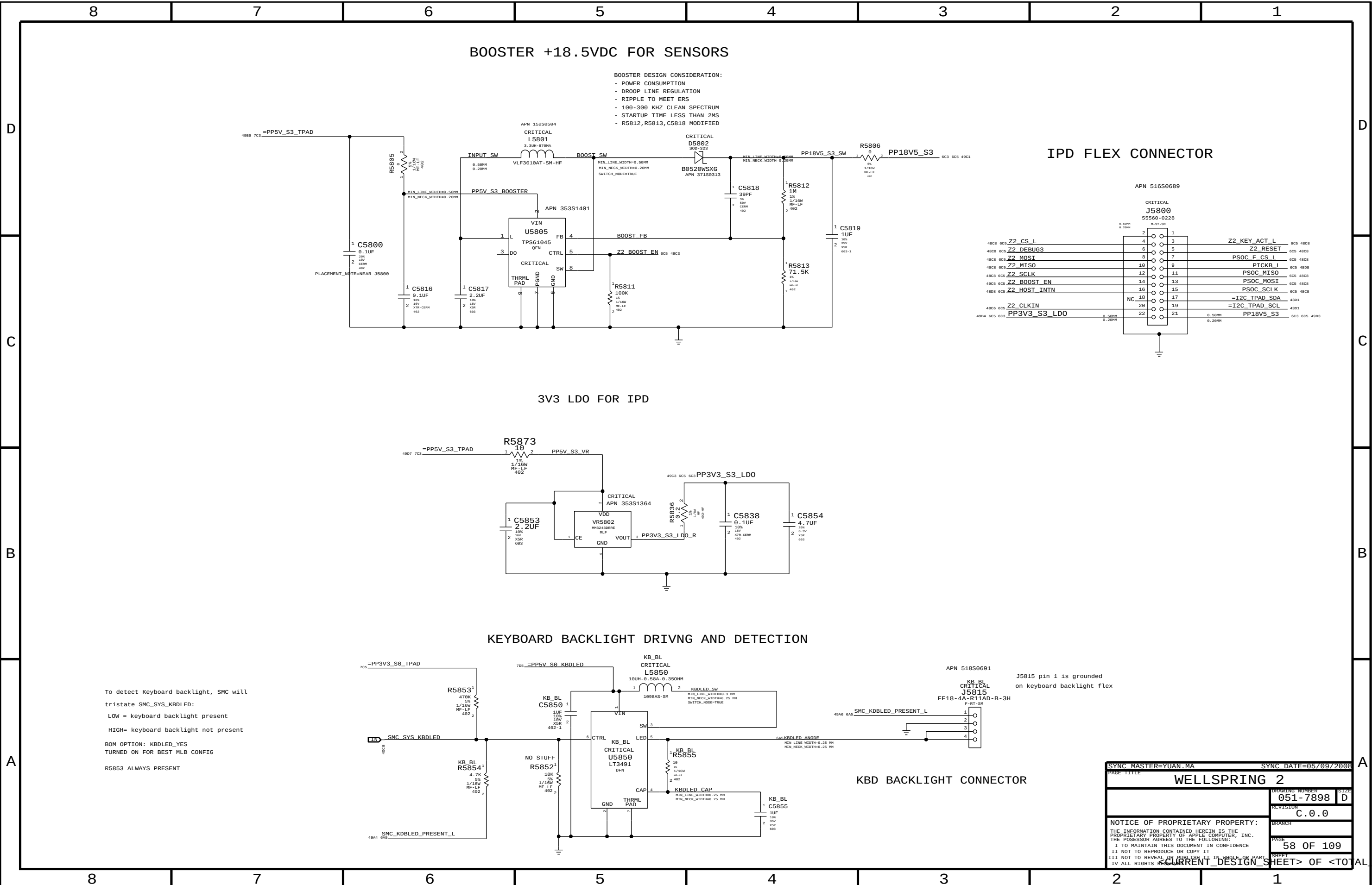
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BOOSTER +18.5VDC FOR SENSORS

BOOSTER DESIGN CONSIDERATION:

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IPD FLEX CONNECTOR

3V3 LDO FOR IPD

KEYBOARD BACKLIGHT DRIVING AND DETECTION

KBD BACKLIGHT CONNECTOR

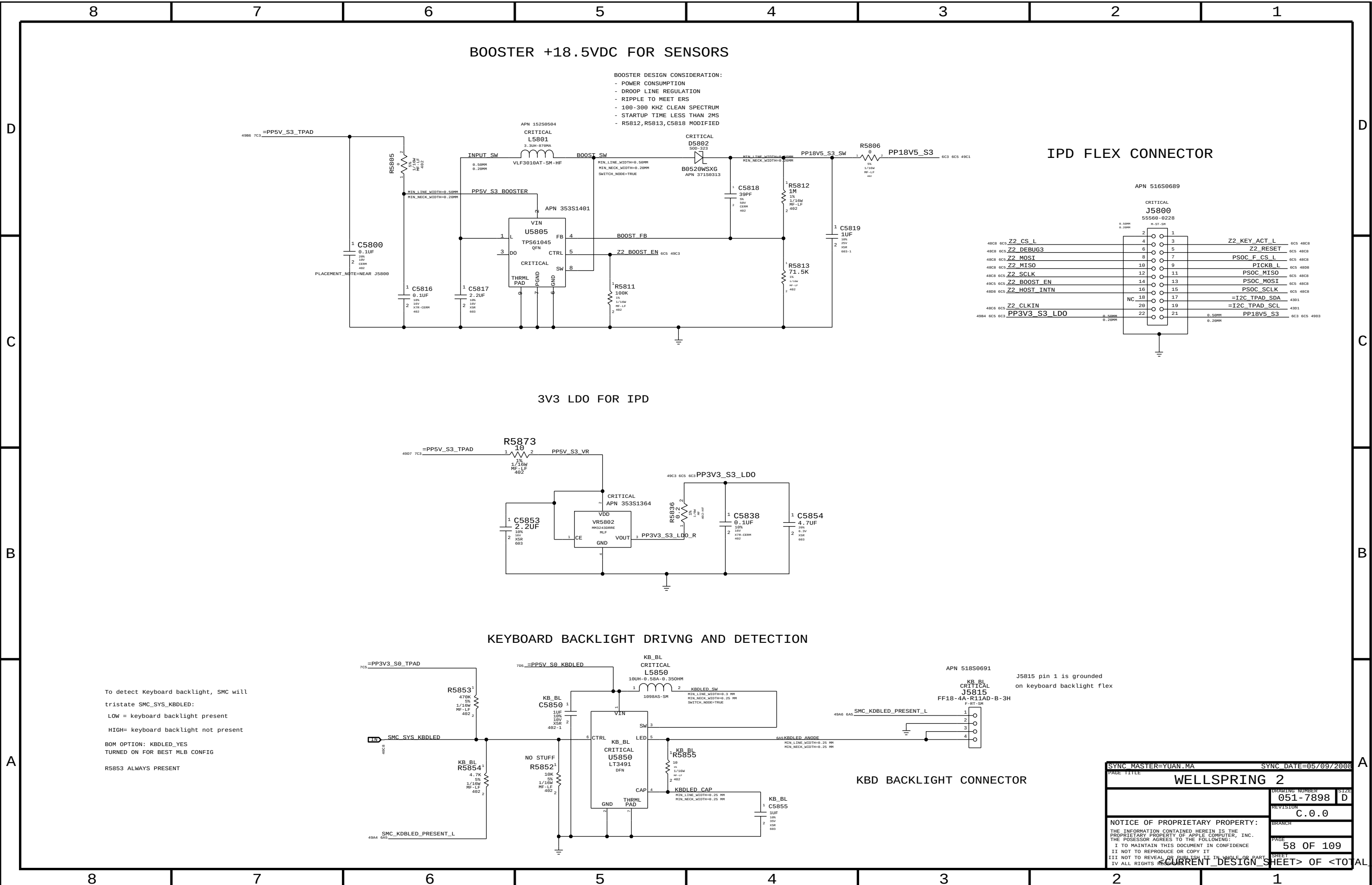
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HIGH= keyboard backlight not present
BOM OPTION: KBDLED_YES
TURNED ON FOR BEST MLB CONFIG
R5853 ALWAYS PRESENT

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8 7 6 5 4 3 2 1

BOOSTER +18.5VDC FOR SENSORS

BOOSTER DESIGN CONSIDERATION:

- POWER CONSUMPTION
- DROOP LINE REGULATION
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IPD FLEX CONNECTOR

3V3 LDO FOR IPD

KEYBOARD BACKLIGHT DRIVING AND DETECTION

KBD BACKLIGHT CONNECTOR

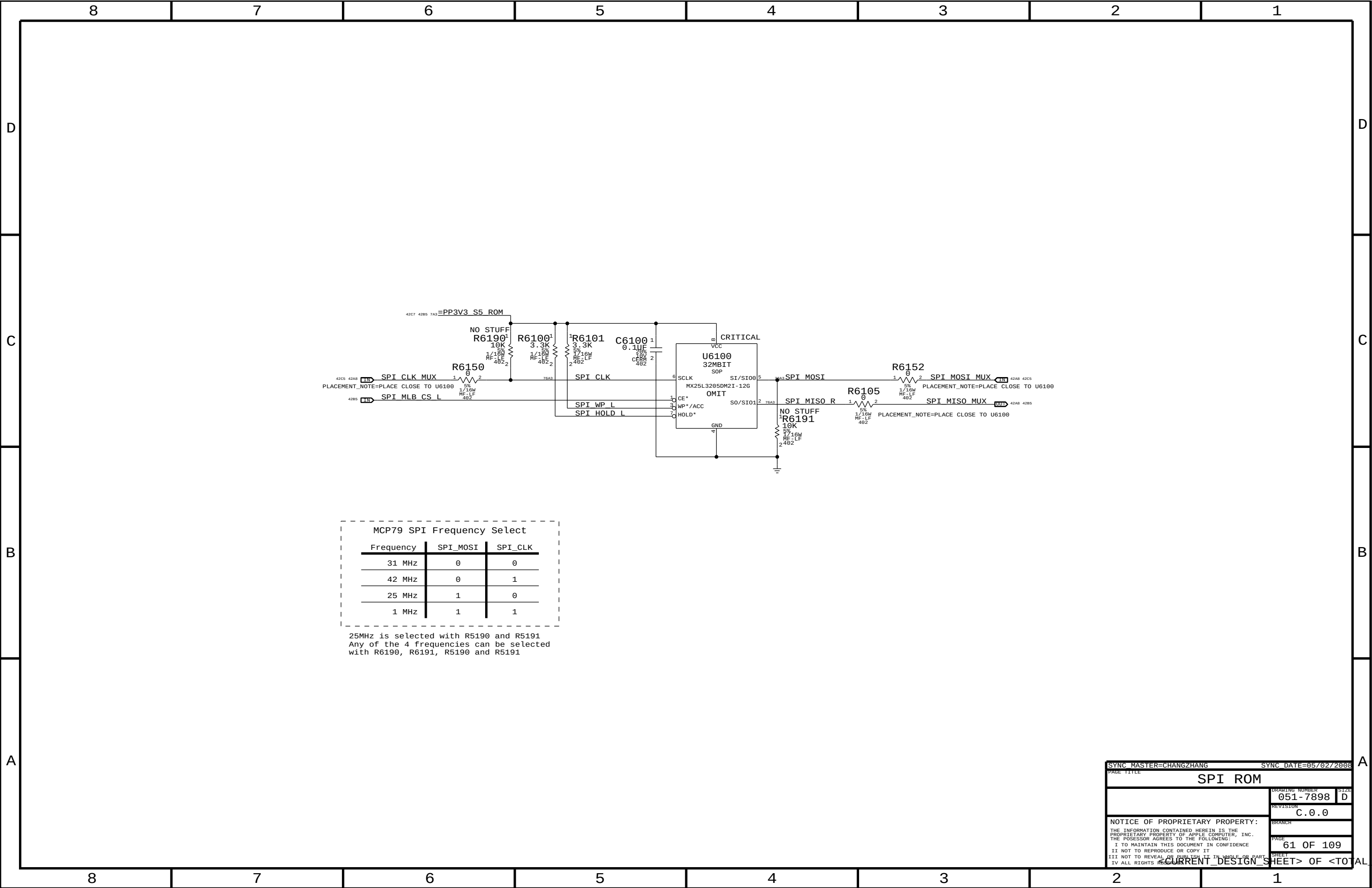
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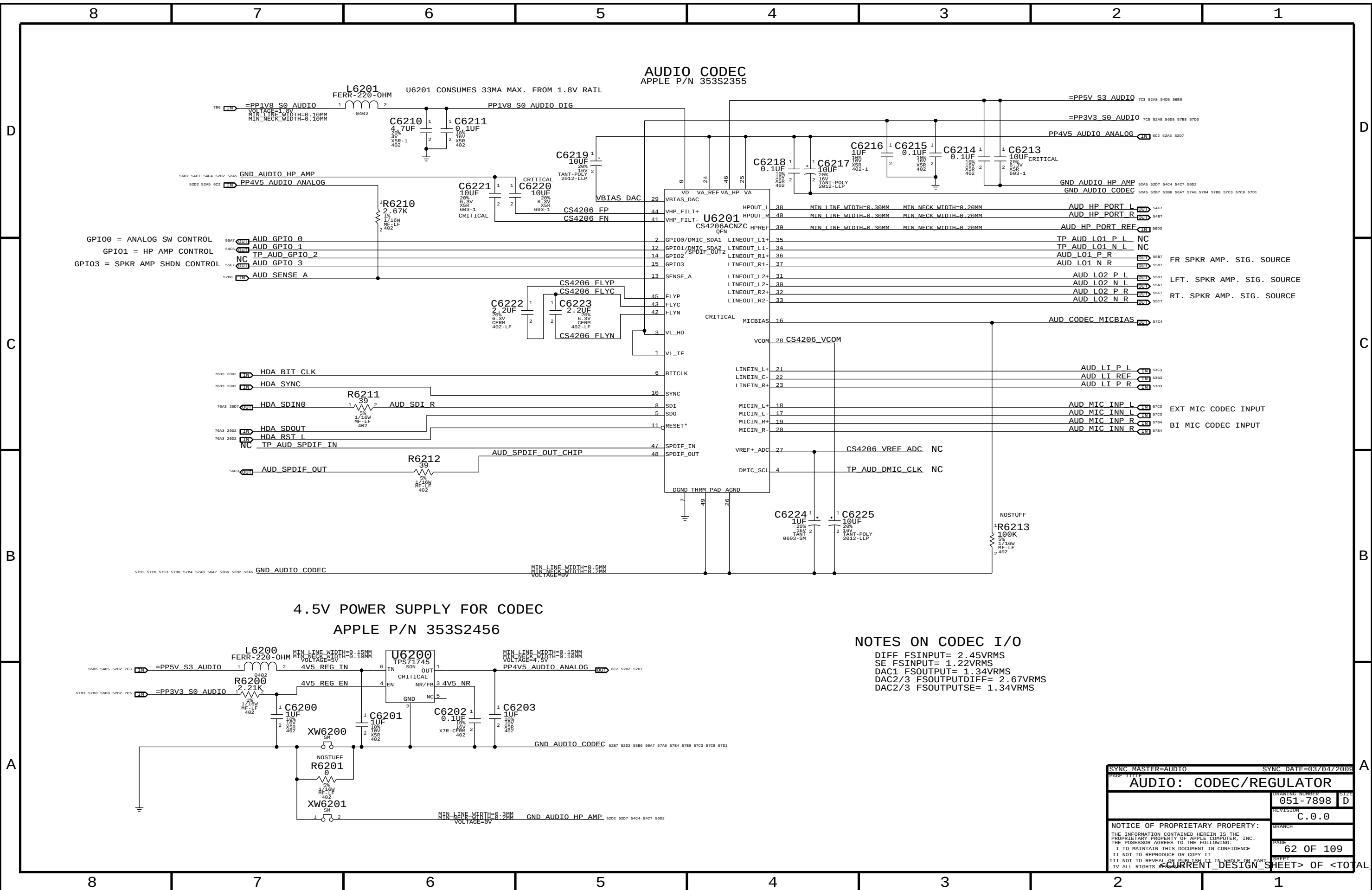
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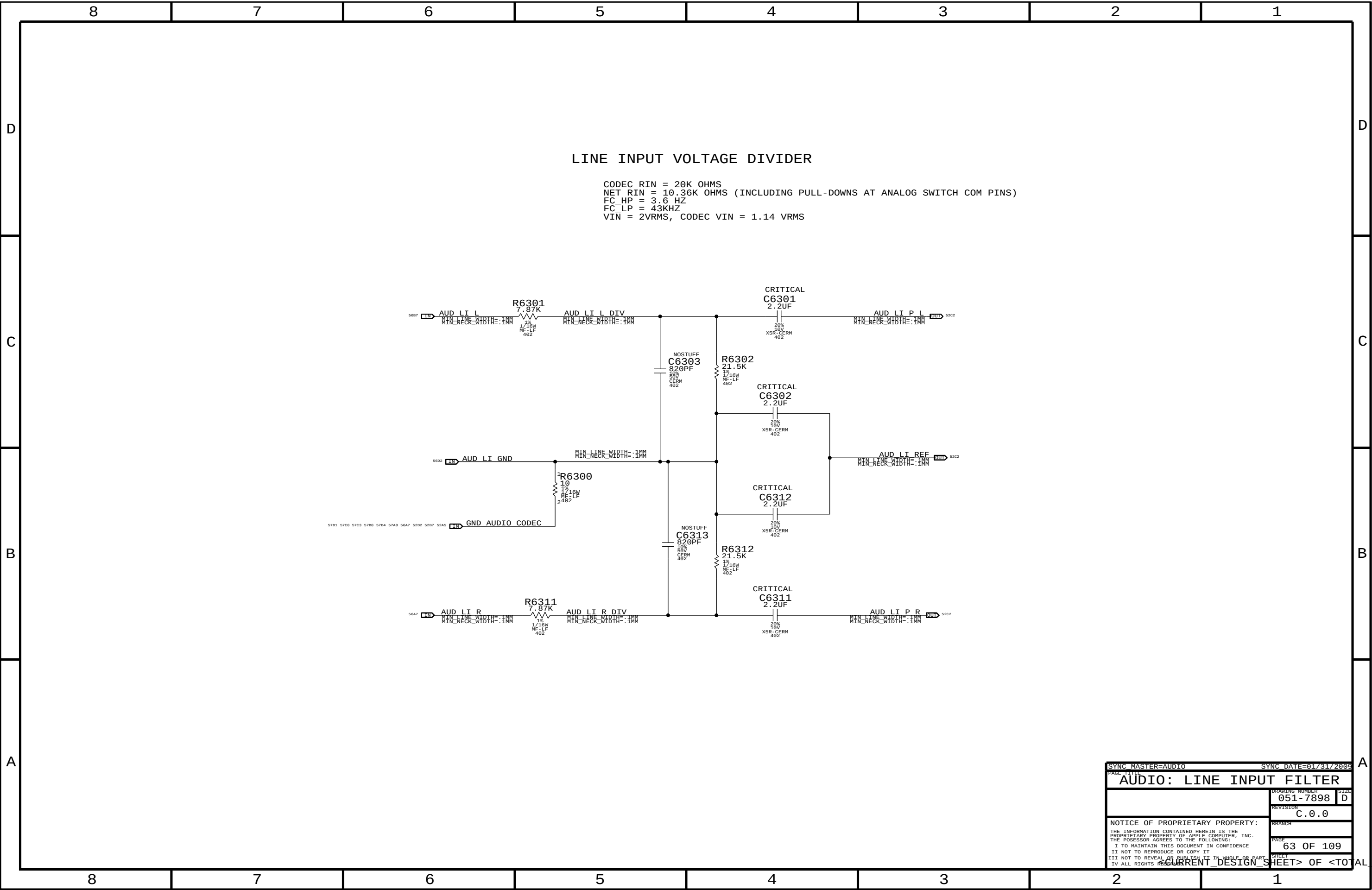
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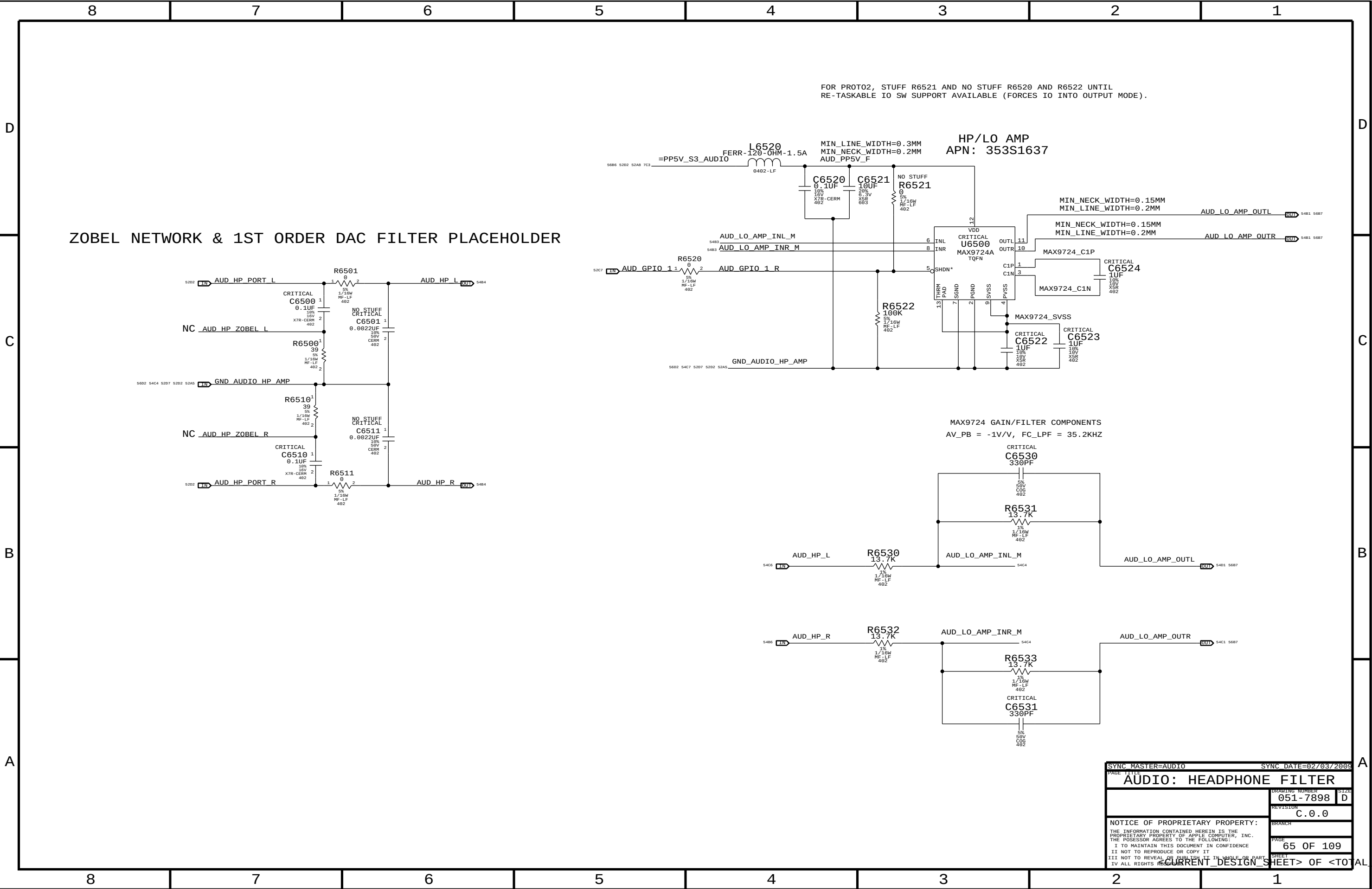
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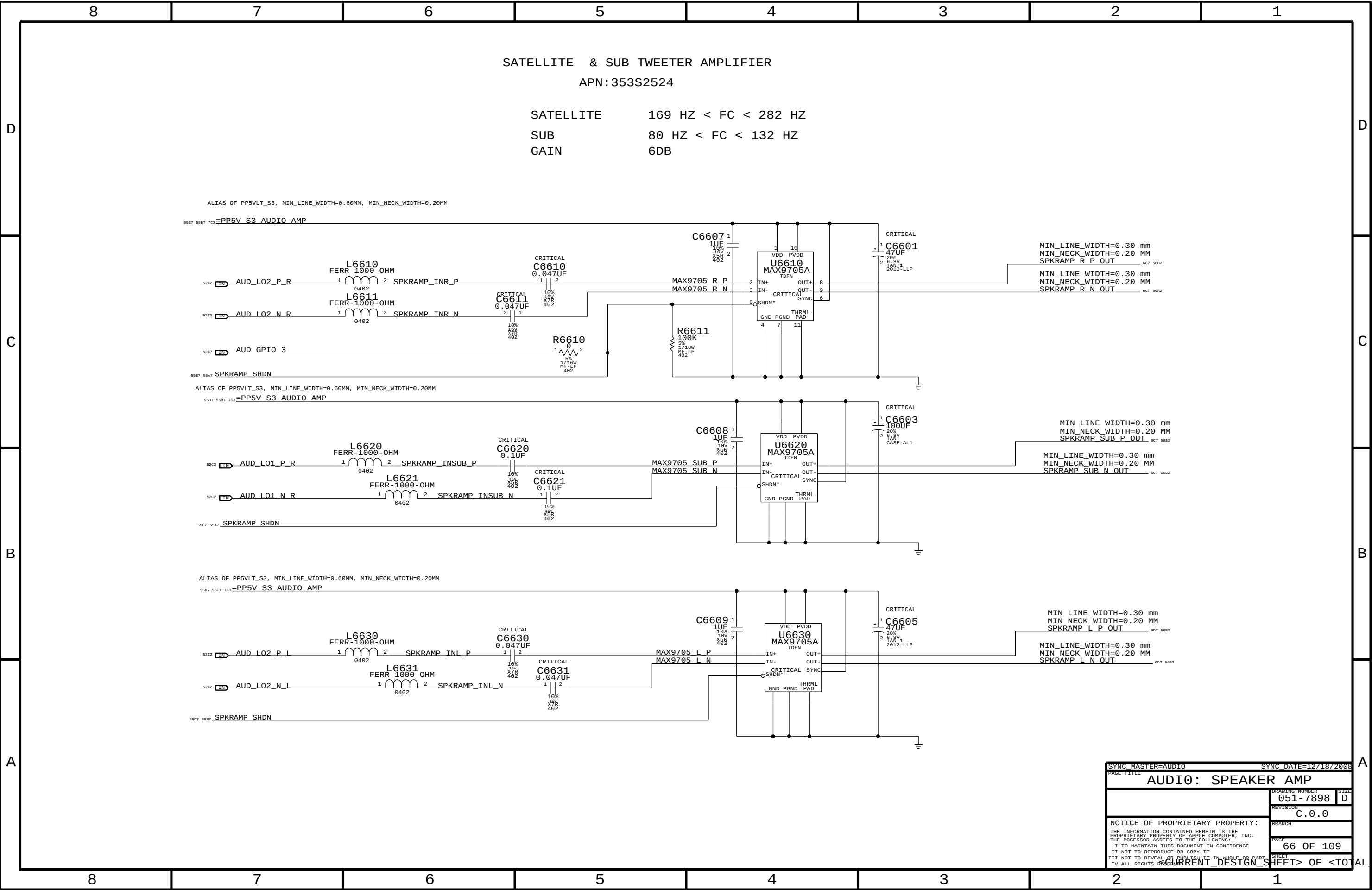
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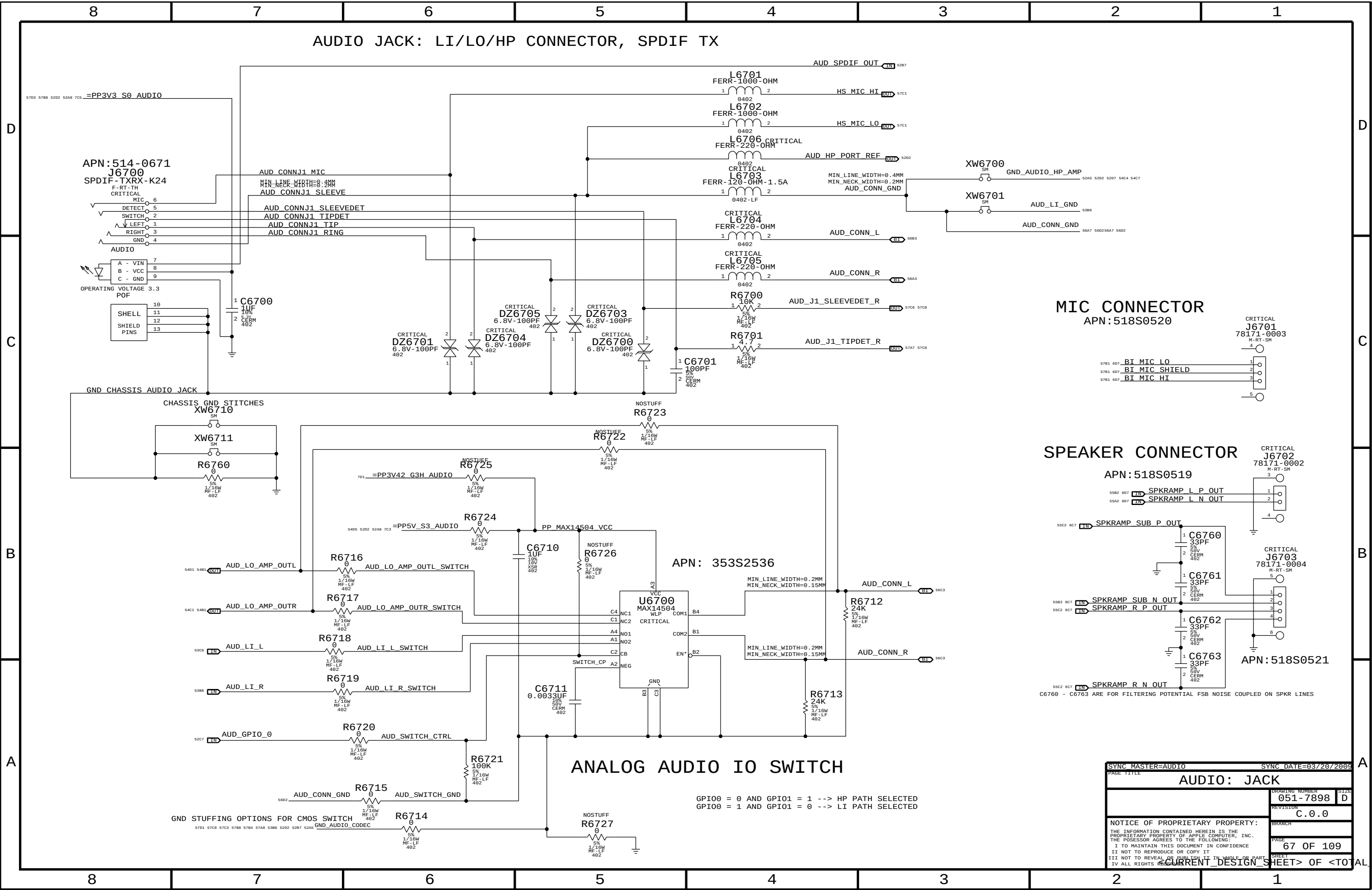








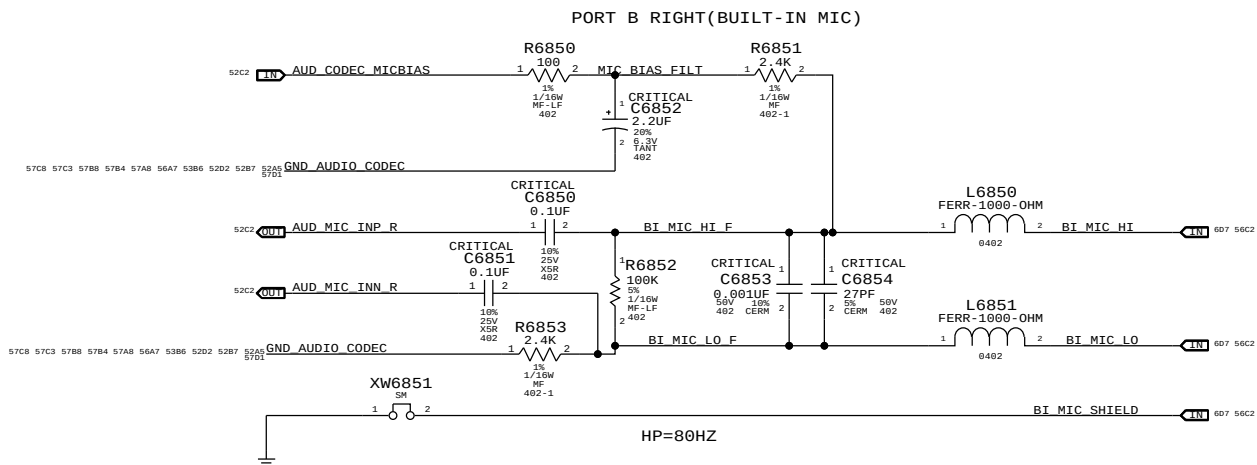
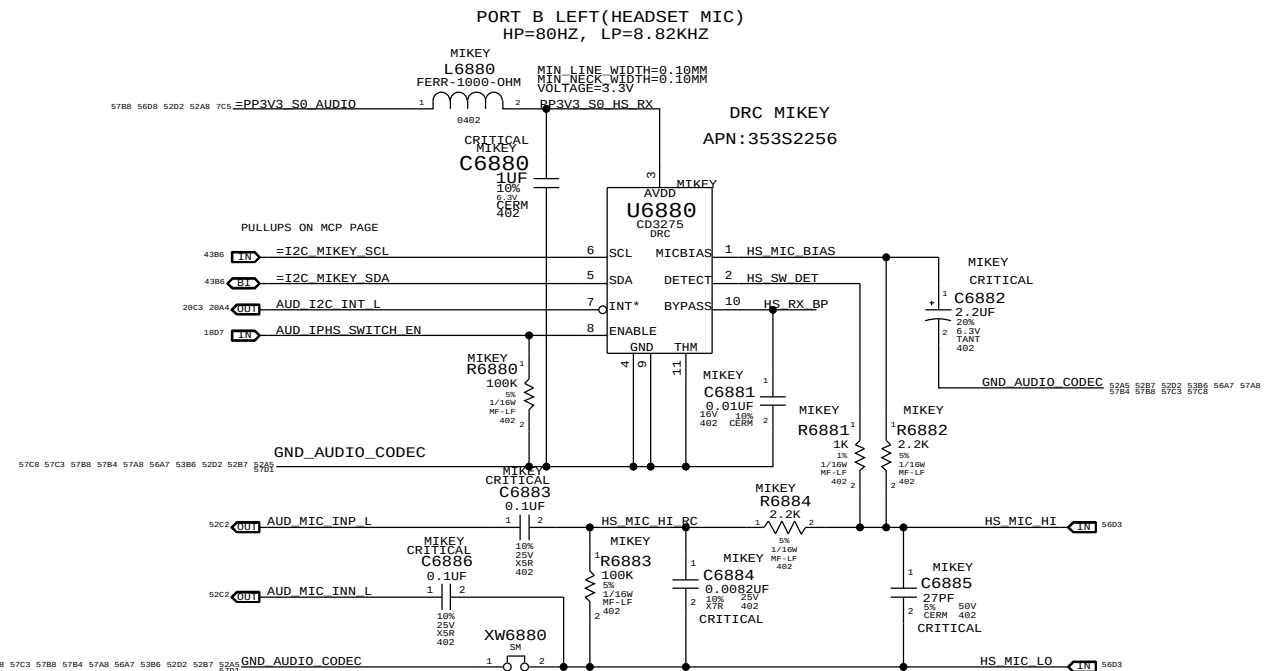
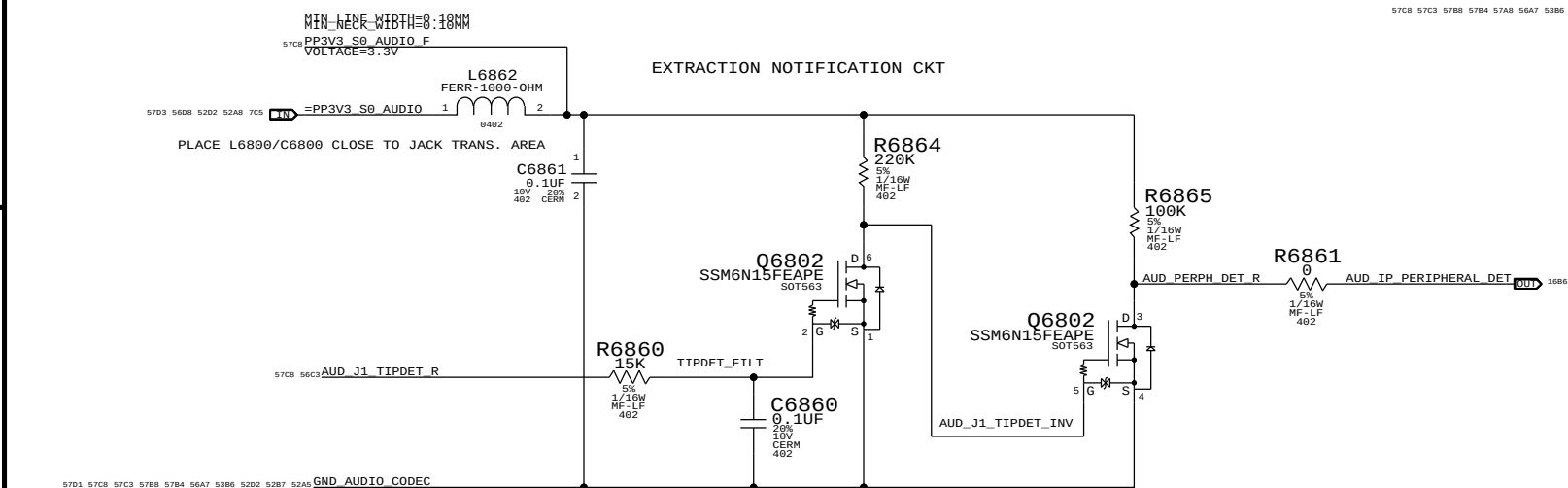
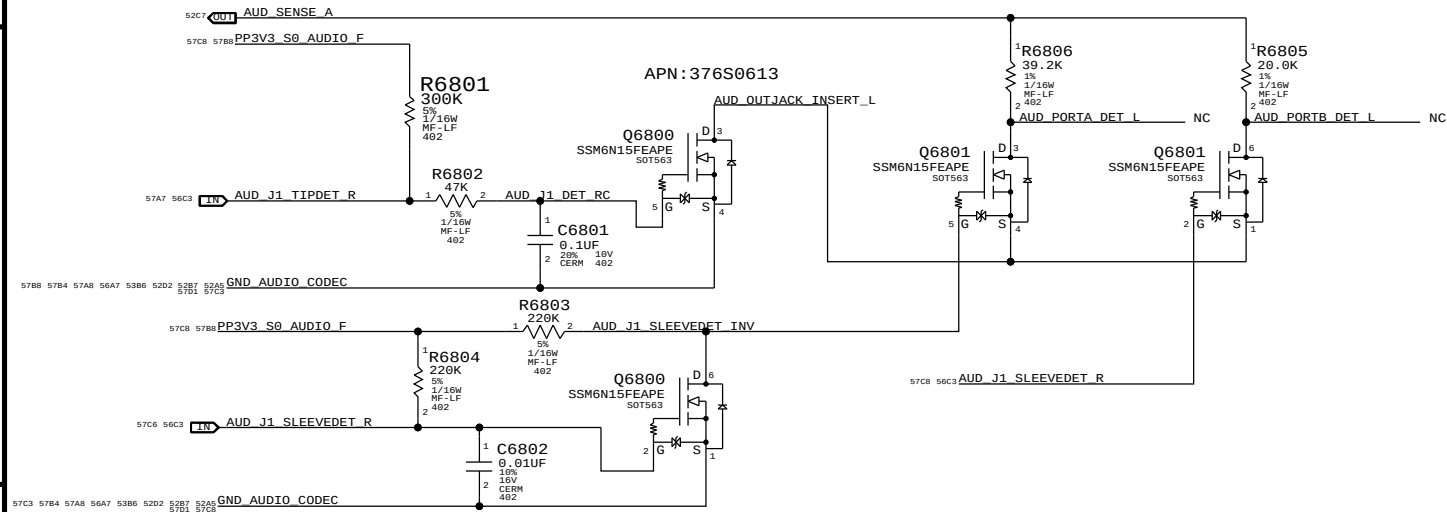
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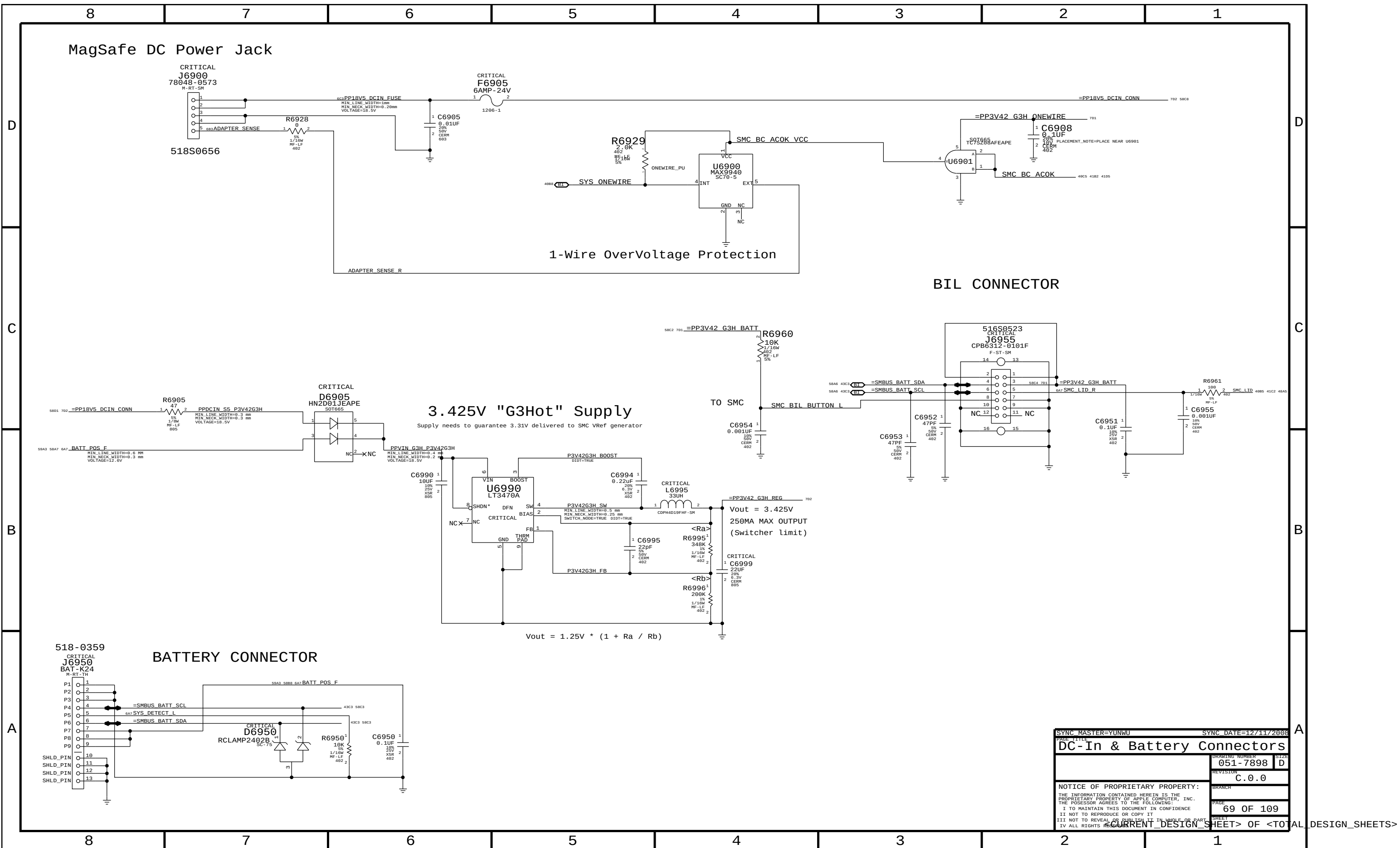
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FUNCTION	VOLUME	CONVERTER	PIN COMPLEX	MUTE CONTROL	DET ASSIGNMENT
HP/LINE OUT	0X02 (2)	0X02 (2)	0X09 (9, A)	GPIO_0 AND GPIO_1	0X09 (A)
LINE IN	0X05 (5)	0X05 (5)	0X0C (12)	GPIO_0 AND GPIO_1	0X09 (A) AND UI ELEMENT
SATELLITES	0X04 (4)	0X04 (4)	0X0B (11)	GPIO_3	N/A
SUB	0X03 (3)	0X03 (03)	0X0A (10)	GPIO_3	N/A
SPDIF OUT	N/A	0X08 (8)	0X10 (16)	N/A	0X0D (B)

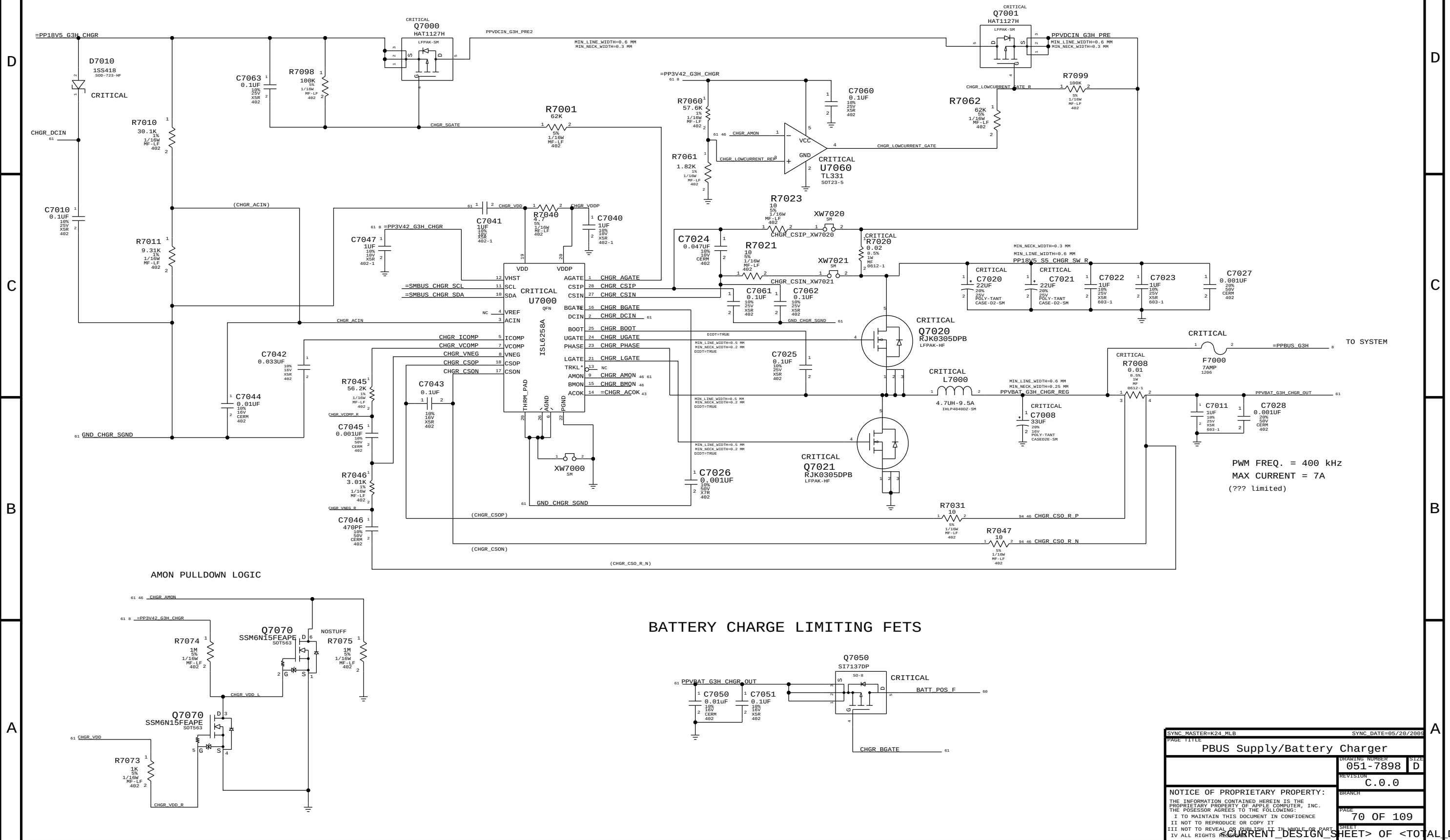
FUNCTION	CONVERTER	PIN COMPLEX	VREF	DET ASSIGNMENT
BUILT-IN MIC	0X06 (6)	0X0D (13, B, RIGHT)	MIC_BIAS (80%)	N/A
HEADSET MIC	0X06 (6)	0X0D (13, V22, B, LEFT)	MIKEY	MIKEY



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PBUS SUPPLY / BATTERY CHARGER

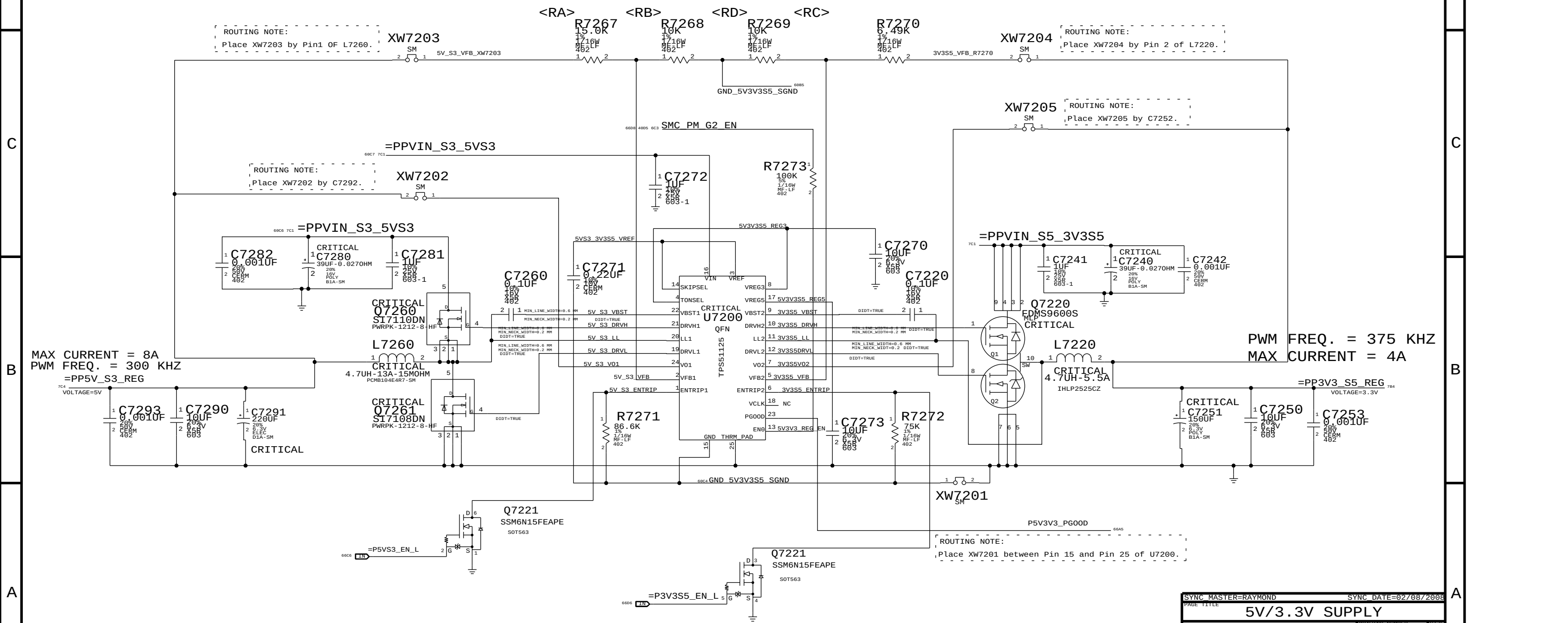


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PAGE TITLE		PBUS Supply/Battery Charger	
DRAWING NUMBER		051-7898	D
REVISION		C.0.0	
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5V_S3/3.3V_S5 POWER SUPPLY

$$V_{OUT} = (2 * RA / RB) + 2$$

$$V_{OUT} = (2 * RC / RD) + 2$$



MAX CURRENT = 8A
PWM FREQ. = 300 KHZ
=PP5V_S3_REG

PWM FREQ. = 375 KHZ
MAX CURRENT = 4A

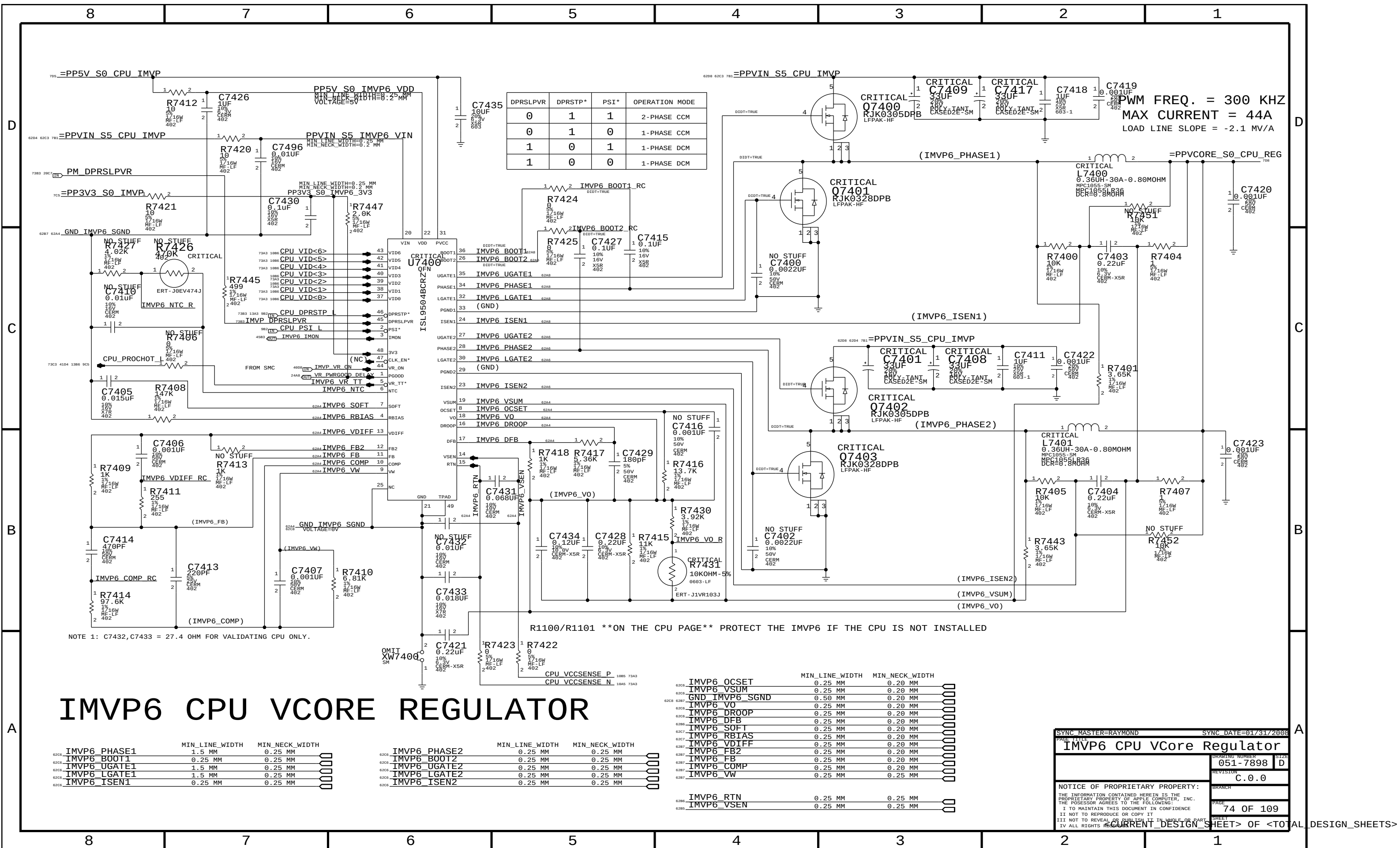
SEPERATED MASTER PG000 FOR BOTH 5V AND 3V3.

SYNC MASTER=RAYMOND		SYNC DATE=02/08/2008	
PAGE TITLE			
5V/3.3V SUPPLY			
		DRAWING NUMBER	SIZE
		051-7898	D
		REVISION	C.0.0
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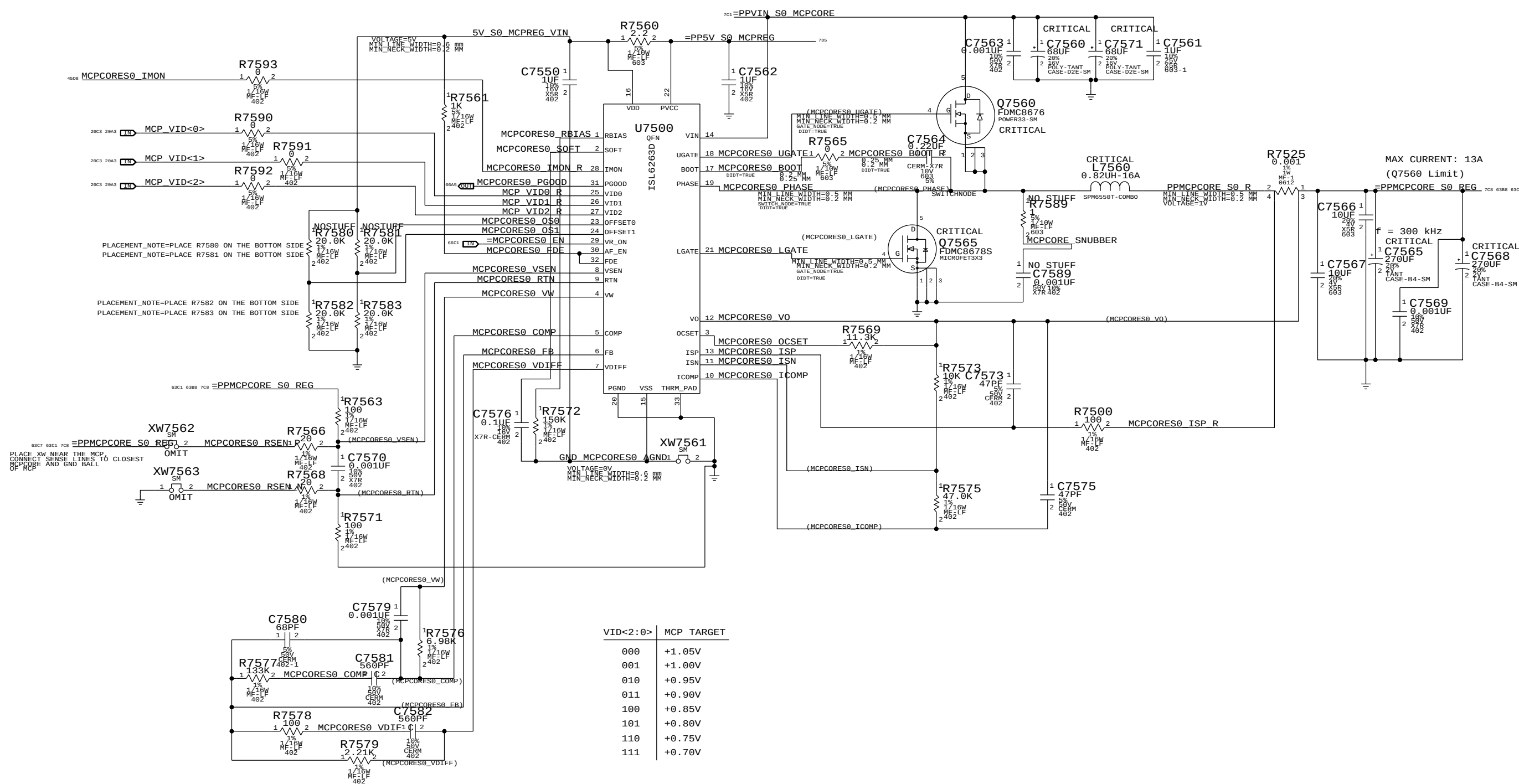
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BA



MCP VCORE POWER SUPPLY



SYNC MASTER=K19 MLB

SYNC DATE=12/10/2008

MCP CORE REGULATOR

DRAWING NUMBER
051-7898

REVISION
C.0.0

BRANCH

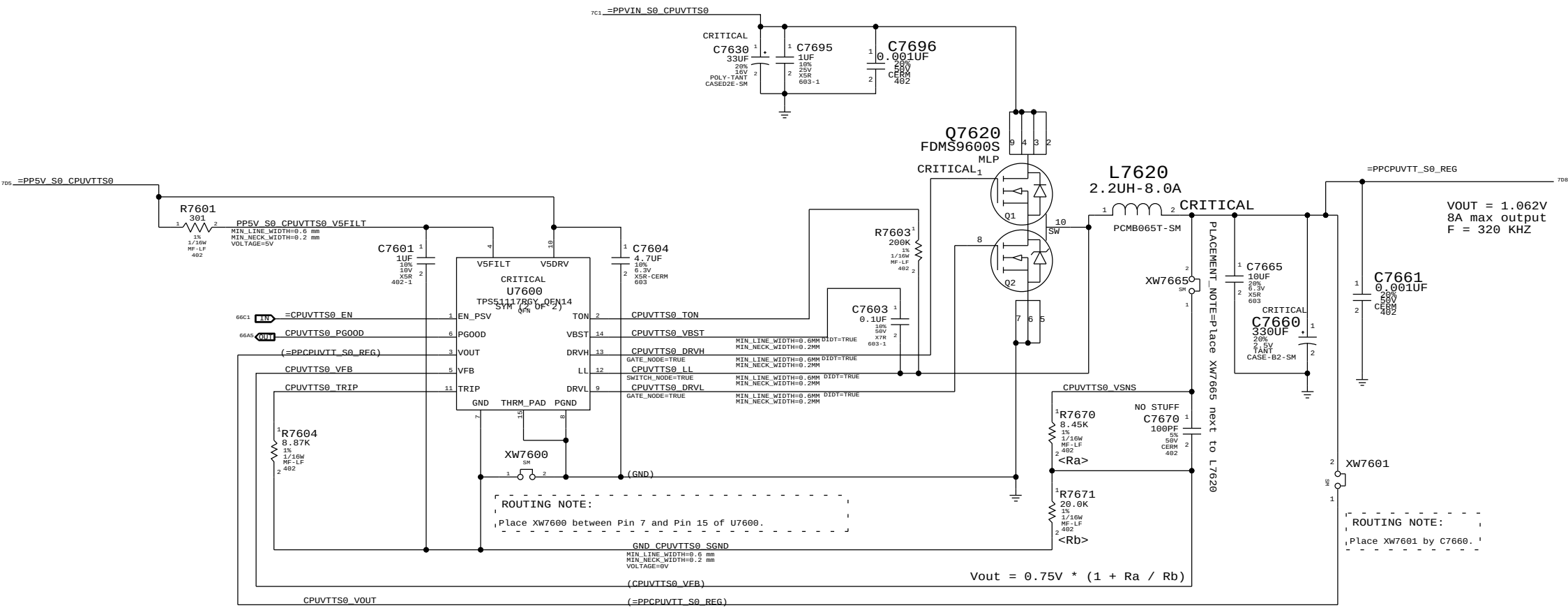
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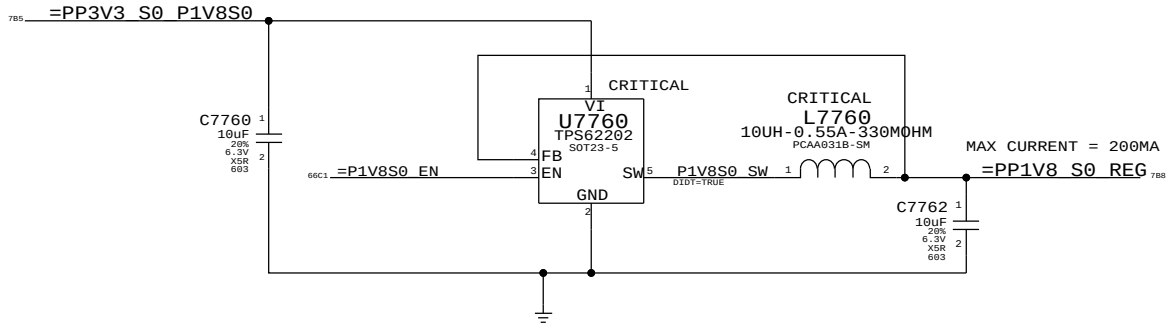
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CPUVTT POWER SUPPLY

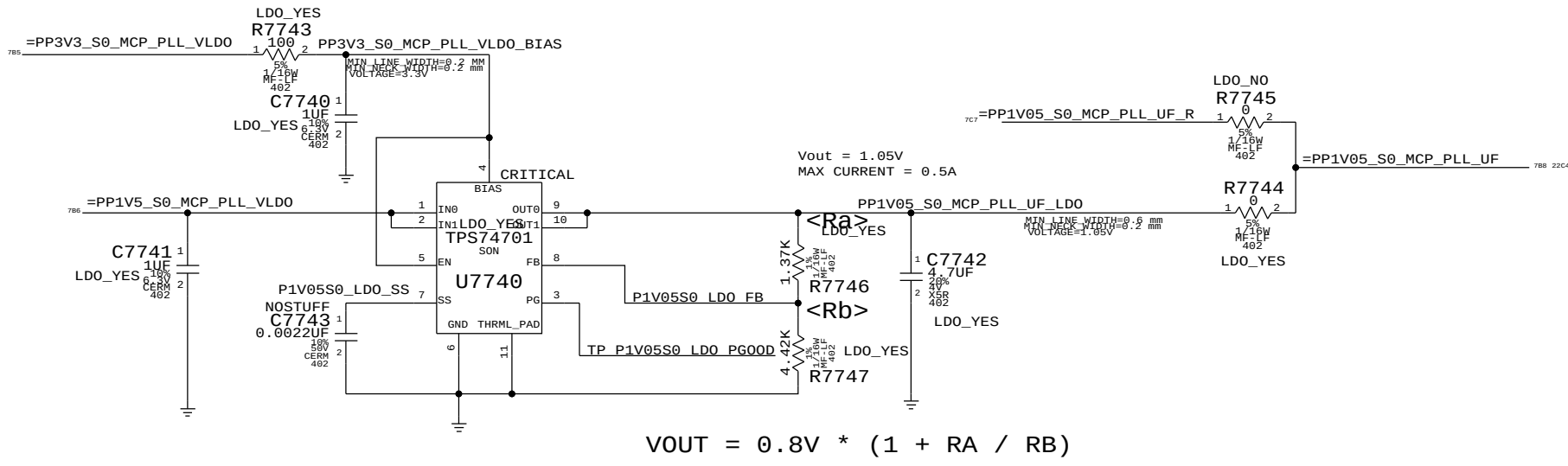


SYNC MASTER=RAYMOND		SYNC DATE=02/08/2008	
PAGE TITLE			
CPU VTT(1.05V)		SUPPLY	
DRAWING NUMBER		SIZE	
051-7898		D	
REVISION		C.0.0	
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1.8V S0 SWITCHER

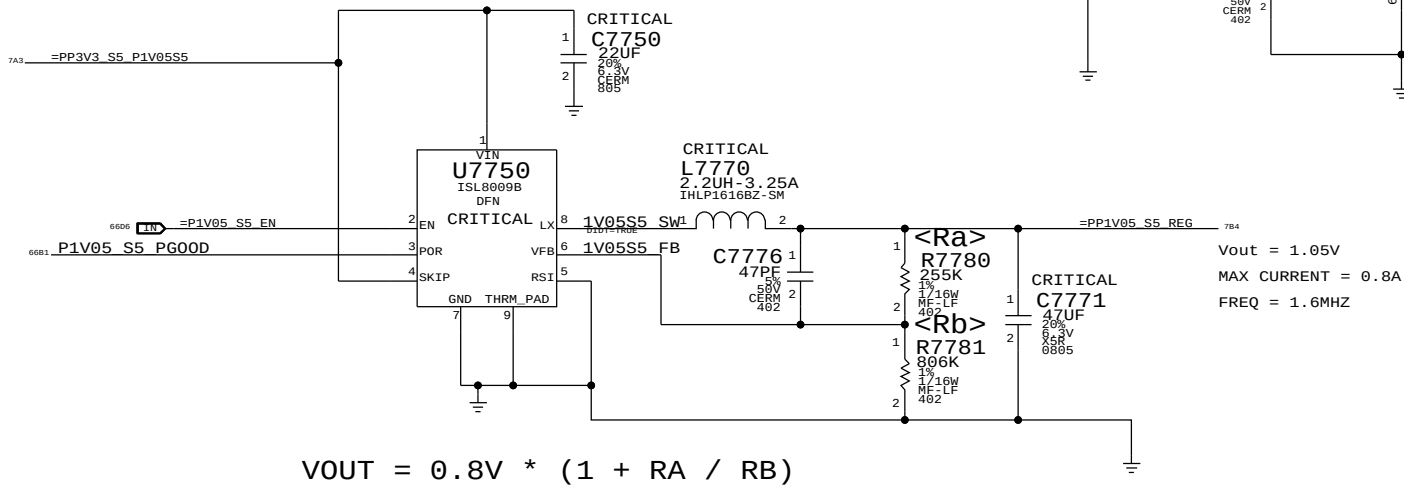


1.05V S0 PLL LDO



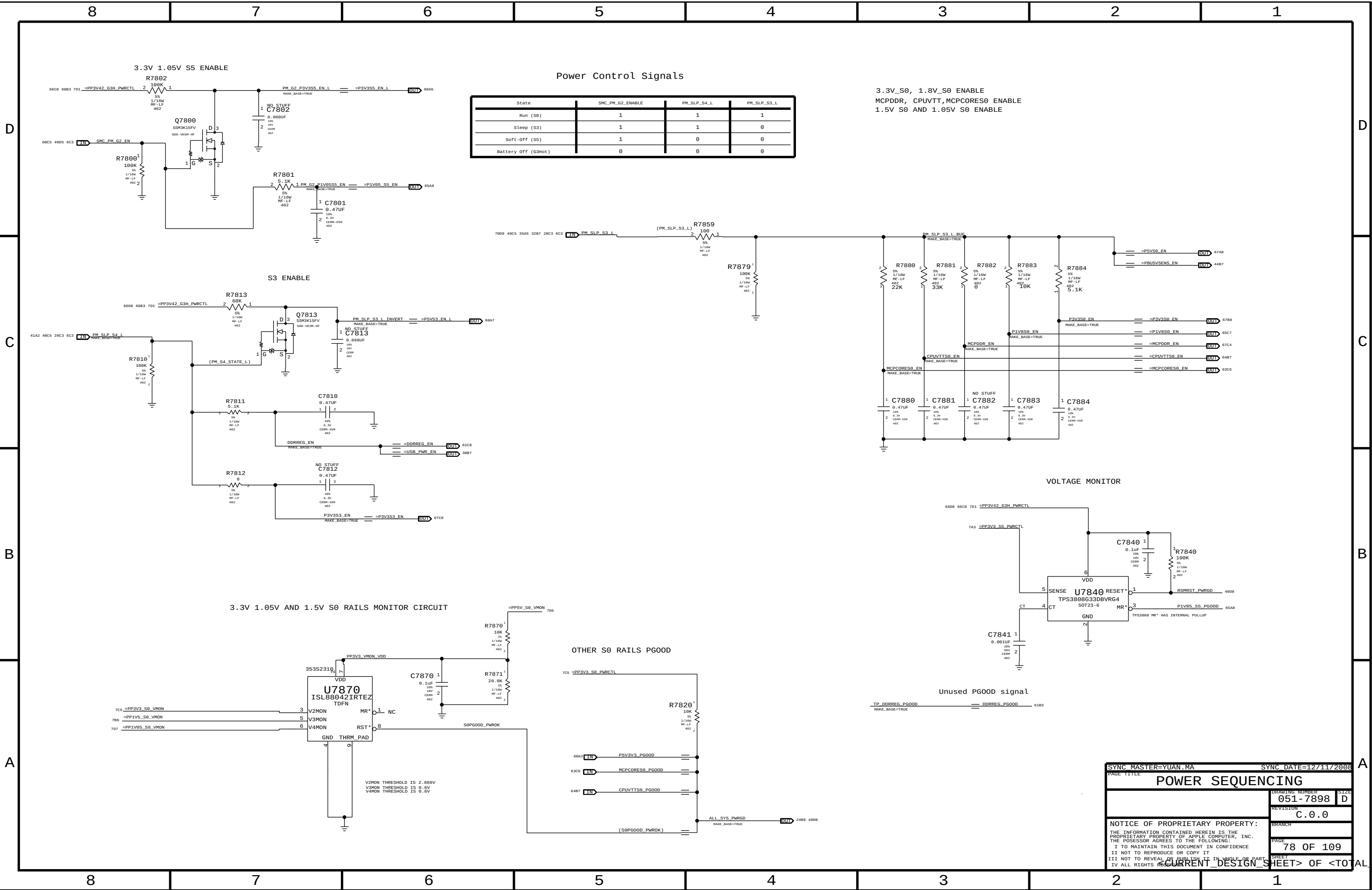
$V_{OUT} = 0.8V * (1 + R_A / R_B)$

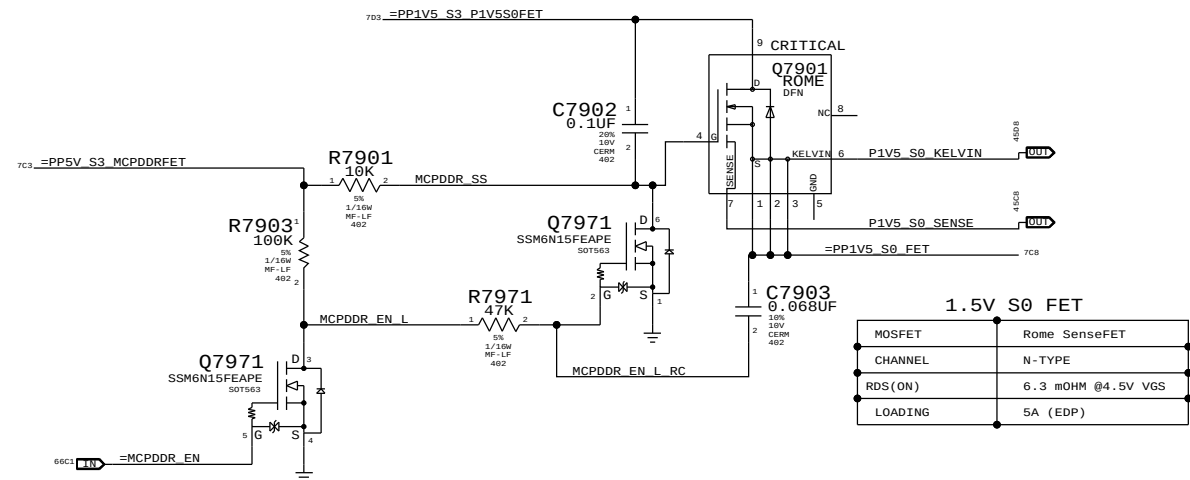
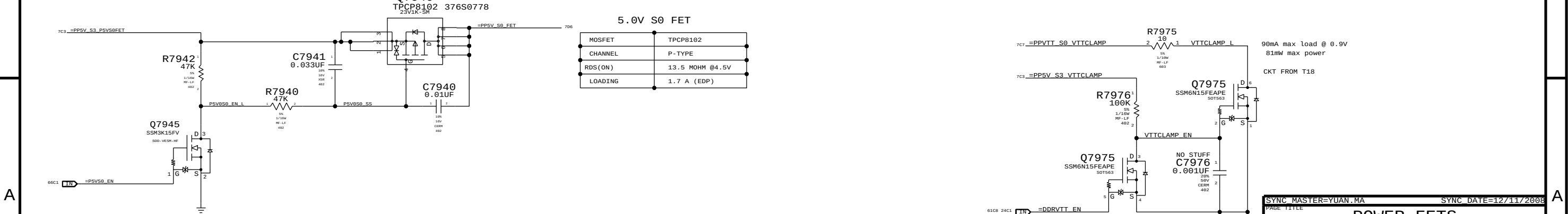
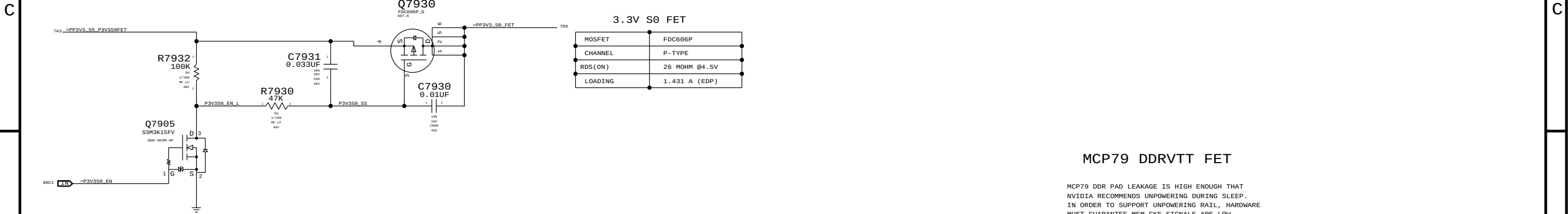
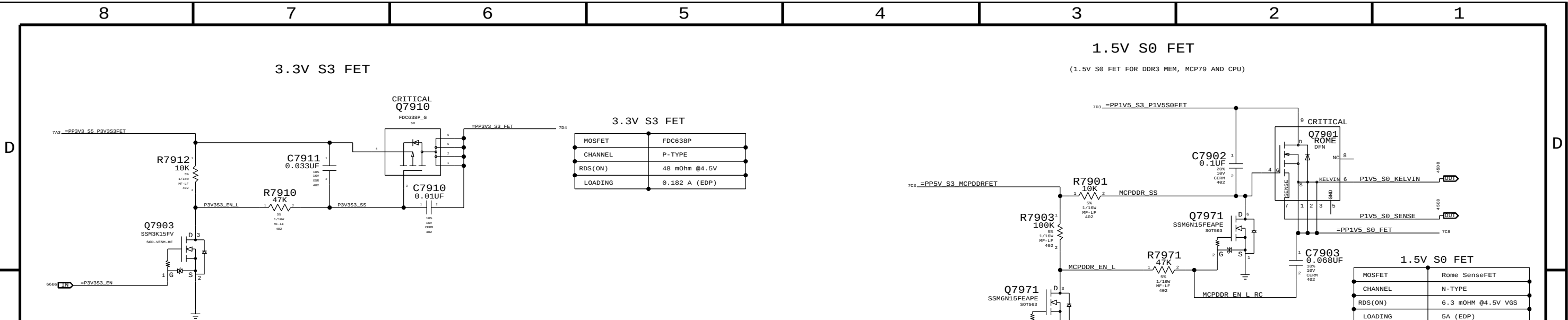
MCP 1.05V S5 (AUXC) SUPPLY



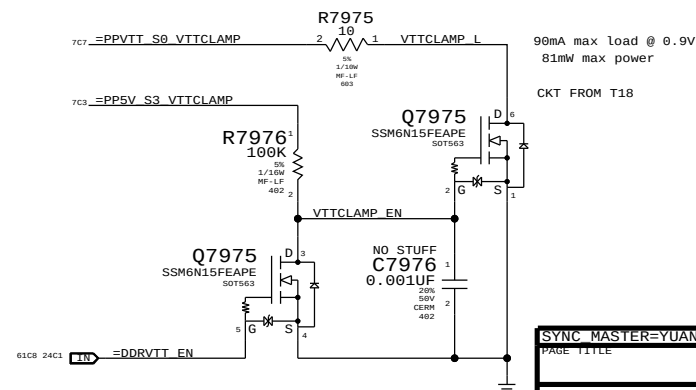
$V_{OUT} = 0.8V * (1 + R_A / R_B)$

SYNC MASTER=RAYMOND		SYNC DATE=01/23/2008	
PAGE TITLE			
MISC POWER SUPPLIES			
		DRAWING NUMBER	SHEET
		051-7898	D
		REVISION	
		C.0.0	
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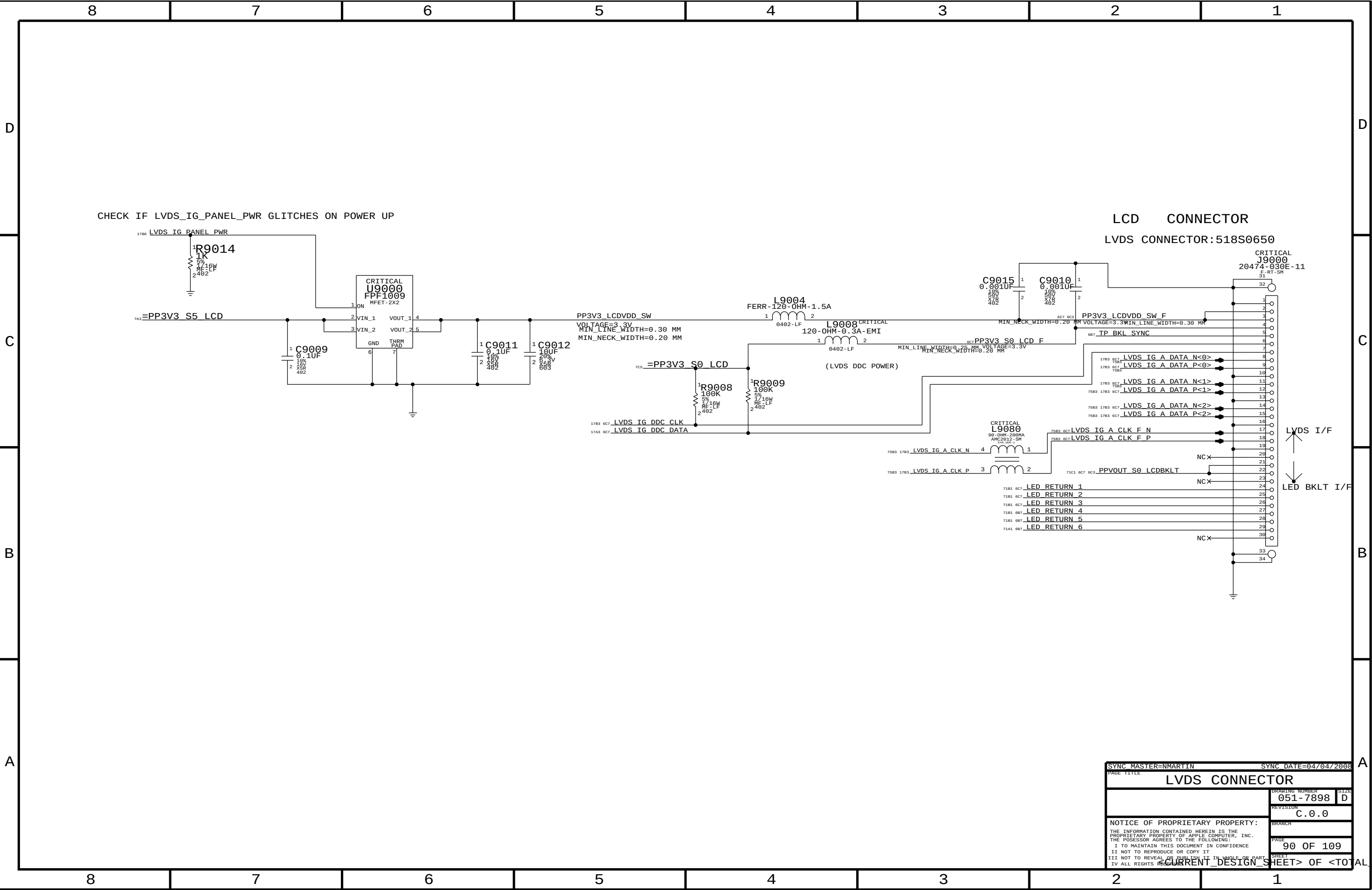




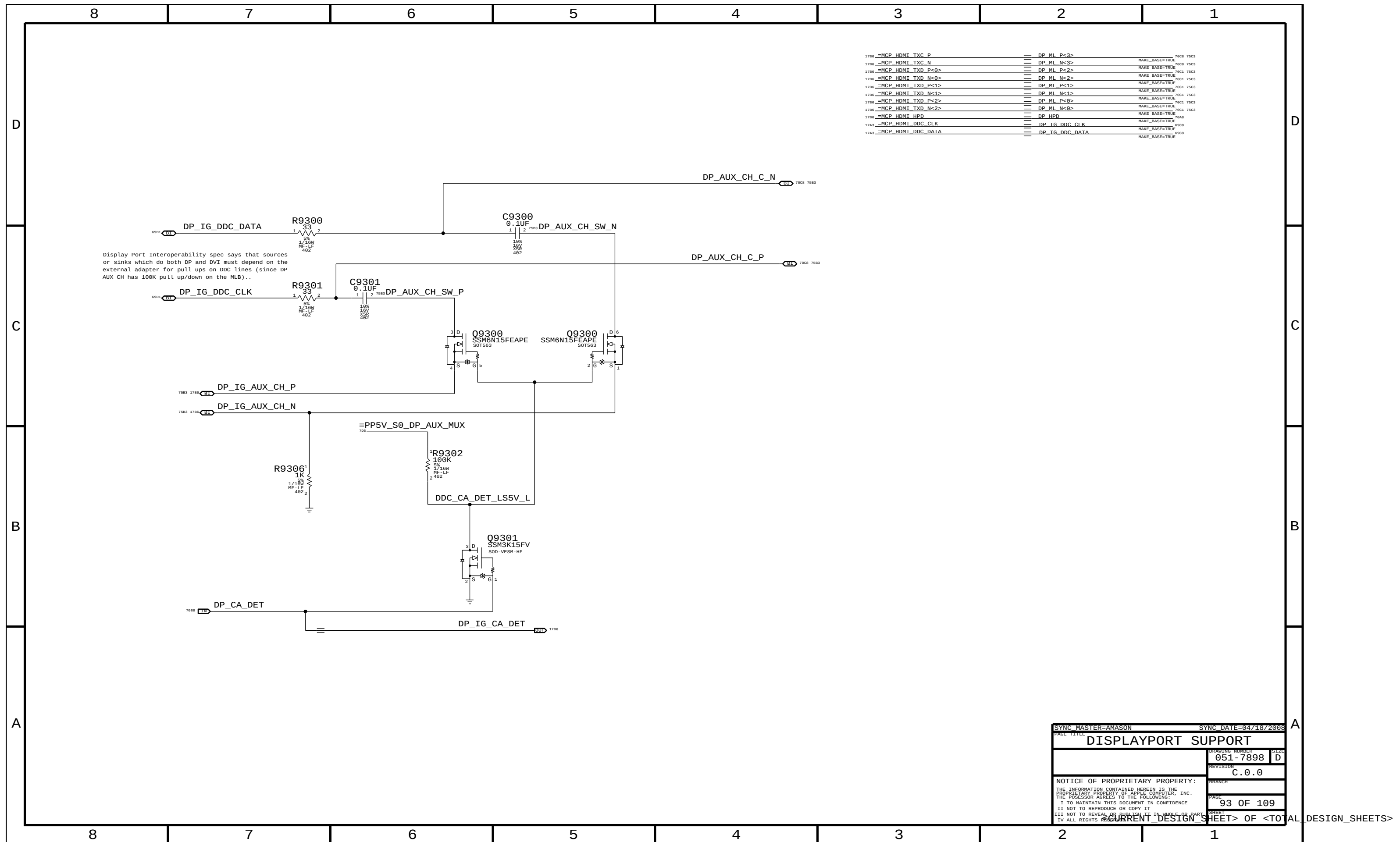
MCPT9 DDR PAD LEAKAGE IS HIGH ENOUGH THAT
NVIDIA RECOMMENDS UNPOWERING DURING SLEEP.
IN ORDER TO SUPPORT UNPOWERING RAIL, HARDWARE
MUST GUARANTEE MEM_CKE SIGNALS ARE LOW
BEFORE RAIL IS TURNED OFF, AND REMAINS LOW
UNTIL AFTER RAIL TURNS BACK ON OR DIMMS
WILL EXIT SELF-REFRESH PREMATURELY.
MEM_VTT_EN OUTPUT FROM MCPT9 USED TO ENABLE CLAMP
ON VTT RAIL, WHICH PULLS ALL CKE SIGNALS
LOW THROUGH VTT TERMINATION RESISTORS.



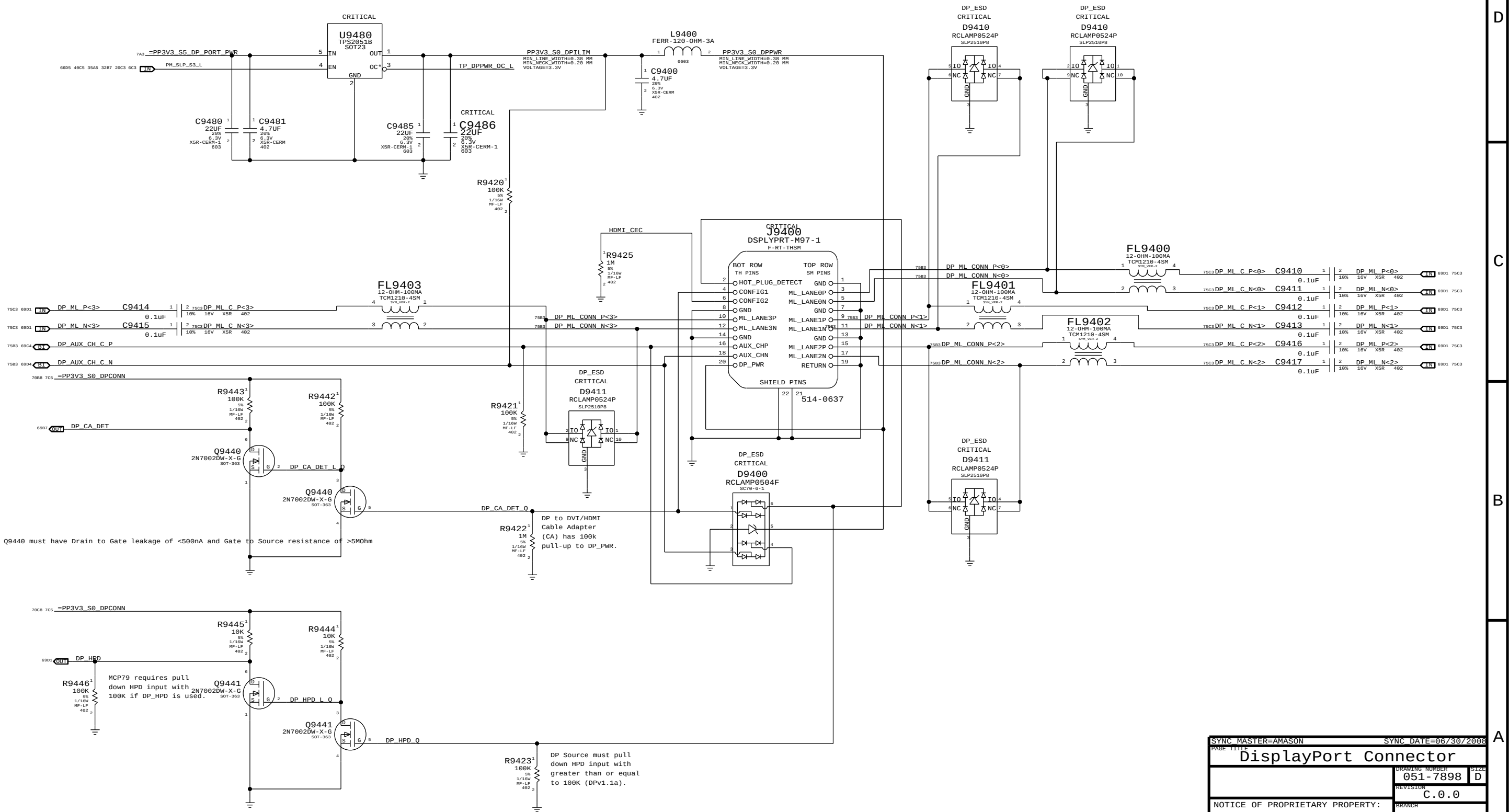
SYNCH MASTER=YUAN.MA		SYNCH DATE=12/11/2006	
PAGE TITLE			
POWER FETS			
		DRAWING NUMBER	SIZE
		051-7898	D
		REVISION	
		C.0.0	
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SYNC MASTER=NMARTIN		SYNC DATE=04/04/2008	
PAGE TITLE			
LVDS CONNECTOR			
	DRAWING NUMBER	051-7898	SIZE D
	REVISION	C.0.0	
	BRANCH		
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Port Power Switch

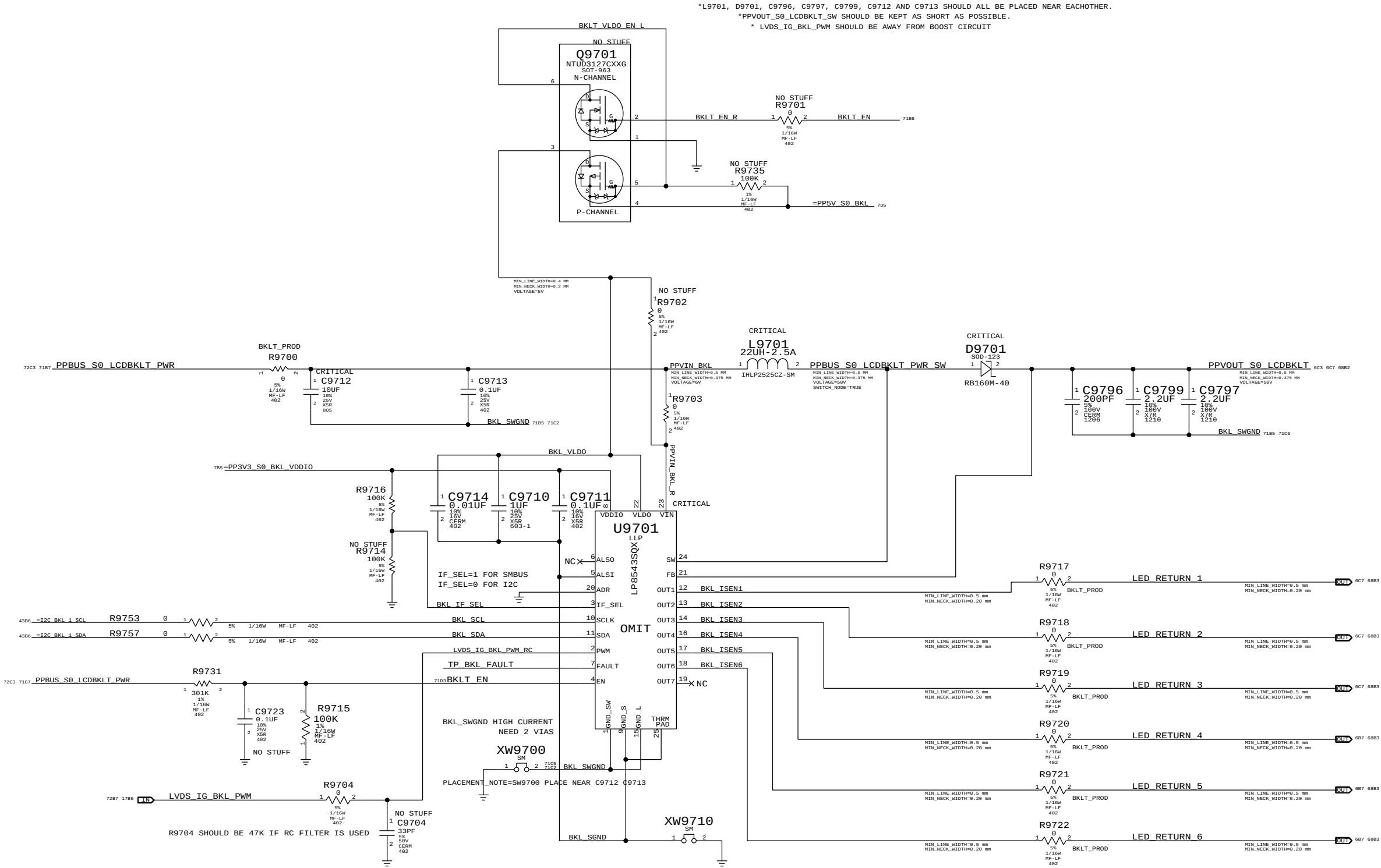


SYNC MASTER=AMASON		SYNC DATE=06/30/2008	
PAGE TITLE			
DisplayPort Connector		DRAWING NUMBER	
		051-7898 D	
		REVISION	
		C.0.0	
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PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
103S0198	6	RES, THIN FLIM, 1/16W, 10.2 OHM, 0.1, 0402, SM	R9717, R9718, R9719, R9720, R9721, R9722		BKLT_ENG
116S0005	1	RES, 1/16W, 0.1 OHM, 1%, 0402, SM	R9700		BKLT_ENG

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
353S2670	1	IC, LP8543, WHT LED BKLT CTRLR, QFN24, PROD	U9701	CRITICAL	

SYNC_MASTER=KIRAN		SYNC_DATE=12/05/2008	
PAGE TITLE			
LCD BACKLIGHT DRIVER			
		DRAWING NUMBER	SIZE
		051-7898	D
		REVISION	
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*L9701, D9701, C9796, C9797, C9799, C9712 AND C9713 SHOULD ALL BE PLACED NEAR EACHOTHER.
*PPVOUT_S0_LCDBKLT_SW SHOULD BE KEPT AS SHORT AS POSSIBLE.
* LVDS_IG_BKL_PWM SHOULD BE AWAY FROM BOOST CIRCUIT

D

C

10

B

A

A

D

CA

PCI Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_PCI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCI	*	=STANDARD	?
CLK_PCI	*	8 MIL	?

SOURCE: MCP79 Interface DG (DG-03328-001_v00), Section 2.8.

LPC Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
LPC	*	6 MIL	?
CLK_LPC	*	8 MIL	?

SOURCE: MCP79 Interface DG (DG-03328-001_v00), Section 2.9.1.

USB 2.0 Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MCP_USB_RBIAS	*	=STANDARD	8 MIL	8 MIL	=STANDARD	=STANDARD	=STANDARD
USB_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB	*	=2x_DIELECTRIC	?	USB	TOP,BOTTOM	=4x_DIELECTRIC	?

SOURCE: MCP79 Interface DG (DG-03328-001_v00), Section 2.10.1.

SMBus Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMB_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMB	*	=2x_DIELECTRIC	?

SOURCE: MCP79 Interface DG (DG-03328-001_v00), Section 2.11.1.

HD Audio Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
HDA	*	=2x_DIELECTRIC	?
MCP_HDA_COMP	*	8 MIL	?

SOURCE: MCP79 Interface DG (DG-03328-001_v00), Section 2.12.1.

SIO Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_SLOW_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_SLOW	*	8 MIL	?

SOURCE: MCP79 Interface DG (DG-03328-001_v00), Section 2.13.

SPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPI_5SS	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SPI	*	8 MIL	?

SOURCE: MCP79 Interface DG (DG-03328-001_v0D), Section 2.14.

ELECTRICAL_CONSTRAINT_SET		NET_TYPE			
		PHYSICAL	SPACING		
	MCP_DEBUG	PCT 55S	PCT	MCP_DEBUG<7..0>	12C3 1807
	PCT_AD	PCT 55S	PCT	PCT_AD<23..8>	
	PCT_AD24	PCT 55S	PCT	PCT_AD<24>	
	PCT_AD	PCT 55S	PCT	PCT_AD<31..25>	
	PCT_AD	PCT 55S	PCT	PCT_PAR	
	PCT_C_BE_I	PCT 55S	PCT	PCT_C_BE_I<3..0>	
	PCT_CNTL	PCT 55S	PCT	PCT_IROY_L	
	PCT_CNTL	PCT 55S	PCT	PCT_DEVSEL_L	
	PCT_CNTL	PCT 55S	PCT	PCT_PERR_L	
	PCT_CNTL	PCT 55S	PCT	PCT_SERR_L	
	PCT_CNTL	PCT 55S	PCT	PCT_STOP_L	
	PCT_CNTL	PCT 55S	PCT	PCT_TRDY_L	
	PCT_CNTL	PCT 55S	PCT	PCT_FRAME_L	
	PCT_REQ0_L	PCT 55S	PCT	PCT_REQ0_L	
	PCT_GNT0_I	PCT 55S	PCT	PCT_GNT0_L	1802 1807
	PCT_REQ1_I	PCT 55S	PCT	PCT_REQ1_L	
	PCT_GNT1_I	PCT 55S	PCT	PCT_GNT1_L	1802 1807
	PCT_INTW_L	PCT 55S	PCT	PCT_INTW_L	
	PCT_INTX_L	PCT 55S	PCT	PCT_INTX_L	
	PCT_INTY_L	PCT 55S	PCT	PCT_INTY_L	
	PCT_INTZ_L	PCT 55S	PCT	PCT_INTZ_L	
	MCP_PCT_CLK2	CLK PCT 55S	CLK_PCT	PCI_CLK33M MCP_R	18C5
	MCP_PCT_CLK2	CLK PCT 55S	CLK_PCT	PCI_CLK33M MCP	18C5
	LPC_AD	LPC 55S	LPC	LPC_AD<3..0>	18B3 40C8 4203
	LPC_FRAME_I	LPC 55S	LPC	LPC_FRAME_L	18C3 40C8 4203
	LPC_RESET_I	LPC 55S	LPC	LPC_RESET_L	18C3 2404
	MCP_LPC_CLK0	CLK LPC 55S	CLK_LPC	LPC_CLK33M SMC_R	18B3 24B4
	MCP_LPC_CLK0	CLK LPC 55S	CLK_LPC	LPC_CLK33M SMC	24B1 40C8
	MCP_LPC_CLK0	CLK LPC 55S	CLK_LPC	LPC_CLK33M LPCPLUS	24B1 4203
	USB_EXTA	USB 90D	USB	USB_EXTA_P	1903 38A8
	USB_EXTA	USB 90D	USB	USB_EXTA_N	1903 38A8
	USB_EXTA	USB 90D	USB	USB_EXTA_MXED_P	38C4
	USB_EXTA	USB 90D	USB	USB_EXTA_MXED_N	38C4
	USB_EXTA	USB 90D	USB	CONN_USB_EXTA_P	38C3
	USB_EXTA	USB 90D	USB	CONN_USB_EXTA_N	38C3
	USB_CAMERA	USB 90D	USB	USB_CAMERA_P	1903 29B5
	USB_CAMERA	USB 90D	USB	USB_CAMERA_N	1903 29B5
	USB_CAMERA	USB 90D	USB	USB_CAMERA_CONN_P	605 29B7
	USB_CAMERA	USB 90D	USB	USB_CAMERA_CONN_N	605 29B7
	USB_BT	USB 90D	USB	USB_BT_P	1903 29B5
	USB_BT	USB 90D	USB	USB_BT_N	19C3 29B5
	USB_BT	USB 90D	USB	CONN_USB2_BT_P	605 29B7
	USB_BT	USB 90D	USB	CONN_USB2_BT_N	605 29B7
	USB_TPAD	USB 90D	USB	USB_TPAD_P	1903 48B8
	USB_TPAD	USB 90D	USB	USB_TPAD_N	1903 48B8
	USB_TPAD	USB 90D	USB	USB_TPAD_R_P	48B7
	USB_TPAD	USB 90D	USB	USB_TPAD_R_N	48B7
	USB_IR	USB 90D	USB	USB_IR_P	1903 3907
	USB_IR	USB 90D	USB	USB_IR_N	1903 3907
	USB_EXTB	USB 90D	USB	USB_EXTB_P	19C3 38A4
	USB_EXTB	USB 90D	USB	USB_EXTB_N	19C3 38B4
	USB_EXTB	USB 90D	USB	CONN_USB_EXTB_P	38B3
	USB_EXTB	USB 90D	USB	CONN_USB_EXTB_N	38B3
	USB_SD	USB 90D	USB	USB_CARDREADER_P	19C3 30C7
	USB_SD	USB 90D	USB	USB_CARDREADER_N	19C3 30C7
	MCP_USB_RB1AS	MCP_USB_RB1AS		MCP_USB_RB1AS_GND	19C4
	SMBUS_MCP_0_CLK	SMB 55S	SMB	SMBUS_MCP_0_CLK	12B6 26C3 43B8
	SMBUS_MCP_0_DATA	SMB 55S	SMB	SMBUS_MCP_0_DATA	12B6 26C3 43B8
	SMBUS_MCP_1_CLK	SMB 55S	SMB	SMBUS_MCP_1_CLK	26C3 43B8
	SMBUS_MCP_1_DATA	SMB 55S	SMB	SMBUS_MCP_1_DATA	26C3 43B8
	HDA_BIT_CLK	HDA 55S	HDA	HDA_BIT_CLK	2602 52C7
	HDA_BIT_CLK_R	HDA 55S	HDA	HDA_BIT_CLK_R	26A7 26D4
	HDA_SYNC	HDA 55S	HDA	HDA_SYNC	2602 52C7
	HDA_SYNC_R	HDA 55S	HDA	HDA_SYNC_R	26A7 26D4
	HDA_RST_L	HDA 55S	HDA	HDA_RST_R_L	26A7 26D4
	HDA_RST_L	HDA 55S	HDA	HDA_RST_L	2602 52C7
	HDA_SD1N0	HDA 55S	HDA	HDA_SD1N0	2607 52C7
	HDA_SD1N0	HDA 55S	HDA	HDA_SD1N_CODEC	
	HDA_SDOUT	HDA 55S	HDA	HDA_SDOUT	2602 52C7
	HDA_SDOUT	HDA 55S	HDA	HDA_SDOUT_R	26A7 26D4
	MCP_HDA_PULLDN_COMP		MCP_HDA_COMP	MCP_HDA_PULLDN_COMP	26C7
	MCP_SUS_CLK	CLK_SLOW 55S	CLK_SLOW	PM_CLK32K_SUSCLK_R	26B3 24B4
	MCP_SUS_CLK	CLK_SLOW 55S	CLK_SLOW	PM_CLK32K_SUSCLK	24B1 40C5
	SPT_CLK	SPT 55S	SPT	SPT_CLK_R	26B3 42A5 42C6
	SPT_CLK	SPT 55S	SPT	SPT_CLK	51C5
	SPT_CLK	SPT 55S	SPT	SPT_ALT_CLK	42C5 42D3
	SPT_MOST	SPT 55S	SPT	SPT_MOST_I	26B3 42A5 42C7
	SPT_MOST	SPT 55S	SPT	SPT_MOST	51C4
	SPT_MOST	SPT 55S	SPT	SPT_ALT_MOST	42C5 42D5
	SPT_MISO	SPT 55S	SPT	SPT_MISO	26B3 42A5 42B7
	SPT_MISO	SPT 55S	SPT	SPT_MISO_R	51C4
	SPT_MISO	SPT 55S	SPT	SPT_ALT_MISO	42B5 42D5
	SPT_CS0	SPT 55S	SPT	SPT_CS0_R_L	26B3 42B7
	SPT_CS0	SPT 55S	SPT	SPT_CS0_L	
		SPT 55S	SPT	SPT_CS1_R_L	
		SPT 55S	SPT	SPT_CS1_R_L_USE_MLB	

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MCP RGMII (Ethernet) Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MCP_MII_COMP	*	=STANDARD	7.5 MIL	7.5 MIL	=STANDARD	=STANDARD	=STANDARD
ENET_MII_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MCP_BUF0_CLK	*	=3:1_SPACING	?
ENET_MII	*	12 MIL	?

SOURCE: MCP73 Interface DG (DG-02974-001_v01), Sections 2.7.2 & 2.7.4

88E1116R (Ethernet PHY) Constraints

[illegible]

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENET_MDI	*	25 MIL	?

SOURCE: MCP73 Interface DG (DG-02974-001_v01), Section 2.7.4

ELECTRICAL_CONSTRAINT_SET		NET_TYPE			
		PHYSICAL	SPACING		
0000	MCP_MII_COMP	MCP_MII_COMP		MCP_MII_COMP_VDD	17C6
0000	MCP_MII_COMP	MCP_MII_COMP		MCP_MII_COMP_GND	17C6
0000	MCP_CLK25M_BUF0	ENET_MII_55S	MCP_BUF0_CLK	RTL25M_BUF0_R	17C3 32A5
0000		ENET_MII_55S	MCP_BUF0_CLK	RTL25M_CLK25M_CKXTAL1	31B6 32A3
0000	ENET_INTR_L	ENET_MII_55S	ENET_MII	ENET_INTR_L	
0000	ENET_MDIO	ENET_MII_55S	ENET_MII	ENET_MDIO	17C3 31B6
0000	ENET_MDC	ENET_MII_55S	ENET_MII	ENET_MDC	17D3 31B6
0000	ENET_PWDOWN_L	ENET_MII_55S	ENET_MII	ENET_PWDOWN_L	
0000		ENET_MII_55S	ENET_MII	ENET_CLK125M_RXCLK_R	31C4
0000	ENET_RXCLK	ENET_MII_55S	ENET_MII	ENET_CLK125M_RXCLK	17D6 31C1
0000		ENET_MII_55S	ENET_MII	ENET_RXD_R<3..0>	31C4
0000	ENET_RXD	ENET_MII_55S	ENET_MII	ENET_RXD<0>	17D6 31C1
0000	ENET_RXD_STRAP	ENET_MII_55S	ENET_MII	ENET_RXD<3..1>	17D6 31C1
0000	ENET_RXD	ENET_MII_55S	ENET_MII	ENET_RX_CTRL	17D6 31B1
0000		ENET_MII_55S	ENET_MII	ENET_RXCTL_R	31B4
0000		ENET_MII_55S	ENET_MII	ENET_CLK125M_TXCLK_R	31C6
0000	ENET_TXCLK	ENET_MII_55S	ENET_MII	ENET_CLK125M_TXCLK	17D3 31C8
0000	ENET_TXD0	ENET_MII_55S	ENET_MII	ENET_TXD<0>	17D3 31C6
0000	ENET_TXD	ENET_MII_55S	ENET_MII	ENET_TXD<3..1>	17D3 31C6
0000	ENET_TXD	ENET_MII_55S	ENET_MII	ENET_TX_CTRL	17D3 31B6
0000		ENET_MII_55S	ENET_MII	ENET_RESET_L	17C3 31B7
0000	ENET_MDI	ENET_MDI_10B0	ENET_MDI	ENET_MDI_P<3..0>	31B3 33B0 33C8
0000		ENET_MDI_10B0	ENET_MDI	ENET_MDI_N<3..0>	31B3 33B0 33C8
0000		ENET_MDI_10B0	ENET_MDI	ENET_MDI_TRAN_P<3..0>	33B4 33C4 33C5
0000		ENET_MDI_10B0	ENET_MDI	ENET_MDI_TRAN_N<3..0>	33B4 33C4 33C5

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PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
1T01_DIFFPAIR	*	=STANDARD	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.1 MM

SMC SMBus Net Properties

NET_TYPE				
ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING		
SMBUS_SMC_A_S3_SCL	SMB_55G	SMB	SMBUS_SMC_A_S3_SCL	6C5 605 4302
SMBUS_SMC_A_S3_SDA	SMB_55G	SMB	SMBUS_SMC_A_S3_SDA	6C5 605 4302
SMBUS_SMC_B_S0_SCL	SMB_55G	SMB	SMBUS_SMC_B_S0_SCL	43C2
SMBUS_SMC_B_S0_SDA	SMB_55G	SMB	SMBUS_SMC_B_S0_SDA	43C2
SMBUS_SMC_0_S0_SCL	SMB_55G	SMB	SMBUS_SMC_0_S0_SCL	4305
SMBUS_SMC_0_S0_SDA	SMB_55G	SMB	SMBUS_SMC_0_S0_SDA	4305
SMBUS_SMC_BSA_SCL	SMB_55G	SMB	SMBUS_SMC_BSA_SCL	6A7 43C5
SMBUS_SMC_BSA_SDA	SMB_55G	SMB	SMBUS_SMC_BSA_SDA	6A7 43C5
SMBUS_SMC_MGMT_SCL	SMB_55G	SMB	SMBUS_SMC_MGMT_SCL	43B5
SMBUS_SMC_MGMT_SDA	SMB_55G	SMB	SMBUS_SMC_MGMT_SDA	43B5

SMBus Charger Net Properties

NET_TYPE				
ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING		
CHGR_CSI	1T01_DIFFPAIR		CHGR_CSI_P	
CHGR_CSI	1T01_DIFFPAIR		CHGR_CSI_N	
CHGR_CS0	1T01_DIFFPAIR		CHGR_CS0_P	
CHGR_CS0	1T01_DIFFPAIR		CHGR_CS0_N	

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